Further, the structure of power system is shown by a single line diagram. The complete 3-phase circuit is seldom necessary to convey even the most detailed information about the system. In fact, the complete diagram is more likely to hide than to clarify the information we are seeking from the system viewpoint.

7.3 Comparison of D.C. and A.C. Transmission

The electric power can be transmitted either by means of d.c. or a.c. Each system has its own merits and demerits. It is, therefore, desirable to discuss the technical advantages and disadvantages of the two systems for transmission of electric power.

1. D.C. transmission. For some years past, the transmission of electric power by d.c. has been receiving the active consideration of engineers due to its numerous advantages.

Advantages. The high voltage d.c. transmission has the following advantages over high voltage a.c. transmission :

(*i*) It requires only two conductors as compared to three for a.c. transmission.

(*ii*) There is no inductance, capacitance, phase displacement and surge problems in d.c. transmission.

(*iii*) Due to the absence of inductance, the voltage drop in a d.c. transmission line is less than the a.c. line for the same load and sending end voltage. For this reason, a d.c. transmission line has better voltage regulation.

(*iv*) There is no skin effect in a d.c. system. Therefore, entire cross-section of the line conductor is utilised.

(*v*) For the same working voltage, the potential stress on the insulation is less in case of d.c. system than that in a.c. system. Therefore, a d.c. line requires less insulation.

(*vi*) A d.c. line has less corona loss and reduced interference with communication circuits.

(*vii*) The high voltage d.c. transmission is free from the dielectric losses, particularly in the case of cables.

(*viii*) In d.c. transmission, there are no stability problems and synchronising difficulties.

Disadvantages

(*i*) Electric power cannot be generated at high d.c. voltage due to commutation problems.

(*ii*) The d.c. voltage cannot be stepped up for transmission of power at high voltages.

(*iii*) The d.c. switches and circuit breakers have their own limitations.

2. A.C. transmission. Now-a-days, electrical energy is almost exclusively generated, transmitted and distributed in the form of a.c.

Advantages

(*i*) The power can be generated at high voltages.

(*ii*) The maintenance of a.c. sub-stations is easy and cheaper.

(*iii*) The a.c. voltage can be stepped up or stepped down by transformers with ease and efficiency. This permits to transmit power at high voltages and distribute it at safe potentials.

Disadvantages

(*i*) An a.c. line requires more copper than a d.c. line.

(*ii*) The construction of a.c. transmission line is more complicated than a d.c. transmission line.

(*iii*) Due to skin effect in the a.c. system, the effective resistance of the line is increased.

(*iv*) An a.c. line has capacitance. Therefore, there is a continuous loss of power due to charging current even when the line is open.

Conclusion. From the above comparison, it is clear that high voltage d.c. transmission is superior to high voltage a.c. transmission. Although at present, transmission of electric power is carried by a.c., there is an increasing interest in d.c. transmission. The introduction of mercury arc rectifiers and thyratrons have made it possible to convert a.c. into d.c. and *vice-versa* easily and efficiently. Such devices can operate upto 30 MW at 400 kV in single units. The present day trend is towards a.c. for generation and distribution and high voltage d.c. for transmission.

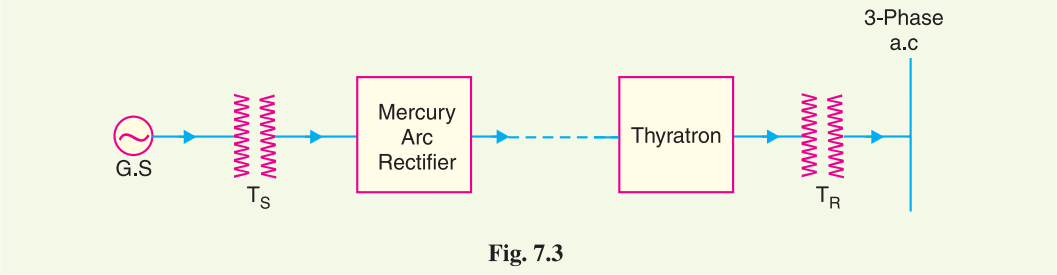


Fig. 7.3

Fig. 7.3 shows the single line diagram of high voltage d.c. transmission. The electric power is generated as a.c. and is stepped up to high voltage by the sending end transformer T_S. The a.c. power at high voltage is fed to the mercury arc rectifiers which convert a.c. into d.c. The transmission of electric power is carried at high d.c. voltage. At the receiving end, d.c. is converted into a.c. with the help of thyratrons. The a.c. supply is stepped down to low voltage by receiving end transformer T_R for distribution.

7.4 Advantages of High Transmission Voltage

The transmission of electric power is carried at high voltages due to the following reasons :

***(i)* Reduces volume of conductor material.** Consider the transmission of electric power by a three-phase line.

Let P = power transmitted in watts
V = line voltage in volts
$\cos\phi$ = power factor of the load
l = length of the line in metres
R = resistance per conductor in ohms
ρ = resistivity of conductor material
a = area of X-section of conductor

$$\text{Load current, } I = \frac{P}{\sqrt{3}\, V \cos\phi}$$

$$\text{Resistance/conductor, } R = \rho l/a$$

$$\text{Total power loss, } W = 3I^2 R = 3\left(\frac{P}{\sqrt{3}\, V\cos\phi}\right)^2 \times \frac{\rho l}{a}$$

$$= \frac{P^2 \rho l}{V^2 \cos^2\phi\, a}$$

$$\therefore \quad \text{Area of X-section, } a = \frac{P^2 \rho l}{W V^2 \cos^2\phi}$$

Total volume of conductor material required

$$= 3\, a\, l = 3\left(\frac{P^2 \rho\, l}{W V^2 \cos^2\phi}\right) l$$

$$= \frac{3P^2\rho l^2}{WV^2\cos^2\phi} \qquad ...(i)$$

It is clear from exp. (*i*) that for given values of P, l, ρ and W, the volume of conductor material required is inversely proportional to the square of transmission voltage and power factor. In other words, the greater the transmission voltage, the lesser is the conductor material required.

(*ii*) Increases transmission efficiency

$$\text{Input power} = P + \text{Total losses}$$

$$= P + \frac{P^2\rho l}{V^2\cos^2\phi\, a}$$

Assuming J to be the current density of the conductor, then,

$$a = I/J$$

$$\therefore \quad \text{Input power} = P + \frac{P^2\rho l J}{V^2\cos^2\phi\, I} = P + \frac{P^2\rho l J}{V^2\cos^2\phi} \times \frac{1}{I}$$

$$= P + \frac{P^2\rho l J}{V^2\cos^2\phi} \times \frac{*\sqrt{3}V\cos\phi}{P}$$

$$= P + \frac{\sqrt{3}\,P\,J\,\rho l}{V\cos\phi} = P\left[1 + \frac{\sqrt{3}\;J\rho l}{V\cos\phi}\right]$$

$$\text{Transmission efficiency} = \frac{\text{Output power}}{\text{Input power}} = \frac{P}{P\left[1 + \frac{\sqrt{3}\;J\rho l}{V\cos\phi}\right]} = \frac{1}{\left[1 + \frac{\sqrt{3}\;J\rho l}{V\cos\phi}\right]}$$

$$= {**}\left[1 - \frac{\sqrt{3}\;J\rho l}{V\cos\phi}\right] \text{ approx.} \qquad ...(ii)$$

As J, ρ and l are constants, therefore, transmission efficiency increases when the line voltage is increased.

(*iii*) Decreases percentage line drop

$$\text{Line drop} = IR = I \times \frac{\rho l}{a}$$

$$= I \times \rho l \times J/I = \rho l J \qquad [\because a = I/J]$$

$$\%\text{age line drop} = \frac{J\rho l}{V} \times 100 \qquad ...(iii)$$

As J, ρ and l are constants, therefore, percentage line drop decreases when the transmission voltage increases.

Limitations of high transmission voltage. From the above discussion, it might appear advisable to use the highest possible voltage for transmission of power in a bid to save conductor material. However, it must be realised that high transmission voltage results in

(*i*) the increased cost of insulating the conductors

(*ii*) the increased cost of transformers, switchgear and other terminal apparatus.

Therefore, there is a limit to the higher transmission voltage which can be economically employed in a particular case. This limit is reached when the saving in cost of conductor material due to

---

* $I = \dfrac{P}{\sqrt{3}V\cos\phi}$

** Binomial theorem.

higher voltage is offset by the increased cost of insulation, transformer, switchgear etc. Hence, the choice of proper transmission voltage is essentially a question of economics. The reader may find further discussion on this topic later in this chapter.

7.5 Various Systems of Power Transmission

It has already been pointed out that for transmission of electric power, 3-phase, 3-wire a.c. system is universally adopted. However, other systems can also be used for transmission under special circumstances. The different possible systems of transmission are :

1. **D.C. system**
   - ***(i)*** D.C. two-wire.
   - ***(ii)*** D.C. two-wire with mid-point earthed.
   - ***(iii)*** D.C. three-wire.
2. **Single-phase A.C. system**
   - ***(i)*** Single-phase two-wire.
   - ***(ii)*** Single-phase two-wire with mid-point earthed.
   - ***(iii)*** Single-phase three-wire.
3. **Two-phase A.C. system**
   - ***(i)*** Two-phase four-wire.
   - ***(ii)*** Two-phase three wire.
4. **Three-phase A.C. system**
   - ***(i)*** Three-phase three-wire.
   - ***(ii)*** Three-phase four-wire.

From the above possible systems of power transmission, it is difficult to say which is the best system unless and until some method of comparison is adopted. Now, the cost of conductor material is one of the most important charges in a system. Obviously, the best system for transmission of power is that for which the volume of conductor material required is minimum. Therefore, the volume of conductor material required forms the basis of comparison between different systems.

While comparing the amount of conductor material required in various systems, the proper comparison shall be on the basis of equal maximum stress on the *dielectric. There are two cases :

(*i*) *When transmission is by overhead system.* In the overhead system, the maximum disruptive stress** exists between the conductor and the earth. Therefore, the comparison of the system in this case has to be made on the basis of maximum voltage between conductor and earth.

(*ii*) *When transmission is by underground system.* In the underground system, the chief stress on the insulation is between conductors. Therefore, the comparison of the systems in this case should be made on the basis of maximum potential difference between conductors.

7.6 Comparison of Conductor Material in Overhead System

In comparing the relative amounts of conductor material necessary for different systems of transmission, similar conditions will be assumed in each case *viz*.,

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* In long transmission lines, the voltage is only limited by the problem of insulating the conductors against disruptive discharge. Therefore, comparison should be on the basis of equality of maximum potential difference *i.e*., equal maximum stress on the dielectric.

** In overhead system, insulation between conductors whether at the supports or intermediate points is always provided by suitably spacing the conductors. Therefore, electric dischrage cannot occur between conductors. However, the insulation has to be provided between the conductor and supporting structure. Therefore, maximum stress is between conductor and earth.

(*i*) same power (*P watts*) transmitted by each system.

(*ii*) the distance (*l metres*) over which power is transmitted remains the same.

(*iii*) the line losses (*W watts*) are the same in each case.

(*iv*) the maximum voltage between any conductor and earth (V_m) is the same in each case.

1. Two-wire d.c. system with one conductor earthed

In the 2-wire d.c. system, one is the outgoing or positive wire and the other is the return or negative wire as shown in Fig. 7.4. The load is connected between the two wires.

Max. voltage between conductors $= V_m$

Power to be transmitted $= P$

$\therefore$ Load current, $I_1 = P/V_m$

If R_1 is the resistance of each line conductor, then,

$$R_1 = \rho\, l/a_1$$

where a_1 is the area of X-section of the conductor.

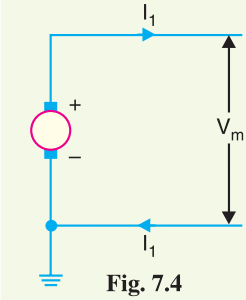


Fig. 7.4

$$\text{Line losses,} \quad W = 2I_1^2 R_1 = 2\left(\frac{P}{V_m}\right)^2 \rho \frac{l}{a_1}$$

$$\therefore \quad \text{Area of X-section, } a_1 = \frac{2P^2\rho l}{W V_m^2}$$

Volume of conductor material required

$$= 2a_1 l = 2\left(\frac{2P^2\rho l}{W V_m^2}\right) l = \frac{4P^2\rho l^2}{W V_m^2}$$

It is a usual practice to make this system as the basis for comparison with other systems. Therefore, volume of conductor material required in this system shall be taken as the basic quantity *i.e.*

$$\frac{4P^2\rho l^2}{W V_m^2} = K \text{ (say)}$$

2. Two-wire d.c. system with mid-point earthed. Fig. 7.5 shows the two-wire d.c. system with mid-point earthed. The maximum voltage between any conductor and earth is V_m so that maximum voltage between conductors is $2V_m$.

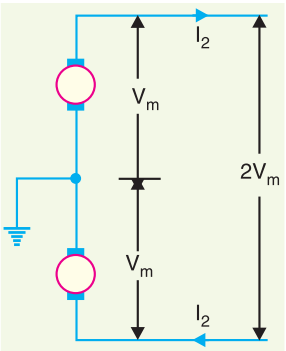


Fig. 7.5

$$\text{Load current,} \quad I_2 = P/2V_m$$

Let a_2 be the area of X-section of the conductor.

$$\text{Line losses, } W = 2I_2^2 R_2 = 2\left(\frac{P}{2V_m}\right)^2 \times \frac{\rho l}{a_2}$$

$[\because R_2 = \rho\, l/a_2]$

$$\therefore \quad W = \frac{P^2\rho l}{2a_2 V_m^2}$$

$$\therefore \quad \text{Area of X-section, } a_2 = \frac{P^2\rho l}{2W V_m^2}$$

$\therefore$ Volume of conductor material required

$$= 2a_2 l = 2\left(\frac{P^2\rho l}{2W V_m^2}\right) l = \frac{P^2\rho l^2}{W V_m^2}$$

$$= \frac{K}{4} \qquad \left[\because K = \frac{4P^2\rho l^2}{W V_m^2}\right]$$

Hence, the volume of conductor material required in this system is *one-fourth* of that required in a two-wire d.c. system with one conductor earthed.

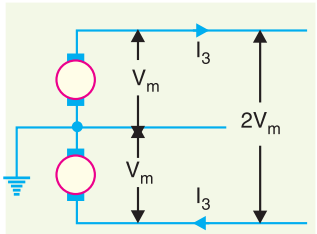


Fig. 7.6

3. Three-wire d.c. system. In a 3-wire d.c. system, there are two outers and a middle or neutral wire which is earthed at the generator end as shown in Fig. 7.6. If the load is balanced, the current in the neutral wire is zero. Assuming balanced loads,

Load current, $I_3 = P/2V_m$

Let a_3 be the area of X-section of each outer wire.

$$\text{Line losses, } W = 2I_3^2 R_3 = 2\left(\frac{P}{2V_m}\right)^2 \times \rho\frac{l}{a_3} = \frac{P^2\rho l}{2V_m^2 a_3}$$

$$\therefore \quad \text{Area of X-section, } a_3 = \frac{P^2\rho l}{2W V_m^2}$$

Assuming the area of X-section of neutral wire to be half that of the outer wire,

Volume of conductor material required

$$= 2{\cdot}5\, a_3 l = 2{\cdot}5\left(\frac{P^2\rho l}{2W V_m^2}\right) l = \frac{2{\cdot}5}{2}\left(\frac{P^2\rho l^2}{W V_m^2}\right)$$

$$= \frac{5}{16} K \qquad \left[\because K = \frac{4P^2\rho l^2}{W V_m^2}\right]$$

Hence the volume of conductor material required in this system is 5/16th of what is required for a 2-wire d.c. system with one conductor earthed.

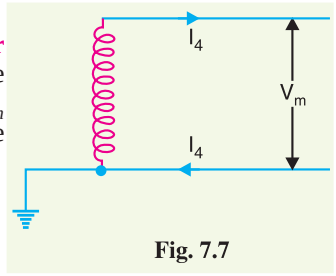


Fig. 7.7

4. Single phase 2-wire a.c. system with one conductor earthed. Fig. 7.7 shows a single phase 2-wire a.c. system with one conductor earthed. The maximum voltage between conductors is V_m so that r.m.s. value of voltage between them is $V_m / \sqrt{2}$. Assuming the load power factor to be cos ϕ,

$$\text{Load current, } I_4 = \frac{P}{\left(V_m/\sqrt{2}\right)\cos\phi} = \frac{\sqrt{2}\,P}{V_m \cos\phi}$$

Let a_4 be the area of X-section of the conductor.

$$\therefore \quad \text{Line losses, } W = 2I_4^2 R_4 = 2\left(\frac{\sqrt{2}\,P}{V_m\cos\phi}\right)^2 \times \frac{\rho l}{a_4} = \frac{4P^2\rho l}{\cos^2\phi\, V_m^2 a_4}$$

$$\therefore \quad \text{Area of X-section, } a_4 = \frac{4P^2\rho l}{\cos^2\phi\, W V_m^2}$$

Volume of conductor material required

$$= 2a_4 l = 2\left(\frac{4P^2\rho l}{V_m^2 W\cos^2\phi}\right) l$$

$$= \frac{2}{\cos^2\phi} \times \frac{4P^2\rho l^2}{W V_m^2}$$

$$= \frac{2K}{\cos^2\phi} \qquad \left[\because K = \frac{4P^2\rho l^2}{W V_m^2}\right]$$

Hence, the volume of conductor material required in this system is $2/\cos^2\phi$ times that of 2-wire d.c. system with the one conductor earthed.

5. Single phase 2-wire system with mid-point earthed. Fig. 7.8 shows a single phase a.c. system with mid-point earthed. The two wires possess equal and opposite voltages to earth (*i.e.*, V_m). Therefore, the maximum voltage between the two wires is $2V_m$. The r.m.s. value of voltage between conductors is $= 2V_m / \sqrt{2} = \sqrt{2}V_m$. Assuming the power factor of the load to be $\cos\phi$,

$$\text{Load current, } I_5 = \frac{P}{\sqrt{2}\, V_m \cos\phi}$$

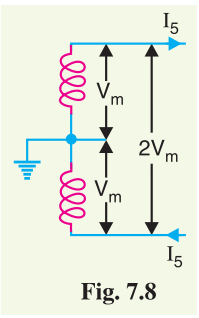


Fig. 7.8

Let a_5 be the area of X-section of the conductor.

$$\text{Line losses, } W = 2 I_5^2 R_5 = 2\left(\frac{P}{\sqrt{2}\, V_m \cos\phi}\right)^2 R_5$$

$$\therefore \quad W = \frac{P^2 \rho\, l}{a_5\, V_m^2 \cos^2\phi}$$

$$\therefore \quad \text{Area of X-section, } a_5 = \frac{P^2 \rho\, l}{W\, V_m^2 \cos^2\phi}$$

Volume of conductor material required

$$= 2\, a_5\, l = 2\left(\frac{P^2 \rho\, l}{W\, V_m^2 \cos^2\phi}\right) l = \frac{2\, P^2 \rho\, l^2}{W\, V_m^2 \cos^2\phi}$$

$$= \frac{2}{\cos^2\phi} \times \frac{P^2 \rho l^2}{W\, V_m^2}$$

$$= \frac{K}{2\cos^2\phi} \qquad \left[\because K = \frac{4\, P^2 \rho\, l^2}{W\, V_m^2}\right]$$

Hence the volume of conductor material required in this system is $1/2 \cos^2\phi$ times that of 2-wire d.c. system with one conductor earthed.

6. Single phase, 3-wire system. The single phase 3-wire system is identical in principle with 3-wire d.c. system. The system consists of two outers and neutral wire taken from the mid-point of the phase winding as shown in Fig. 7.9. If the load is balanced, the current through the neutral wire is zero. Assuming balanced load,

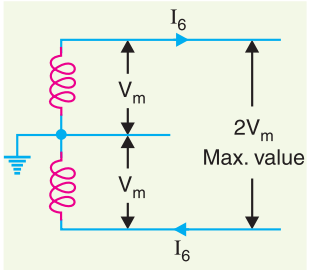


Fig. 7.9

Max. voltage between conductors $= 2\, V_m$

R.M.S.value of voltage between conductors $= 2V_m / \sqrt{2} = \sqrt{2}V_m$

If the p.f of the load is $\cos\phi$, then,

$$\text{Load current, } I_6 = \frac{P}{\sqrt{2}\, V_m \cos\phi}$$

Let a_6 be the area of X-section of each outer conductor.

$$\text{Line losses, } W = 2 I_6^2 R_6 = 2\left(\frac{P}{\sqrt{2}\, V_m \cos\phi}\right)^2 \times \frac{\rho\, l}{a_6}$$

$$= \frac{P^2 \rho\, l}{a_6\, V_m^2 \cos^2\phi}$$

$$\therefore \quad \text{Area of X-section, } a_6 = \frac{P^2\, \rho\, l}{W\, V_m^2 \cos^2\phi}$$

Assuming the area of X-section of neutral wire to be half that of the outer wire,

Volume of conductor material required

$$= 2{\cdot}5\, a_6\, l = 2{\cdot}5 \left(\frac{P^2\, \rho\, l}{W\, V_m^2\, \cos^2 \phi} \right) l = \frac{2 \cdot 5\, P^2\, \rho\, l^2}{W\, V_m^2 \cos^2 \phi}$$

$$= \frac{2 \cdot 5}{\cos^2 \phi} \times \frac{P^2\, \rho\, l^2}{W\, V_m^2}$$

$$= \frac{5K}{8 \cos^2 \phi} \qquad \left[\because K = \frac{4P^2\, \rho\, l^2}{W\, V_m^2} \right]$$

Hence, the volume of conductor material required in this system is 5/8 $\cos^2 \phi$ times that required in a 2-wire d.c. system with one conductor earthed.

7. Two phase, 4-wire a.c. system. As shown in Fig. 7.10, the four wires are taken from the ends of the two-phase windings and the mid-points of the two windings are connected together. This system can be considered as two independent single phase systems, each transmitting one half of the total power.

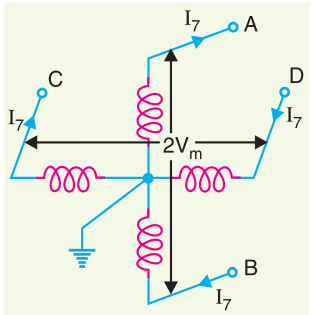


Fig. 7.10

Max. voltage between outers A and $B = 2V_m$

R.M.S. value of voltage $= 2V_m / \sqrt{2} = \sqrt{2}\, V_m$

Power supplied per phase (*i.e.*, by outers A and B) $= P/2$

Assuming p.f. of the load to be $\cos \phi$,

$$\text{Load current, } I_7 = \frac{P/2}{\sqrt{2}\, V_m \cos \phi} = \frac{P}{2\sqrt{2}\, V_m \cos \phi}$$

Let a_7 be the area of X-section of one conductor.

$$\text{Line losses, } W = 4\, I_7^2\, R_7 = 4 \left(\frac{P}{2\sqrt{2}\, V_m \cos \phi} \right)^2 \times \frac{\rho\, l}{a_7}$$

$$\therefore \quad W = \frac{P^2\, \rho\, l}{2a_7\, V_m^2 \cos^2 \phi}$$

$$\therefore \quad \text{Area of X-section, } a_7 = \frac{P^2\, \rho\, l}{2\, W\, V_m^2 \cos^2 \phi}$$

$\therefore$ Volume of conductor material required

$$= 4\, a_7\, l$$

$$= 4 \left(\frac{P^2\, \rho\, l}{2\, W\, V_m^2 \cos^2 \phi} \right) l = \frac{4P^2 \rho\, l^2}{2\, W\, V_m^2 \cos^2 \phi}$$

$$= \frac{1}{2 \cos^2 \phi} \times \frac{4P^2\, \rho\, l^2}{W\, V_m^2}$$

$$= \frac{K}{2 \cos^2 \phi} \qquad \left[\because K = \frac{4P^2\, \rho\, l^2}{W\, V_m^2} \right]$$

Hence, the volume of conductor material required for this system is $1/2 \cos^2 \phi$ times that of 2-wire d.c. system with one conductor earthed.

8. Two-phase, 3-wire system. Fig. 7.11 shows two-phase, 3-wire a.c. system. The third or neutral wire is taken from the junction of two-phase windings whose voltages are in quadrature with each other. Obviously, each phase transmits one half of the total power. The R.M.S. voltage between

outgoing conductor and neutral

$$= V_m/\sqrt{2}$$

$$\therefore \quad \text{Current in each outer, } I_8 = \frac{P/2}{\frac{V_m}{\sqrt{2}}\cos\phi} = \frac{P}{\sqrt{2}\,V_m\cos\phi}$$

$$\text{Current in neutral* wire} = \sqrt{I_8^2 + I_8^2} = \sqrt{2}I_8$$

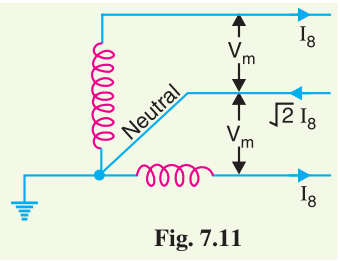


Fig. 7.11

Assuming the current density to be constant, the area of X-section of the neutral wire will be $\sqrt{2}$ ** times that of either of the outers.

$$\therefore \quad \text{Resistance of neutral wire} = \frac{R_8}{\sqrt{2}} = \frac{\rho l}{\sqrt{2}\,a_8}$$

$$\text{Line losses, } W = 2I_8^2 R_8 + \left(\sqrt{2}\,I_8\right)^2 \frac{R_8}{\sqrt{2}} = I_8^2 R_8\left(2+\sqrt{2}\right)$$

$$= \left(\frac{P}{\sqrt{2}\,V_m\cos\phi}\right)^2 \times \frac{\rho l}{a_8}\left(2+\sqrt{2}\right)$$

$$\therefore \quad W = \frac{P^2\rho l}{2\,a_8\,V_m^2\cos^2\phi}\left(2+\sqrt{2}\right)$$

$$\therefore \quad \text{Area of X-section, } a_8 = \frac{P^2\,\rho\,l}{2W\,V_m^2\cos^2\phi}\left(2+\sqrt{2}\right)$$

Volume of conductor material required

$$= 2\,a_8\,l + \sqrt{2}\,a_8\,l = a_8\,l\left(2+\sqrt{2}\right)$$

$$= \frac{P^2\,\rho\,l^2}{2W\,V_m^2\cos^2\phi}\left(2+\sqrt{2}\right)^2$$

$$= \frac{1\cdot457}{\cos^2\phi}K \qquad \left[\because K = \frac{4P^2\,\rho\,l^2}{W\,V_m^2}\right]$$

Hence, the volume of conductor material required for this system is $1\cdot457/\cos^2\phi$ times that of 2-wire d.c. system with one conductor earthed.

9. 3-Phase, 3-wire system. This system is almost universally adopted for transmission of electric power. The 3-phase, 3-wire system may be star connected or delta connected. Fig. 7.12 shows 3-phase, 3-wire star† connected system. The neutral point N is earthed.

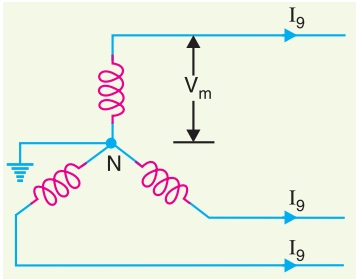


Fig. 7.12

$$\text{R.M.S. voltage per phase} = V_m/\sqrt{2}$$

$$\text{Power transmitted per phase} = P/3$$

* Current in the neutral wire is the phasor sum of currents in the outer wires. Now, the currents in the outers are in quadrature (*i.e.*, 90° apart) with each other.

** Since the neutral wire carries $\sqrt{2}$ times the current in each of the outers, its X-section must be increased in the same ratio to maintain the same current density.

† The same result will be obtained if Δ-connected system is considered.

Load current per phase, $I_9 = \dfrac{P/3}{\left(V_m/\sqrt{2}\cos\phi\right)} = \dfrac{\sqrt{2}\,P}{3\,V_m\cos\phi}$

Let a_9 be the area of X-section of each conductor.

$$\text{Line losses, } W = 3\,I_9^2\,R_9 = 3\left(\frac{\sqrt{2}\,P}{3\,V_m\cos\phi}\right)^2\frac{\rho\,l}{a_9} = \frac{2\,P^2\,\rho\,l}{3\,a_9\,V_m^2\cos^2\phi}$$

$$\therefore \quad \text{Area of X-section, } a_9 = \frac{2\,P^2\,\rho\,l}{3\,W\,V_m^2\cos^2\phi}$$

Volume of conductor material required

$$= 3\,a_9\,l = 3\left(\frac{2\,P^2\,\rho\,l}{3\,W\,V_m^2\cos^2\phi}\right)l = \frac{2\,P^2\,\rho\,l^2}{W\,V_m^2\cos^2\phi}$$

$$= \frac{0{\cdot}5\,K}{\cos^2\phi} \qquad \left[\because K = \frac{4P^2\,\rho\,l^2}{W\,V_m^2}\right]$$

Hence, the volume of conductor material required for this system is $0{\cdot}5/\cos^2\phi$ times that required for 2-wire d.c. system with one conductor earthed.

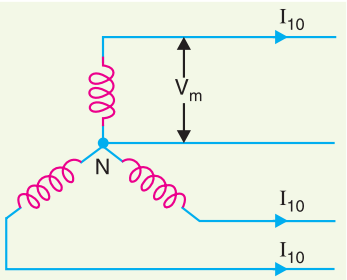


Fig. 7.13

10. 3-phase, 4-wire system. In this case, 4th or neutral wire is taken from the neutral point as shown in Fig. 7.13. The area of X-section of the neutral wire is generally one-half that of the line conductor. If the loads are balanced, then current through the neutral wire is zero. Assuming balanced loads and p.f. of the load as $\cos\phi$,

$$\text{Line losses, } W = \text{Same as in 3 phase, 3-wire}$$

$$= \frac{2\,P^2\,\rho\,l}{3\,a_{10}\,V_m^2\cos^2\phi}$$

$$\therefore \quad \text{Area of X-section, } a_{10} = \frac{2\,P^2\,\rho\,l}{3\,W\,V_m^2\cos^2\phi}$$

As the area of X-section of neutral wire is one-half that of any line conductor,

$\therefore$ Volume of conductor material required

$$= 3{\cdot}5\,a_{10}\,l = 3{\cdot}5\left(\frac{2\,P^2\,\rho\,l}{3\,W\,V_m^2\cos^2\phi}\right)\times l$$

$$= \frac{7\,P^2\,\rho\,l^2}{3\,W\,V_m^2\cos^2\phi} = \frac{7}{3\cos^2\phi}\times\frac{P^2\rho\,l^2}{W\,V_m^2}$$

$$= \frac{7\,K}{12\cos^2\phi} \qquad \left[\because K = \frac{4P^2\,\rho\,l^2}{W\,V_m^2}\right]$$

Hence, the volume of conductor material required for this system is $7/12\cos^2\phi$ times that required for 2-wire d.c. system with one conductor earthed.

7.7 Comparison of Conductor Material in Underground System

In an underground transmission using multi-core* belted type cables, the chief stress on the insulation is usually between conductors. Under such situations, comparison is made on the basis of maximum voltage between conductors. Again assumptions made are :

(i) The same power (P watts) is transmitted by each system.

(ii) The distance (l metres) over which power is transmitted remains the same.

(iii) The line losses (W watts) are the same in each case.

(iv) The maximum voltage between conductors (V_m) is the same in each case.

1. Two-wire d.c. system. If V_m denotes the maximum potential difference between the conductors, it will also be the working voltage in this case.

$$\text{Load current, } I_1 = P/V_m$$

$$\text{Line losses, } W = 2I_1^2 R_1 = 2\left(\frac{P}{V_m}\right)^2 \frac{\rho l}{a_1}$$

$$\therefore \quad W = \frac{2P^2 \rho l}{a_1 V_m^2}$$

$$\therefore \quad \text{Area of X-section, } a_1 = \frac{2P^2 \rho l}{W V_m^2}$$

$\therefore$ Volume of conductor material required

$$= 2 a_1 l = 2\left(\frac{2P^2 \rho l}{W V_m^2}\right) l = \frac{4 P^2 \rho l^2}{W V_m^2} = K \text{ (say)}$$

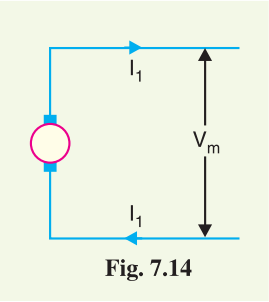


Fig. 7.14

This volume will be taken as the basic quantity and comparison shall be made for other systems *i.e.*,

$$\frac{4 P^2 \rho l^2}{W V_m^2} = K$$

2. Two-wire d.c. system with mid point earthed

$$\text{Load current, } I_2 = P/V_m$$

$$\text{Line losses, } W = 2I_2^2 R_2 = 2\left(\frac{P}{V_m}\right)^2 \rho \frac{l}{a_2}$$

$$\therefore \quad W = \frac{2P^2 \rho l}{V_m^2 a_2}$$

$$\therefore \quad \text{Area of X-section, } a_2 = \frac{2P^2 \rho l}{W V_m^2}$$

Volume of conductor material required

$$= 2 a_2 l = 2\left(\frac{2P^2 \rho l}{W V_m^2}\right) l = \frac{4P^2 \rho l^2}{W V_m^2} = K$$

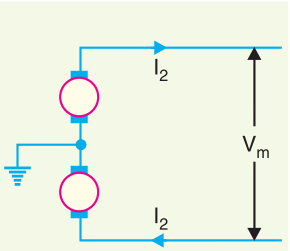


Fig. 7.15

* When the underground transmission is by single-core or multi-core cables of S.L. type, the stress is from conductor to earth. Under such conditions, maximum voltage between conductor and earth forms basis of comparison of various systems.

Hence, the volume of conductor material required in this system is the same as that for 2-wire d.c. system.

3. Three wire d.c. system. The maximum voltage between the outers is V_m. Assuming balanced load, the current through the neutral wire will be zero.

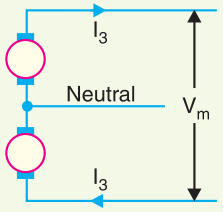


Fig. 7.16

$$\text{Load current, } I_3 = P/V_m$$

$$\text{Line losses, } W = 2\,I_3^2\,R_3 = 2\left(\frac{P}{V_m}\right)^2 \rho\,\frac{l}{a_3}$$

$$\therefore \quad W = \frac{2P^2\,\rho\,l}{V_m^2\,a_3}$$

$$\therefore \quad \text{Area of } X\text{-section, } a_3 = \frac{2P^2\,\rho\,l}{W\,V_m^2}$$

Assuming the area of X-section of neutral wire to be half of that of either outers,

Volume of conductor material required

$$= 2{\cdot}5\,a_3 l = 2{\cdot}5\left(\frac{2P^2\,\rho\,l}{W\,V_m^2}\right) l = \frac{5P^2\,\rho\,l^2}{W\,V_m^2}$$

$$= 1{\cdot}25\,K \qquad \left[\because K = \frac{4P^2\,\rho l^2}{W\,V_m^2}\right]$$

Hence, the volume of conductor material required in the system is 1·25 times that required for 2-wire d.c. system.

4. Single phase, 2-wire a.c. system. As the maximum voltage between conductors is V_m, therefore, r.m.s. value of voltage is $V_m/\sqrt{2}$. Assuming the p.f. of the load to be cos ϕ,

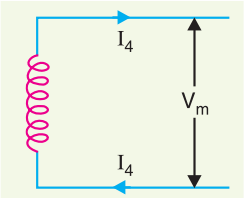


Fig. 7.17

$$\text{Load current, } I_4 = \frac{P}{V_m/\sqrt{2}\,\cos\phi} = \frac{\sqrt{2}P}{V_m\cos\phi}$$

$$\text{Line losses, } W = 2I_4^2 R_4$$

$$= 2\left(\frac{\sqrt{2}P}{V_m\cos\phi}\right)^2 \rho\,\frac{l}{a_4} = \frac{4P^2\,\rho\,l}{a_4\,V_m^2\cos^2\phi}$$

$$\therefore \quad \text{Area of } X\text{-section, } a_4 = \frac{4P^2\,\rho\,l}{W\,V_m^2\cos^2\phi}$$

Volume of conductor material required

$$= 2a_4\,l = 2\left(\frac{4P^2\,\rho\,l}{W\,V_m^2\cos^2\phi}\right) l = \frac{8P^2\,\rho\,l^2}{W\,V_m^2\cos^2\phi}$$

$$= \frac{2}{\cos^2\phi} \times \frac{4P^2\,\rho\,l^2}{W\,V_m^2}$$

$$= \frac{2\,K}{\cos^2\phi} \qquad \left[\because K = \frac{4P^2\,\rho l^2}{W\,V_m^2}\right]$$

Hence, the volume of conductor material required for this system is $2/\cos^2\phi$ times that required in a 2-wire d.c. system.

5. Single phase, 2-wire system with mid-point earthed. The maximum value of voltage between the outers is V_m. Therefore, the r.m.s. value of voltage is $V_m/\sqrt{2}$. If the p.f.of the load is cos ϕ,

$$\text{Load current, } I_5 = \frac{\sqrt{2}P}{V_m \cos\phi}$$

$$\text{Line losses, } W = 2 I_5^2 R_5 = 2\left(\frac{\sqrt{2}P}{V_m \cos\phi}\right)^2 \rho \frac{l}{a_5}$$

$$\therefore \quad W = \frac{4P^2 \rho l}{a_5 V_m^2 \cos^2\phi}$$

$$\therefore \quad \text{Area of } X\text{-section, } a_5 = \frac{4P^2 \rho l}{W V_m^2 \cos^2\phi}$$

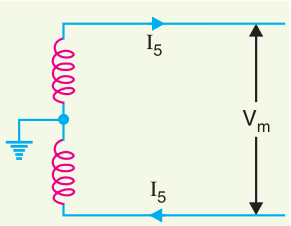


Fig. 7.18

Volume of conductor material required

$$= 2 a_5 l = 2\left(\frac{4P^2 \rho l}{W V_m^2 \cos^2\phi}\right) l = \frac{8P^2 \rho l^2}{W V_m^2 \cos^2\phi}$$

$$= \frac{2}{\cos^2\phi} \times \frac{4P^2 \rho l^2}{W V_m^2}$$

$$= \frac{2K}{\cos^2\phi} \qquad \left[\because K = \frac{4P^2 \rho l^2}{W V_m^2}\right]$$

Hence, the volume of conductor material required in this system is $2/\cos^2\phi$ times that required in 2-wire d.c. system.

6. Single phase, 3-wire system. If the load is considered balanced, the system reduces to a single phase 2-wire except that a neutral wire is provided in addition. Assuming the area of *X*-section of the neutral wire to be half of either of the outers,

Volume of conductor material required

$$= 2{\cdot}5 * a_4 l$$

$$= 2{\cdot}5\left(\frac{4P^2 \rho l}{W V_m^2 \cos^2\phi}\right) l$$

$$= \frac{10P^2 \rho l^2}{W V_m^2 \cos^2\phi}$$

$$= \frac{2{\cdot}5}{\cos^2\phi} \times \frac{4P^2 \rho l^2}{W V_m^2}$$

$$= \frac{2{\cdot}5K}{\cos^2\phi} \qquad \left[\because K = \frac{4P^2 \rho l^2}{W V_m^2}\right]$$

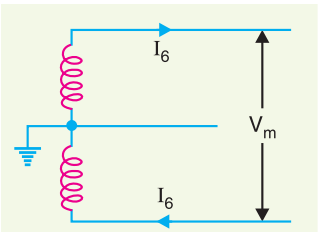


Fig. 7.19

Hence, the volume of conductor material required in this system is $2{\cdot}5/\cos^2\phi$ times that required in 2-wire d.c. system.

7. Two-phase, 4-wire system. This system can be considered as two independent single phase systems, each transmitting one-half of the total power. It is clear form Fig. 7.20 that voltage across outers (*AB* or *CD*) is twice that of single phase 2-wire (refer back to Fig. 7.17). Therefore, current (I_7) in each conductor will be half that in single-phase 2-wire system. Consequently, area of *X*-section of each conductor is also half but as there are four wires, so volume of conductor

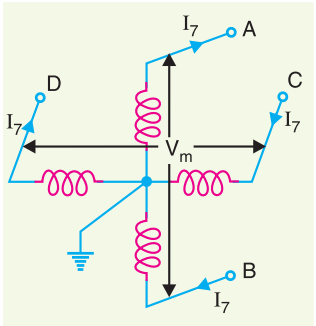


Fig. 7.20

* Area of *X*-section of conductor for single phase 2-wire system.

material used is the same as in a single phase, 2-wire system *i.e.*

Volume of conductor material required.

$$= \frac{2\,K}{\cos^2 \phi}$$

Hence, volume of conductor material required in this system is $2 / \cos^2 \phi$ times that required in 2-wire d.c. system.

8. Two-phase, 3-wire system. Fig. 7.21 shows two phase, 3-wire a.c. system. Let us suppose that maximum voltage between the outers is V_m. Then maximum voltage between either outer and neutral wire is $V_m / \sqrt{2}$ *.

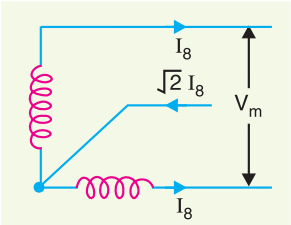


Fig. 7.21

R.M.S. voltage between outer and neutral wire

$$= \frac{V_m/\sqrt{2}}{\sqrt{2}} = \frac{V_m}{2}$$

Current in each outer, $I_8 = \dfrac{P/2}{V_m/2 \cos \phi} = \dfrac{P}{V_m \cos \phi}$

Current in neutral wire $= \sqrt{I_8^2 + I_8^2} = \sqrt{2} I_8$

Assuming the current density to be constant, the area of X-section of neutral wire will be $\sqrt{2}$ times that of either of the outers.

$\therefore$ Resistance of neutral wire $= \dfrac{R_8}{\sqrt{2}} = \dfrac{\rho\, l}{\sqrt{2}\, a_8}$

$$\text{Line losses, } W = 2I_8^2 R_8 + (\sqrt{2}\, I_8)^2 \frac{R_8}{\sqrt{2}} = I_8^2 R_8 (2 + \sqrt{2})$$

$$= \left(\frac{P}{V_m \cos \phi}\right)^2 \times \rho \frac{l}{a_8} \left(2 + \sqrt{2}\right)$$

$$\therefore \quad W = \frac{P^2 \rho l}{a_8 V_m^2 \cos^2 \phi} \left(2 + \sqrt{2}\right)$$

$$\therefore \quad \text{Area of X-section, } a_8 = \frac{P^2 \rho l}{W V_m^2 \cos^2 \phi} \left(2 + \sqrt{2}\right)$$

Volume of conductor material required

$$= 2\, a_8 l + \sqrt{2}\, a_8 l = a_8 l\, (2 + \sqrt{2})$$

$$= \frac{P^2 \rho\, l^2}{W V_m^2 \cos^2 \phi} \left(2 + \sqrt{2}\right)^2$$

$$= \frac{2 \cdot 194\, K}{\cos^2 \phi} \qquad \left[\because K = \frac{4 P^2 \rho\, l^2}{W V_m^2}\right]$$

Hence, the volume of conductor material required in this system is $2 \cdot 914 / \cos^2 \phi$ times that required in 2-wire d.c. system.

9. 3-Phase, 3-wire system. Suppose that the maximum value of voltage between the conductors is V_m. Then maximum voltage between each phase and neutral is $V_m / \sqrt{3}$. Therefore, r.m.s. value of voltage per phase

---

* The voltages in two phase windings are 90° out of phase.

$$= \frac{V_m}{\sqrt{3}} \times \frac{1}{\sqrt{2}} = \frac{V_m}{\sqrt{6}}$$

Power transmitted per phase $= P/3$

$$\therefore \quad \text{Load current per phase, } I_9 = \frac{P/3}{V_m/\sqrt{6}\cos\phi} = \frac{\sqrt{6}\,P}{3\,V_m\cos\phi}$$

$$\text{Line losses, } W = 3\,I_9^2\,R_9$$

$$= 3\left(\frac{\sqrt{6}\,P}{3V_m\cos\phi}\right)^2 \times \rho\frac{l}{a_9}$$

$$\therefore \quad W = \frac{2P^2\,\rho\,l}{a_9 V_m^2\cos^2\phi}$$

$$\therefore \quad \text{Area of } X\text{-section, } a_9 = \frac{2P^2\,\rho l}{W\,V_m^2\cos^2\phi}$$

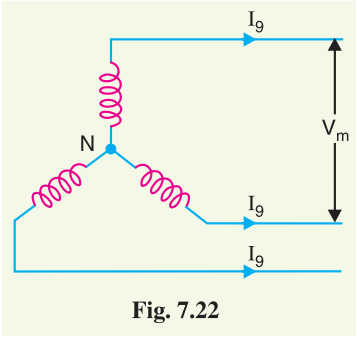


Fig. 7.22

Volume of conductor material required

$$= 3\,a_9\,l = 3\left(\frac{2P^2\rho\,l}{W\,V_m^2\cos^2\phi}\right)l = \frac{6P^2\,\rho\,l^2}{W\,V_m^2\cos^2\phi}$$

$$= \frac{1{\cdot}5}{\cos^2\phi} \times \frac{4P^2\,\rho\,l^2}{W\,V_m^2}$$

$$= \frac{1{\cdot}5\,K}{\cos^2\phi} \qquad \left[\because K = \frac{4\,P^2\,\rho\,l^2}{W\,V_m^2}\right]$$

Hence, the volume of conductor material required in this system is $1{\cdot}5/\cos^2\phi$ times that required in 2-wire d.c. system.

10. 3-phase, 4-wire system. Fig. 7.23 shows the 3-phase, 4-wire system. If the loads are balanced, then neutral wire carries no current. Consequently, the system reduces to a 3-phase, 3-wire system except that there is additional neutral wire. Assuming the area of X-section of the neutral wire to be half that of line conductor,

Volume of conductor material required

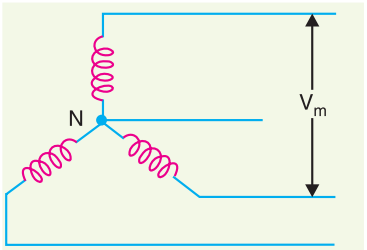


Fig. 7.23

$$= 3{\cdot}5\,a_9\,l$$

$$= 3{\cdot}5\left(\frac{2P^2\rho\,l}{W\,V_m^2\cos^2\phi}\right)l$$

$$= \frac{7P^2\rho\,l^2}{W\,V_m^2\cos^2\phi} = \frac{7}{\cos^2\phi} \times \frac{P^2\,\rho\,l^2}{W\,V_m^2}$$

$$= \frac{1{\cdot}75\,K}{\cos^2\phi} \qquad \left[\because K = \frac{4\,P^2\,\rho\,l^2}{W\,V_m^2}\right]$$


Underground transmission system.

Hence the volume of conductor material required in this system is $1{\cdot}75/\cos^2\phi$ times that required in 2-wire d.c. system.

7.8 Comparison of Various Systems of Transmission

Below is given the table which shows the ratio of conductor-material in any system compared with that in the corresponding 2-wire d.c. system. Cos ϕ is the power factor in an a.c. system.

System	Same maximum voltage to earth	Same maximum voltage between conductors
1. D.C. system		
(i) Two-wire	1	1
(ii) Two-wire mid-point earthed	0·25	1
(iii) 3-wire	0·3125	1·25
2. Single phase system		
(i) 2 wire	$2/\cos^2\phi$	$2/\cos^2\phi$
(ii) 2-wire with mid-point earthed	$\frac{0\cdot5}{\cos^2\phi}$	$\frac{2}{\cos^2\phi}$
(iii) 3-wire	$\frac{0\cdot625}{\cos^2\phi}$	$\frac{2\cdot5}{\cos^2\phi}$
3. Two-phase system		
(i) 2-phase, 4-wire	$\frac{0\cdot5}{\cos^2\phi}$	$\frac{2}{\cos^2\phi}$
(ii) 2-phase, 3-wire	$\frac{1\cdot457}{\cos^2\phi}$	$\frac{2\cdot914}{\cos^2\phi}$
4. Three-phase system		
(i) 3-phase, 3-wire	$\frac{0\cdot5}{\cos^2\phi}$	$\frac{1\cdot5}{\cos^2\phi}$
(ii) 3-phase, 4-wire	$\frac{0\cdot583}{\cos^2\phi}$	$\frac{1\cdot75}{\cos^2\phi}$

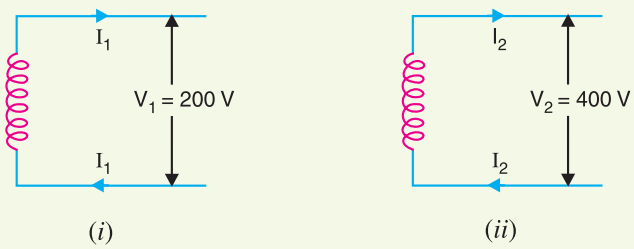
The following points may be noted from the above table :

(i) There is a great saving in conductor material if d.c. system is adopted for transmission of electric power. However, due to technical difficulties, d.c. system is not used for transmission.

(ii) Considering the a.c. system, the 3-phase a.c. system is most suitable for transmission due to two reasons. Firstly, there is considerable saving in conductor material. Secondly, this system is convenient and efficient.

Example 7.1 *What is the percentage saving in feeder copper if the line voltage in a 2-wire d.c. system is raised from 200 volts to 400 volts for the same power transmitted over the same distance and having the same power loss ?*

Solution. Fig. 7.24 (*i*) shows 200 volts system, whereas Fig. 7.24 (*ii*) shows 400 volts system. Let P be the power delivered and W be power loss in both cases. Let v_1 and a_1 be the volume and area of X-section for 200 V system and v_2 and a_2 for that of 400 V system.



(*i*) (*ii*)

Fig. 7.24

Now, $P = V_1 I_1 = 200\, I_1$ ...(*i*)

And $P = V_2 I_2 = 400\, I_2$ ...(*ii*)

As same power is delivered in both cases,

$\therefore \quad 200 I_1 = 400\, I_2$ or $I_2 = (200/400)\, I_1 = 0{\cdot}5 I_1$

Power loss in 200 *V* system, $W_1 = 2 I_1^2 R_1$

Power loss in 400 *V* system, $W_2 = 2 I_2^2 R_2 = 2(0{\cdot}5\, I_1)^2 R_2 = 0{\cdot}5 I_1^2 R_2$

As power loss in the two cases is the same,

$\therefore \quad W_1 = W_2$

or $2 I_1^2 R_1 = 0{\cdot}5\, I_1^2 R_2$

or $R_2/R_1 = 2/0{\cdot}5 = 4$

or $a_1/a_2 = 4$

or $v_1/v_2 = 4$

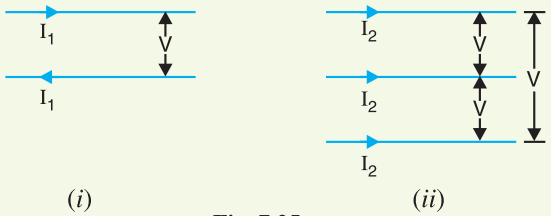
or $v_2/v_1 = 1/4 = 0{\cdot}25$

$\therefore$ % age saving in feeder copper $= \dfrac{v_1 - v_2}{v_1} \times 100 = \left(\dfrac{v_1}{v_1} - \dfrac{v_2}{v_1}\right) \times 100$

$= (1 - 0{\cdot}25) \times 100 = \mathbf{75\%}$

Example 7.2 *A d.c. 2-wire system is to be converted into a.c. 3-phase, 3-wire system by the addition of a third conductor of the same cross-section as the two existing conductors. Calculate the percentage additional load which can now be supplied if the voltage between wires and the percentage loss in the line remain unchanged. Assume a balanced load of unity power factor.*

Solution. Fig. 7.25 (*i*) shows the 2-wire d.c. system, whereas Fig. 7.25 (*ii*) shows the 3-phase, 3-wire system. Suppose *V* is the voltage between conductors for the two cases. Let *R* be the resistance per conductor in each case.



(*i*) (*ii*)

Fig. 7.25

Two-wire d.c. system. Referring to Fig. 7.25 (*i*),

Power supplied, $P_1 = V I_1$

Power loss, $W_1 = 2 I_1^2 R$

Percentage power loss $= \dfrac{2 I_1^2 R}{V I_1} \times 100$ ...(*i*)

3-phase, 3-wire a.c. system. Referring to Fig. 7.25 (*ii*),

Power supplied, $P_2 = \sqrt{3}\, V I_2$

Power loss, $W_2 = 3 I_2^2 R$

Percentage power loss $= \dfrac{3 I_2^2 R}{\sqrt{3}\, V I_2} \times 100$ ...(*ii*)

As the percentage power loss in the two cases is the same,

$$\therefore \quad \text{exp. } (i) = \text{exp. } (ii)$$

$$\text{or} \quad \frac{2 I_1^2 R}{V I_1} \times 100 = \frac{3 I_2^2 R}{\sqrt{3} V I_2} \times 100$$

$$\text{or} \quad 2 I_1 = \sqrt{3} I_2$$

$$\text{or} \quad I_2 = \frac{2}{\sqrt{3}} I_1$$

$$\text{Now,} \quad \frac{P_2}{P_1} = \frac{\sqrt{3} V I_2}{V I_1} = \frac{\sqrt{3} V \times \frac{2}{\sqrt{3}} I_1}{V I_1} = 2$$

$$\therefore \quad P_2 = 2 P_1$$

i.e. additional power which can be supplied at unity p.f. by 3-phase, 3-wire a.c. system = **100%**.

Example 7.3. *A d.c. 3-wire system is to be converted into a 3-phase, 4-wire system by adding a fourth wire equal in X-section to each outer of the d.c. system. If the percentage power loss and voltage at the consumer's terminals are to be the same in the two cases, find the extra power at unity power factor that can be supplied by the a.c. system. Assume loads to be balanced.*

Solution. Fig. 7.26 (*i*) shows the 3-wire d.c. system, whereas Fig. 7.26 (*ii*) shows 3-phase, 4-wire system. Suppose that V is consumer's terminal voltage (*i.e.*, between conductor and neutral) in the two cases. Let R be the resistance per conductor in each case.

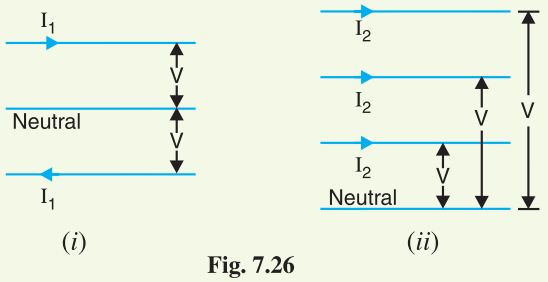


Fig. 7.26

3-wire d.c. system. Refer to Fig. 7.26 (*i*). As the loads are balanced, therefore, neutral wire carries no current. Consequently, there is no power loss in the neutral wire.

$$\text{Power supplied, } P_1 = 2 V I_1$$

$$\text{Power loss, } W_1 = 2 I_1^2 R$$

$$\text{Percentage power loss} = \frac{2 I_1^2 R}{2 V I_1} \times 100 \qquad ...(i)$$

3-phase, 4-wire a.c. system. Refer to Fig. 7.26 (*ii*). Since the loads are balanced, the neutral wire carries no current and hence there is no power loss in it.

$$\text{Power supplied, } P_2 = *3 V I_2 \qquad [\because \cos \phi = 1]$$

$$\text{Power loss, } W_2 = 3 I_2^2 R$$

$$\text{Percentage power loss} = \frac{3 I_2^2 R}{3 V I_2} \times 100 \qquad ...(ii)$$

As the percentage power loss in the two cases is the same, therefore, exp. (*i*) is equal to exp. (*ii*) *i.e.*,

* Power per phase = $V I_2$

$\therefore$ Power in 3-phases = $3V I_2$

$$\frac{2 I_1^2 R}{2 V I_1} \times 100 = \frac{3 I_2^2 R}{3 V I_2} \times 100$$

or $$I_1 = I_2$$

Now $$\frac{P_2}{P_1} = \frac{3 V I_2}{2 V I_1} = \frac{3 V I_1}{2 V I_1} = 1{\cdot}5$$

$\therefore$ $$P_2 = 1{\cdot}5 P_1$$

i.e., extra power that can be supplied at unity power factor by 3-phase, 4-wire a.c. system = **50%.**

Example 7.4. *A single phase a.c. system supplies a load of 200 kW and if this system is converted to 3-phase, 3-wire a.c. system by running a third similar conductor, calculate the 3-phase load that can now be supplied if the voltage between the conductors is the same. Assume the power factor and transmission efficiency to be the same in the two cases.*

Solution. Fig. 7.27 (*i*) shows the single phase 2-wire a.c. system, whereas Fig. 7.27 (*ii*) shows 3-phase, 3-wire system. Suppose that V is the voltage between the conductors in the two cases. Let R be the resistance per conductor and cos ϕ the power factor in each case.

Single phase 2-wire system. Referring to Fig. 7.27 (*i*),

Power supplied, $P_1 = V I_1 \cos \phi$

Power loss, $W_1 = 2 I_1^2 R$

$$\% \text{ age power loss} = \frac{2 I_1^2 R}{V I_1 \cos \phi} \times 100 \quad ...(i)$$

3-phase, 3-wire a.c. system. Referring to Fig. 7.27 (*ii*),

Power supplied, $P_2 = \sqrt{3} V I_2 \cos \phi$

Power loss, $W_2 = 3 I_2^2 R$

$$\% \text{ age power loss} = \frac{3 I_2^2 R}{\sqrt{3} V I_2 \cos \phi} \times 100 \quad ...(ii)$$

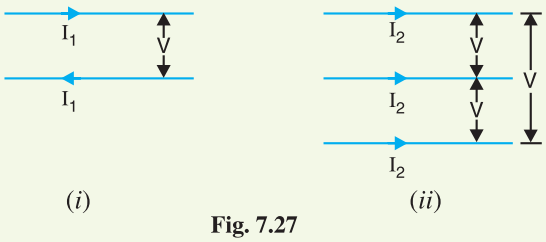


Fig. 7.27

As the transmission efficiency in the two cases is the same, therefore, percentage power loss will also be the same *i.e.*,

exp. (*i*) = exp. (*ii*)

or $$\frac{2 I_1^2 R}{V I_1 \cos \phi} \times 100 = \frac{3 I_2^2 R}{\sqrt{3} V I_2 \cos \phi} \times 100$$

or $$2 I_1 = \sqrt{3}\ I_2$$

or $$I_2 = \frac{2}{\sqrt{3}} I_1$$

Now, $$\frac{P_2}{P_1} = \frac{\sqrt{3} V I_2 \cos \phi}{V I_1 \cos \phi} = \frac{\sqrt{3} V \frac{2}{\sqrt{3}} I_1 \cos \phi}{V I_1 \cos \phi} = 2$$

As the power supplied by single phase, 2-wire (*i.e.*, P_1) is 200 kW,

$\therefore$ Power supplied by 3-phase, 3-wire a.c. system is

$$P_2 = 2P_1 = 2 \times 200 = \mathbf{400\ kW}$$

It may be seen that 3-phase, 3-wire system can supply 100% additional load.

Example 7.5. *A 50 km long transmission line supplies a load of 5 MVA at 0·8 p.f. lagging at 33 kV. The efficiency of transmission is 90%. Calculate the volume of aluminium conductor required for the line when (i) single phase, 2-wire system is used (ii) 3-phase, 3-wire system is used. The specific resistance of aluminium is $2{\cdot}85 \times 10^{-8}\ \Omega$ m.*

Solution.

Power transmitted = MVA $\times \cos\phi = 5 \times 0{\cdot}8 = 4$ MW $= 4 \times 10^6$ W

Line loss, W = 10% of power transmitted $= (10/100) \times 4 \times 10^6 = 4 \times 10^5$ W

Length of line, l = 50 km $= 50 \times 10^3$ m

(*i*) Single phase, 2-wire system

Apparent power $= V I_1$

$$\therefore \quad I_1 = \frac{\text{Apparent power}}{V} = \frac{5 \times 10^6}{33 \times 10^3} = 151{\cdot}5 \text{ A}$$

Suppose a_1 is the area of cross-section of aluminium conductor.

$$\text{Line loss, } W = 2 I_1^2 R_1 = 2 I_1^2 \left(\rho \frac{l}{a_1}\right)$$

$$\therefore \quad \text{Area of X-section, } a_1 = \frac{2 I_1^2 \rho l}{W} = \frac{2 \times (151{\cdot}5)^2 \times \left(2{\cdot}85 \times 10^{-8}\right) 50 \times 10^3}{4 \times 10^5}$$

$$= 1{\cdot}635 \times 10^{-4}\ \text{m}^2$$

Volume of conductor required $= 2 a_1 l = 2 \times (1{\cdot}635 \times 10^{-4}) \times 50 \times 10^3 = \mathbf{16{\cdot}35\ m^3}$

(*ii*) 3-phase, 3-wire system

$$\text{Line current, } I_2 = \frac{\text{Apparent power}}{\sqrt{3}\, V} = \frac{5 \times 10^6}{\sqrt{3} \times 33 \times 10^3} = 87{\cdot}5 \text{ A}$$

Suppose a_2 is the area of cross-section of the conductor in this case.

$$\text{Line loss, } W = 3 I_2^2 R_2 = 3 I_2^2 \left(\rho \frac{l}{a_2}\right)$$

$$\therefore \quad \text{Area of X-section, } a_2 = \frac{3 I_2^2 \rho l}{W} = \frac{3 \times (87{\cdot}5)^2 \times \left(2{\cdot}85 \times 10^{-8}\right) \times 50 \times 10^3}{4 \times 10^5}$$

$$= 0{\cdot}818 \times 10^{-4}\ \text{m}^2$$

Volume of conductor required $= 3 a_2 l = 3 \times (0{\cdot}818 \times 10^{-4}) \times 50 \times 10^3 = \mathbf{12{\cdot}27\ m^3}$

Note that volume of conductor (and hence weight) required is less in case of 3-phase, 3-wire system.

Example 7.6. *A sub-station supplies power at 11 kV, 0·8 p.f. lagging to a consumer through a single phase transmission line having total resistance (both go and return) of 0.15 Ω. The voltage drop in the line is 15%. If the same power is to be supplied to the same consumer by two wire d.c. system by a new line having a total resistance of 0·05 Ω and if the allowable voltage drop is 25%, calculate the d.c. supply voltage.*

Solution :

Single phase system

Let I_1 be the load current. Then,

$$\text{Voltage drop} = I_1R_1 = I_1 \times 0{\cdot}15 \text{ volts} \quad ...(i)$$

$$\text{Also voltage drop} = \frac{15}{100} \times 11000 = 1650 \text{ volts} \quad ...(ii)$$

From eqs. (*i*) and (*ii*), $I_1 \times 0{\cdot}15 = 1650$

$$\therefore \quad I_1 = \frac{1650}{0{\cdot}15} = 11000 \text{ A}$$

$$\text{Power received by consumer} = \text{Apparent power} \times \cos\phi$$

$$= \frac{(11{,}000 \times 11{,}000) \times 0{\cdot}8}{1000} \text{ kW} = 9{\cdot}68 \times 10^4 \text{ kW}$$

D.C. two-wire system. The power to be supplied by d.c. 2-wire system is $9{\cdot}68 \times 10^4$ kW = $9{\cdot}68 \times 10^7$ W. Let V volts be the supply voltage.

$$\therefore \quad \text{Load current, } I_2 = \frac{9{\cdot}68 \times 10^7}{V}$$

$$\text{Voltage drop} = I_2 R_2 = \left(\frac{9{\cdot}68 \times 10^7}{V}\right) \times 0{\cdot}05 \quad ...(iii)$$

$$\text{Allowable voltage drop} = \frac{25}{100} \times V = 0{\cdot}25 \text{ V} \quad ...(iv)$$

$$\therefore \quad \frac{9{\cdot}68 \times 10^7 \times 0{\cdot}05}{V} = 0{\cdot}25 \text{ V}$$

or $$V^2 = \frac{9{\cdot}68 \times 10^7 \times 0{\cdot}05}{0{\cdot}25} = 1936 \times 10^4$$

$$\therefore \quad V = \sqrt{1936 \times 10^4} = \mathbf{4400 \text{ V}}$$

TUTORIAL PROBLEMS

1. What is the percentage saving in copper feeder if the line voltage in a 2-wire d.c. system is raised from 220 V to 500 V for the same power transmitted over the same distance and having the same power loss? **[80·64%]**
2. A single phase load of 5 MW is transmitted by a pair of overhead conductors. If a third conductor of the same cross-section and material be added and 3-phase supply be thus substituted for the original single phase, calculate the 3-phase load which can now be transmitted if the voltage between the conductors and the percentage loss in the lines remains unchanged. **[10 MW]**
3. Electric power of 50 MW is to be transmitted over a 132 KV, 3-phase, 3-wire transmission line. The length of the line is 300 km and the efficiency of transmission is 85%. Aluminium is used for conductor material which has resistivity of 3×10^{-9} Ωm. Calculate the volume of conductor material required for a power factor of 0·8 lagging. **[242 m^3]**
4. A 3-phase, 4-wire system is used for lighting. Compare the amount of copper required with that needed for a 2-wire d.c. system with the same lamp voltage. Assume the same losses and balanced load. The neutral is one-half the cross-section of one of the respective outers. **[Copper for 3ϕ, 4 wire : Copper for 2-wire d.c. = 0·292 : 1]**
5. 30,000 kW at power factor 0·8 lagging is being transmitted over a 220 kV, three-phase transmission line. The length of the line is 275 km and the efficiency of transmission is 90%. Calculate the weight of copper required. Also calculate the weight of copper had the power been transmitted over a single-phase transmission line for the same line voltage and losses. Assume that the resistance of 1 km long conductor and 1 sq. cm is 0·173 Ω and specific gravity of copper is 8·9. **[338×10^3 kg : $450{\cdot}67 \times 10^3$ kg]**

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CHAPTER

13



D.C. Distribution

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Introduction

In the beginning of the electrical age, electricity was generated as a direct current and voltages were low. The resistance losses in the lines made it impracticable to transmit and distribute power for more than a few localities of the city. With the development of the transformer, a.c. has taken over the load formerly supplied by d.c. Now-a-days, electrical energy is generated, transmitted and distributed in the form of a.c. as an economical proposition. The transformer permits the transmission and distribution of a.c. power at high voltages. This has greatly reduced the current in the conductors (and hence their sizes) and the resulting line losses.

However, for certain applications, d.c. supply is absolutely necessary. For example, d.c. supply is required for the operation of variable speed machinery (*e.g.* d.c. motors), electrochemical work and electric traction. For this purpose, a.c. power is converted into d.c. power at the sub-station by using converting machinery *e.g.* mercury are rectifiers, rotary converters and motor-generator sets. The d.c. supply from the

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sub-station is conveyed to the required places for distribution. In this chapter, we shall confine our attention to the various aspects of d.c. distribution.

13.1 Types of D.C. Distributors

The most general method of classifying d.c. distributors is the way they are fed by the feeders. On this basis, d.c. distributors are classified as:

(i) Distributor fed at one end

(ii) Distributor fed at both ends

(iii) Distributor fed at the centre

(iv) Ring distributor.

(i) **Distributor fed at one end.** In this type of feeding, the distributor is connected to the supply at one end and loads are taken at different points along the length of the distributor. Fig. 13.1 shows the single line diagram of a d.c. distributor *A B* fed at the end *A* (also known as *singly fed distributor*) and loads I_1, I_2 and I_3 tapped off at points *C*, *D* and *E* respectively.

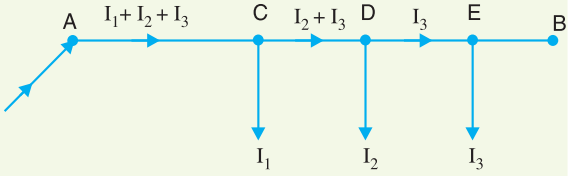


Fig. 13.1

The following points are worth noting in a singly fed distributor :

(a) The current in the various sections of the distributor away from feeding point goes on decreasing. Thus current in section *AC* is more than the current in section *CD* and current in section *CD* is more than the current in section *DE*.

(b) The voltage across the loads away from the feeding point goes on decreasing. Thus in Fig. 13.1, the minimum voltage occurs at the load point *E*.

(c) In case a fault occurs on any section of the distributor, the whole distributor will have to be disconnected from the supply mains. Therefore, continuity of supply is interrupted.

(ii) **Distributor fed at both ends.** In this type of feeding, the distributor is connected to the supply mains at both ends and loads are tapped off at different points along the length of the distributor. The voltage at the feeding points may or may not be equal. Fig. 13.2 shows a distributor *A B* fed at the ends *A* and *B* and loads of I_1, I_2 and I_3 tapped off at points *C*, *D* and *E* respectively. Here, the load voltage goes on decreasing as we move away from one feeding point *say A*, reaches minimum value and then again starts rising and reaches maximum value when we reach the other feeding point *B*. The minimum voltage occurs at some load point and is never fixed. It is shifted with the variation of load on different sections of the distributor.

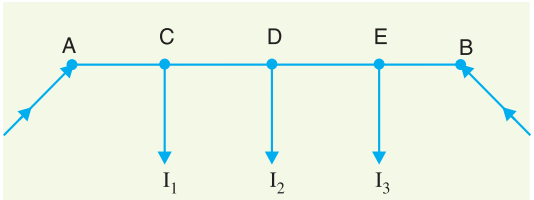


Fig. 13.2

Advantages

(a) If a fault occurs on any feeding point of the distributor, the continuity of supply is maintained from the other feeding point.

(b) In case of fault on any section of the distributor, the continuity of supply is maintained from the other feeding point.

(*c*) The area of X-section required for a doubly fed distributor is much less than that of a singly fed distributor.

(*iii*) **Distributor fed at the centre.** In this type of feeding, the centre of the distributor is connected to the supply mains as shown in Fig. 13.3. It is equivalent to two singly fed distributors, each distributor having a common feeding point and length equal to half of the total length.

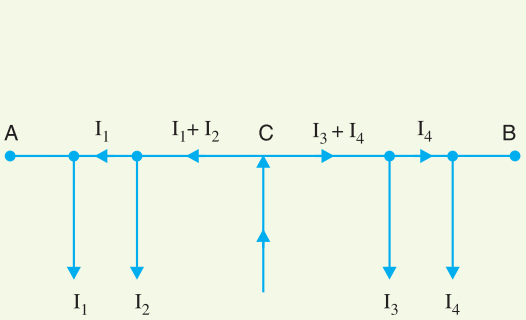


Fig. 13.3

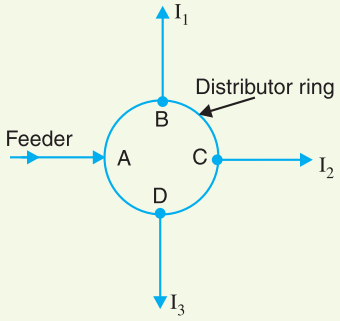


Fig. 13.4

(*iv*) **Ring mains.** In this type, the distributor is in the form of a closed ring as shown in Fig.13.4. It is equivalent to a straight distributor fed at both ends with equal voltages, the two ends being brought together to form a closed ring. The distributor ring may be fed at one or more than one point.

13.2 D.C. Distribution Calculations

In addition to the methods of feeding discussed above, a distributor may have (*i*) concentrated loading (*ii*) uniform loading (*iii*) both concentrated and uniform loading. The concentrated loads are those which act on particular points of the distributor. A common example of such loads is that tapped off for domestic use. On the other hand, distributed loads are those which act uniformly on all points of the distributor. Ideally, there are no distributed loads. However, a nearest example of distributed load is a large number of loads of same wattage connected to the distributor at equal distances.



D.C. Load

In d.c. distribution calculations, one important point of interest is the determination of point of minimum potential on the distributor. The point where it occurs depends upon the loading conditions and the method of feeding the distributor. The distributor is so designed that the minimum potential on it is not less than 6% of rated voltage at the consumer's terminals. In the next sections, we shall discuss some important cases of d.c. distributors separately.

13.3 D.C. Distributor Fed at one End—Concentrated Loading

Fig. 13.5 shows the single line diagram of a 2-wire d.c. distributor AB fed at one end A and having concentrated loads I_1, I_2, I_3 and I_4 tapped off at points C, D, E and F respectively.

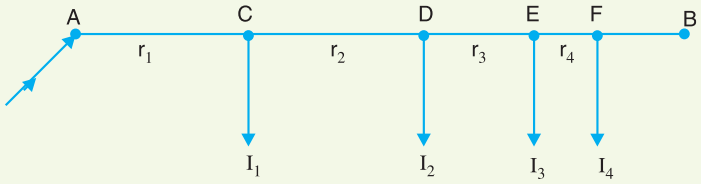


Fig. 13.5

Let r_1, r_2, r_3 and r_4 be the resistances of both wires (go and return) of the sections AC, CD, DE and EF of the distributor respectively.

Current fed from point A = $I_1 + I_2 + I_3 + I_4$
Current in section AC = $I_1 + I_2 + I_3 + I_4$
Current in section CD = $I_2 + I_3 + I_4$
Current in section DE = $I_3 + I_4$
Current in section EF = I_4
Voltage drop in section AC = $r_1 (I_1 + I_2 + I_3 + I_4)$
Voltage drop in section CD = $r_2 (I_2 + I_3 + I_4)$
Voltage drop in section DE = $r_3 (I_3 + I_4)$
Voltage drop in section EF = $r_4 I_4$

$\therefore$ Total voltage drop in the distributor

$$= r_1 (I_1 + I_2 + I_3 + I_4) + r_2 (I_2 + I_3 + I_4) + r_3 (I_3 + I_4) + r_4 I_4$$

It is easy to see that the minimum potential will occur at point F which is farthest from the feeding point A.

Example 13.1. *A 2-wire d.c. distributor cable AB is 2 km long and supplies loads of 100A, 150A,200A and 50A situated 500 m, 1000 m, 1600 m and 2000 m from the feeding point A. Each conductor has a resistance of 0·01 Ω per 1000 m. Calculate the p.d. at each load point if a p.d. of 300 V is maintained at point A.*

Solution. Fig. 13.6 shows the single line diagram of the distributor with its tapped currents.

Resistance per 1000 m of distributor = $2 \times 0{\cdot}01 = 0{\cdot}02\,\Omega$
Resistance of section AC, R_{AC} = $0{\cdot}02 \times 500/1000 = 0{\cdot}01\,\Omega$
Resistance of section CD, R_{CD} = $0{\cdot}02 \times 500/1000 = 0{\cdot}01\,\Omega$
Resistance of section DE, R_{DE} = $0{\cdot}02 \times 600/1000 = 0{\cdot}012\,\Omega$
Resistance of section EB, R_{EB} = $0{\cdot}02 \times 400/1000 = 0{\cdot}008\,\Omega$

Referring to Fig. 13.6, the currents in the various sections of the distributor are :

$I_{EB} = 50$ A ; $I_{DE} = 50 + 200 = 250$ A
$I_{CD} = 250 + 150 = 400$ A ; $I_{AC} = 400 + 100 = 500$ A

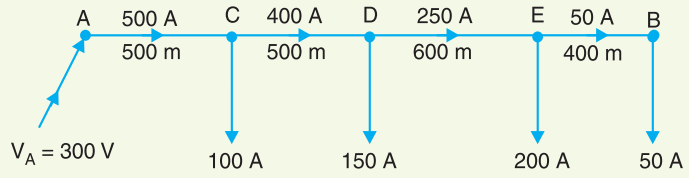


Fig. 13.6

P.D. at load point *C*, V_C = Voltage at A – Voltage drop in AC

$= V_A - I_{AC} R_{AC}$

$= 300 - 500 \times 0{\cdot}01 = \mathbf{295\ V}$

P.D. at load point *D*, $V_D = V_C - I_{CD} R_{CD}$

$= 295 - 400 \times 0{\cdot}01 = \mathbf{291\ V}$

P.D. at load point *E*, $V_E = V_D - I_{DE} R_{DE}$

$= 291 - 250 \times 0{\cdot}012 = \mathbf{288\ V}$

P.D. at load point *B*, $V_B = V_E - I_{EB} R_{EB}$

$= 288 - 50 \times 0{\cdot}008 = \mathbf{287{\cdot}6\ V}$

Example 13.2. *A 2-wire d.c. distributor AB is 300 metres long. It is fed at point A. The various loads and their positions are given below :*

At point	*distance from A in metres*	*concentrated load in amperes*
C	*40*	*30*
D	*100*	*40*
E	*150*	*100*
F	*250*	*50*

If the maximum permissible voltage drop is not to exceed 10 V, find the cross-sectional area of the distributor. Take $\rho = 1{\cdot}78 \times 10^{-8}\ \Omega m$.

Solution. The single line diagram of the distributor along with its tapped currents is shown in Fig. 13.7. Suppose that resistance of 100 m length of the distributor is *r* ohms. Then resistance of various sections of the distributor is :

$R_{AC} = 0{\cdot}4\,r\,\Omega$; $R_{CD} = 0{\cdot}6\,r\,\Omega$; $R_{DE} = 0{\cdot}5\,r\,\Omega$; $R_{EF} = r\,\Omega$

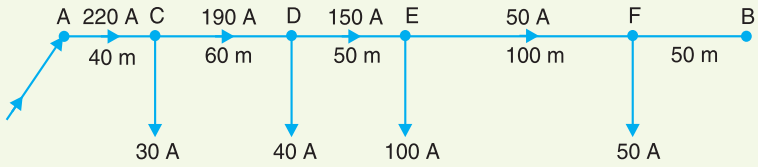


Fig. 13.7

Referring to Fig. 13.7, the currents in the various sections of the distributor are :

$I_{AC} = 220$ A ; $I_{CD} = 190$ A ; $I_{DE} = 150$ A ; $I_{EF} = 50$ A

Total voltage drop over the distributor

$= I_{AC} R_{AC} + I_{CD} R_{CD} + I_{DE} R_{DE} + I_{EF} R_{EF}$

$= 220 \times 0{\cdot}4r + 190 \times 0{\cdot}6r + 150 \times 0{\cdot}5r + 50 \times r$

$= 327\ r$

As the maximum permissible drop in the distributor is 10 V,

$\therefore \quad 10 = 327\ r$

or $r = 10/327 = 0{\cdot}03058\ \Omega$

X-sectional area of conductor $= {}^{*}\dfrac{\rho l}{r/2} = \dfrac{1{\cdot}78\times10^{-8}\times100}{\frac{0{\cdot}03058}{2}} = 116{\cdot}4\times10^{-6}\ \text{m}^2 = \mathbf{1{\cdot}164\ cm^2}$

Example 13.3. *Two tram cars (A & B) 2 km and 6 km away from a sub-station return 40 A and 20 A respectively to the rails. The sub-station voltage is 600 V d.c. The resistance of trolley wire is 0·25 Ω/km and that of track is 0·03 Ω/km. Calculate the voltage across each tram car.*

Solution. The tram car operates on d.c. supply. The positive wire is placed overhead while the rail track acts as the negative wire. Fig. 13.8 shows the single line diagram of the arrangement.

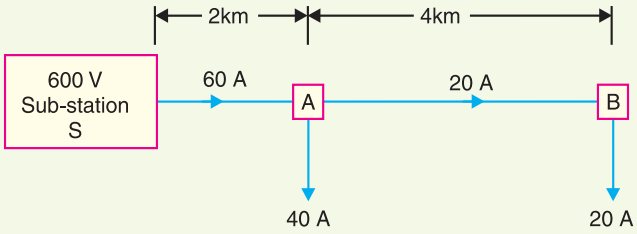


Fig. 13.8

Resistance of trolley wire and track/km

$= 0{\cdot}25 + 0{\cdot}03 = 0{\cdot}28\Omega$

Current in section $SA = 40 + 20 = 60$ A

Current in section $AB = 20$ A

Voltage drop in section $SA = 60\times0{\cdot}28\times2 = 33{\cdot}6$ V

Voltage drop in section $AB = 20\times0{\cdot}28\times4 = 22{\cdot}4$ V

$\therefore$ Voltage across tram $A = 600 - 33{\cdot}6 = \mathbf{566{\cdot}4\ V}$

Voltage across tram $B = 566{\cdot}4 - 22{\cdot}4 = \mathbf{544\ V}$

Example 13.4. *The load distribution on a two-wire d.c. distributor is shown in Fig. 13.9. The cross-sectional area of each conductor is 0·27 cm^2. The end A is supplied at 250 V. Resistivity of the wire is $\rho = 1{\cdot}78\ \mu\Omega$ cm. Calculate (i) the current in each section of the conductor (ii) the two-core resistance of each section (iii) the voltage at each tapping point.*

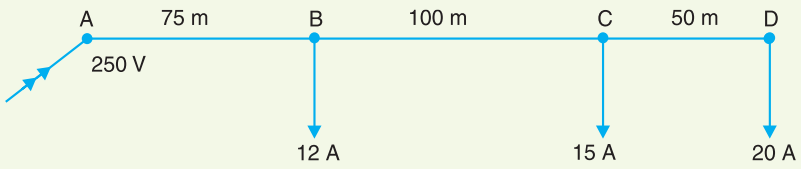


Fig. 13.9

Solution.

(*i*) Currents in the various sections are :

Section CD, $I_{CD} = \mathbf{20\ A}$; section BC, $I_{BC} = 20 + 15 = \mathbf{35\ A}$

Section AB, $I_{AB} = 20 + 15 + 12 = \mathbf{47\ A}$

(*ii*) Single-core resistance of the section of 100 m length

$= \rho\dfrac{l}{a} = 1{\cdot}78\times10^{-6}\times\dfrac{100\times100}{0{\cdot}27} = 0{\cdot}066\,\Omega$

* Note that resistance of each conductor of $l = 100$ m is $r/2$.

The resistances of the various sections are :

$$R_{AB} = 0{\cdot}066 \times 0{\cdot}75 \times 2 = \mathbf{0{\cdot}099\ \Omega}\ ;\ R_{BC} = 0{\cdot}066 \times 2 = \mathbf{0{\cdot}132\ \Omega}$$

$$R_{CD} = 0{\cdot}066 \times 0{\cdot}5 \times 2 = \mathbf{0{\cdot}066\ \Omega}$$

(*iii*) Voltage at tapping point *B* is

$$V_B = V_A - I_{AB} R_{AB} = 250 - 47 \times 0{\cdot}099 = \mathbf{245{\cdot}35\ V}$$

Voltage at tapping point *C* is

$$V_C = V_B - I_{BC} R_{BC} = 245{\cdot}35 - 35 \times 0{\cdot}132 = \mathbf{240{\cdot}73\ V}$$

Voltage at tapping point *D* is

$$V_D = V_C - I_{CD} R_{CD} = 240{\cdot}73 - 20 \times 0{\cdot}066 = \mathbf{239{\cdot}41\ V}$$

TUTORIAL PROBLEMS

1. What should be the minimum cross-sectional area of each conductor in a two-core cable 100 m long to limit the total voltage drop to 4% of the declared voltage of 250V when the conductors carry 60A ? The resistivity of the conductor is 2·845 μΩ cm. **[0·34 cm^2]**
2. A 2-wire d.c. distributor, 500 m long is fed at one of its ends. The cross-sectional area of each conductor is 3·4 cm^2 and the resistivity of copper is 1·7 μΩcm. The distributor supplies 200 A at a distance of 300m from the feeding point and 100 A at the terminus. Calculate the voltage at the feeding end if the voltage at the terminus is to be 230 V. **[241 V]**
3. A 2-wire d.c. distributor *A B* 500 metres long is fed from point *A* and is loaded as under :

Distance from feeding point *A* (in metres)	100	300	400	500
Load (amperes)	20	40	40	50

If the specific resistance of copper is $1{\cdot}7 \times 10^{-8}$ Ωm, what must be the cross-section of each wire in order that the voltage drop in the distributor shall not exceed 10 volts ? **[1·734 cm^2]**

4. A 2-wire d.c. distributor is 250 m long. It is to be loaded as shown in Fig 13.10 at 50 m intervals. If the maximum voltage drop is not to exceed 10V and the resistivity of core material is 0·7 × 2·54 μΩ cm, determine the maximum cross-sectional area of each conductor. **[1·602 cm^2]**

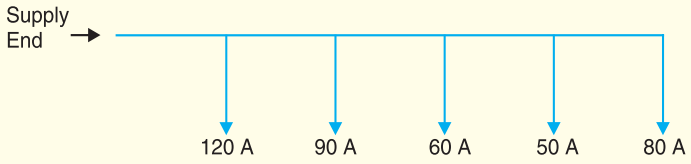


Fig. 13.10

13.4 Uniformly Loaded Distributor Fed at One End

Fig 13.11 shows the single line diagram of a 2-wire d.c. distributor *A B* fed at one end *A* and loaded uniformly with *i* amperes per metre length. It means that at every 1 m length of the distributor, the load tapped is *i* amperes. Let *l* metres be the length of the distributor and *r* ohm be the resistance per metre run.

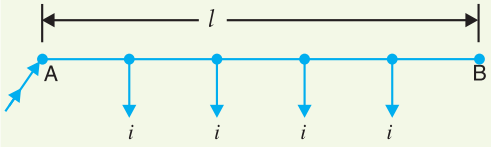


Fig. 13.11

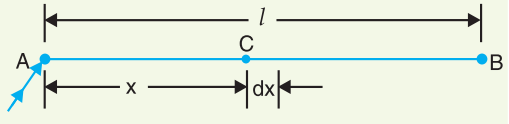


Fig. 13.12

Consider a point *C* on the distributor at a distance *x* metres from the feeding point *A* as shown in Fig. 13.12. Then current at point *C* is

$$= i\,l - i\,x \text{ amperes} = i\,(l - x) \text{ amperes}$$

Now, consider a small length dx near point C. Its resistance is $r\ dx$ and the voltage drop over length dx is

$$dv = i\,(l-x)\,r\,dx = i\,r\,(l-x)\,dx$$

Total voltage drop in the distributor upto point C is

$$v = \int_0^x i\,r\,(l-x)\,dx = i\,r\left(l\,x - \frac{x^2}{2}\right)$$

The voltage drop upto point B (*i.e.* over the whole distributor) can be obtained by putting $x = l$ in the above expression.

$\therefore$ Voltage drop over the distributor AB

$$= i\,r\left(l \times l - \frac{l^2}{2}\right)$$

$$= \tfrac{1}{2}\,i\,r\,l^2 = \tfrac{1}{2}\,(i\,l)\,(r\,l)$$

$$= \tfrac{1}{2}\,I\,R$$

where $i\,l = I$, the total current entering at point A

$r\,l = R$, the total resistance of the distributor

Thus, in a uniformly loaded distributor fed at one end, the total voltage drop is equal to that produced by the whole of the load assumed to be concentrated at the middle point.

Example 13.5. *A 2-wire d.c. distributor 200 metres long is uniformly loaded with 2A/metre. Resistance of single wire is 0·3 Ω/km. If the distributor is fed at one end, calculate :*

(*i*) *the voltage drop upto a distance of 150 m from the feeding point*

(*ii*) *the maximum voltage drop*

Solution.

Current loading, $i = 2$ A/m

Resistance of distributor per metre run,

$$r = 2 \times 0{\cdot}3/1000 = 0{\cdot}0006\,\Omega$$

Length of distributor, $l = 200$ m

(i) Voltage drop upto a distance x metres from feeding point

$$= i\,r\left(l\,x - \frac{x^2}{2}\right)$$ [See Art. 13·4]

Here, $x = 150$ m

$\therefore$ Desired voltage drop $= 2 \times 0{\cdot}0006\left(200 \times 150 - \dfrac{150 \times 150}{2}\right) =$ **22·5 V**

(ii) Total current entering the distributor,

$$I = i \times l = 2 \times 200 = 400\ \text{A}$$

Total resistance of the distributor,

$$R = r \times l = 0{\cdot}0006 \times 200 = 0{\cdot}12\,\Omega$$

$\therefore$ Total drop over the distributor

$$= \frac{1}{2}\,I\,R = \frac{1}{2} \times 400 \times 0{\cdot}12 =$$ **24 V**

Example 13.6. *A uniform 2-wire d.c. distributor 500 metres long is loaded with 0.4 ampere/metre and is fed at one end. If the maximum permissible voltage drop is not to exceed 10 V, find the cross-sectional area of the distributor conductor. Take* $\rho = 1{\cdot}7 \times 10^{-6}\ \Omega$ *cm.*

Solution.

Current entering the distributor, $I = i \times l = 0{\cdot}4 \times 500 = 200$ A

Max. permissible voltage drop $= 10$ V

Let r ohm be the resistance per metre length of the distributor (both wires).

Max. voltage drop $= \frac{1}{2} I R$

or $10 = \frac{1}{2} I r l$ $[\because R = r l]$

or $r = \dfrac{2 \times 10}{I \times l} = \dfrac{2 \times 10}{200 \times 500} = 0{\cdot}2 \times 10^{-3}\ \Omega$

$\therefore$ Area of cross-section of the distributor conductor is

$$a = \frac{\rho l}{r/2} = \frac{1{\cdot}7 \times 10^{-6} \times 100^{*} \times 2}{0{\cdot}2 \times 10^{-3}} = \mathbf{1{\cdot}7\ cm^2}$$

Example 13.7. *A 250 m , 2-wire d.c. distributor fed from one end is loaded uniformly at the rate of 1·6 A/metre. The resistance of each conductor is 0·0002 Ω per metre. Find the voltage necessary at feed point to maintain 250 V (i) at the far end (ii) at the mid-point of the distributor.*

Solution.

Current loading, $i = 1{\cdot}6$A/m

Current entering the distributor, $I = i \times l = 1{\cdot}6 \times 250 = 400$ A

Resistance of the distributor per metre run

$r = 2 \times 0{\cdot}0002 = 0{\cdot}0004\,\Omega$

Total resistance of distributor, $R = r \times l = 0{\cdot}0004 \times 250 = 0{\cdot}1\,\Omega$

(i) Voltage drop over the entire distributor

$$= \frac{1}{2} I R = \frac{1}{2} \times 400 \times 0{\cdot}1 = 20 \text{ V}$$

$\therefore$ Voltage at feeding point $= 250 + 20 = \mathbf{270\ V}$

(ii) Voltage drop upto a distance of x metres from feeding point

$$= i r \left(l x - \frac{x^2}{2} \right)$$

Here $x = l/2 = 250/2 = 125$ m

$\therefore$ Voltage drop $= 1{\cdot}6 \times 0{\cdot}0004 \left(250 \times 125 - \dfrac{(125)^2}{2} \right) = 15$ V

$\therefore$ Voltage at feeding point $= 250 + 15 = \mathbf{265\ V}$

Example 13.8. *Derive an expression for the power loss in a uniformly loaded distributor fed at one end.*

Solution. Fig. 13.13 shows the single line diagram of a 2-wire d.c. distributor AB fed at end A and loaded uniformly with i amperes per metre length.

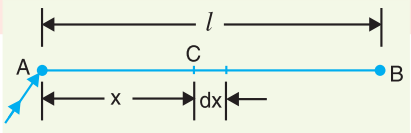


Fig. 13.13

Let l = length of the distributor in metres

r = resistance of distributor (both conductors) per metre run

Consider a small length dx of the distributor at point C at a distance x from the feeding end A. The small length dx will carry current which is tapped in the length CB.

$\therefore$ Current in $dx = i\,l - i\,x = i\,(l - x)$

---

* Because we have assumed that r ohm is the resistance of 1m (= 100 cm) length of the distributor.

Power loss in length dx = (current in length dx)^2 × Resistance of length dx

$= [i\,(l-x)]^2 \times r\,dx$

Total power loss P in the whole distributor is

$$P = \int_0^l [i\,(l-x)]^2\, r\,dx = \int_0^l i^2\left(l^2 + x^2 - 2lx\right) r\,dx$$

$$= i^2 r \int_0^l \left(l^2 + x^2 - 2lx\right) dx = i^2 r \left[l^2 x + \frac{x^3}{3} - \frac{2lx^2}{2}\right]_0^l$$

$$= i^2 r \left[l^3 + \frac{l^3}{3} - l^3\right] = i^2 \times \frac{r\,l^3}{3}$$

$$\therefore \quad P = \frac{i^2\, r\, l^3}{3}$$

Example 13.9. *Calculate the voltage at a distance of 200 m of a 300 m long distributor uniformly loaded at the rate of 0.75 A per metre. The distributor is fed at one end at 250 V. The resistance of the distributor (go and return) per metre is 0·00018 Ω. Also find the power loss in the distributor.*

Solution.

Voltage drop at a distance x from supply end

$$= i\,r\left(l\,x - \frac{x^2}{2}\right)$$

Here i = 0·75 A/m; l = 300 m ; x = 200 m ; r = 0·00018 Ω/m

$$\therefore \quad \text{Voltage drop} = 0{\cdot}75 \times 0{\cdot}00018\left[300 \times 200 - \frac{(200)^2}{2}\right] = 5.4 \text{ V}$$

Voltage at a distance of 200 m from supply end

$$= 250 - 5{\cdot}4 = \mathbf{244{\cdot}6\ V}$$

Power loss in the distributor is

$$P = \frac{i^2\, r\, l^3}{3} = \frac{(0{\cdot}75)^2 \times 0{\cdot}00018 \times (300)^3}{3} = \mathbf{911{\cdot}25\ W}$$

TUTORIAL PROBLEMS

1. A 2-wire d.c. distributor 500 m long is loaded uniformly at the rate of 0·4A/m. If the voltage drop in the distributor is not to exceed 5V, calculate the area of X-section of each conductor required when the distributor is fed at one end. Take resistivity of conductor material as $1{\cdot}7 \times 10^{-8}$ Ωm. **[3·4 cm^2]**
2. A uniformly distributed load on a distributor of length 500 m is rated at 1 A per metre length. The distributor is fed from one end at 220V. Determine the voltage drop at a distance of 400 m from the feeding point. Assume a loop resistance of 2×10^{-5} Ω per metre. **[2·4 V]**
3. A 250 m, 2-wire d.c. distributor fed from one end is loaded uniformly at the rate of 0·8 A per metre. The resistance of each conductor is 0·0002 Ω per metre. Find the necessary voltage at the feeding point to maintain 250 V at the far end of the distributor. **[260 V]**

13.5 Distributor Fed at Both Ends — Concentrated Loading

Whenever possible, it is desirable that a long distributor should be fed at both ends instead of at one end only, since total voltage drop can be considerably reduced without increasing the cross-section of the conductor. The two ends of the distributor may be supplied with ***(i)*** equal voltages ***(ii)*** unequal voltages.

(*i*) Two ends fed with equal voltages. Consider a distributor *A B* fed at both ends with equal voltages *V* volts and having concentrated loads I_1, I_2, I_3, I_4 and I_5 at points *C*, *D*, *E*, *F* and *G* respectively as shown in Fig. 13.14. As we move away from one of the feeding points, *say A*, p.d. goes on decreasing till it reaches the minimum value at some load point, *say E*, and then again starts rising and becomes *V* volts as we reach the other feeding point *B*.

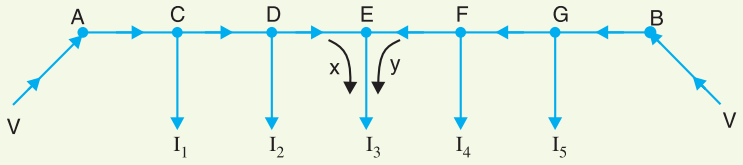


Fig. 13.14

All the currents tapped off between points *A* and *E* (minimum p.d. point) will be supplied from the feeding point *A* while those tapped off between *B* and *E* will be supplied from the feeding point *B*. The current tapped off at point *E* itself will be partly supplied from *A* and partly from *B*. If these currents are *x* and *y* respectively, then,

$$I_3 = x + y$$

Therefore, we arrive at a very important conclusion that at the point of minimum potential, current comes from both ends of the distributor.

Point of minimum potential. It is generally desired to locate the point of minimum potential. There is a simple method for it. Consider a distributor *A B* having three concentrated loads I_1, I_2 and I_3 at points *C*, *D* and *E* respectively. Suppose that current supplied by feeding end *A* is I_A. Then current distribution in the various sections of the distributor can be worked out as shown in Fig. 13.15 (*i*). Thus

$$I_{AC} = I_A; \qquad I_{CD} = I_A - I_1$$
$$I_{DE} = I_A - I_1 - I_2; \qquad I_{EB} = I_A - I_1 - I_2 - I_3$$

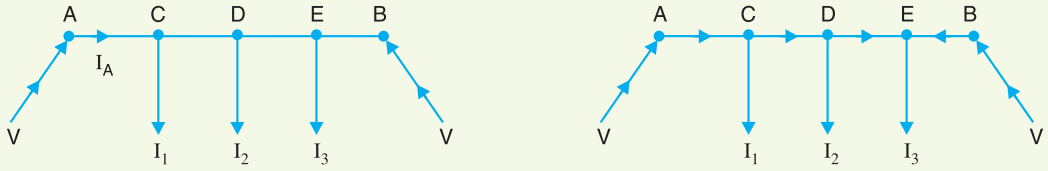


Fig. 13.15

Voltage drop between *A* and *B* = Voltage drop over *A B*

or $V - V = I_A R_{AC} + (I_A - I_1) R_{CD} + (I_A - I_1 - I_2) R_{DE} + (I_A - I_1 - I_2 - I_3) R_{EB}$

From this equation, the unknown I_A can be calculated as the values of other quantities are generally given. Suppose *actual* directions of currents in the various sections of the distributor are indicated as shown in Fig. 13.15 (*ii*). The load point where the currents are coming from both sides of the distributor is the point of minimum potential *i.e.* point *E* in this case

(*ii*) Two ends fed with unequal voltages. Fig. 13.16 shows the distributor *A B* fed with unequal voltages ; end *A* being fed at V_1 volts and end *B* at V_2 volts. The point of minimum potential can be found by following the same procedure as discussed above. Thus in this case,

Voltage drop between *A* and *B* = Voltage drop over *A B*

or $V_1 - V_2$ = Voltage drop over *A B*

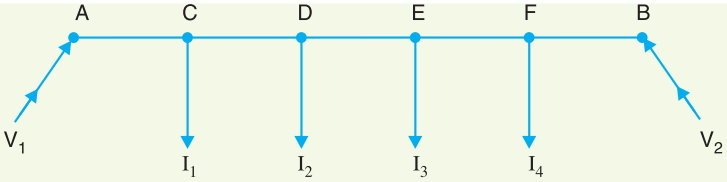


Fig. 13.16

Example 13.10. *A 2-wire d.c. street mains AB, 600 m long is fed from both ends at 220 V. Loads of 20 A, 40 A, 50 A and 30 A are tapped at distances of 100m, 250m, 400m and 500 m from the end A respectively. If the area of X-section of distributor conductor is 1cm^2, find the minimum consumer voltage. Take* $\rho = 1{\cdot}7 \times 10^{-6}\ \Omega\ cm$.

Solution. Fig. 13.17 shows the distributor with its tapped currents. Let I_A amperes be the current supplied from the feeding end A. Then currents in the various sections of the distributor are as shown in Fig. 13.17.

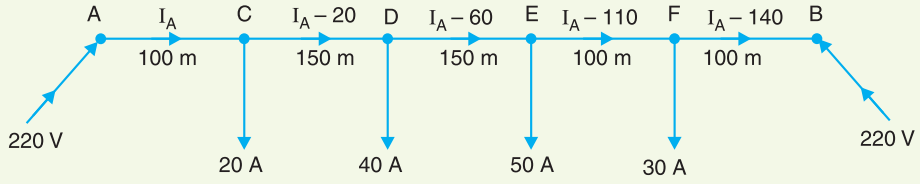


Fig. 13.17

Resistance of 1 m length of distributor

$$= 2 \times \frac{1{\cdot}7 \times 10^{-6} \times 100}{1} = 3{\cdot}4 \times 10^{-4}\ \Omega$$

Resistance of section AC, $R_{AC} = (3{\cdot}4 \times 10^{-4}) \times 100 = 0{\cdot}034\ \Omega$

Resistance of section CD, $R_{CD} = (3{\cdot}4 \times 10^{-4}) \times 150 = 0{\cdot}051\ \Omega$

Resistance of section DE, $R_{DE} = (3{\cdot}4 \times 10^{-4}) \times 150 = 0{\cdot}051\ \Omega$

Resistance of section EF, $R_{EF} = (3{\cdot}4 \times 10^{-4}) \times 100 = 0{\cdot}034\ \Omega$

Resistance of section FB, $R_{FB} = (3{\cdot}4 \times 10^{-4}) \times 100 = 0{\cdot}034\ \Omega$

Voltage at B = Voltage at A – Drop over length AB

or $V_B = V_A - [I_A R_{AC} + (I_A - 20) R_{CD} + (I_A - 60) R_{DE} + (I_A - 110) R_{EF} + (I_A - 140) R_{FB}]$

or $220 = 220 - [0{\cdot}034 I_A + 0{\cdot}051 (I_A - 20) + 0{\cdot}051 (I_A - 60) + 0{\cdot}034 (I_A - 110) + 0{\cdot}034 (I_A - 140)]$

$= 220 - [0{\cdot}204 I_A - 12{\cdot}58]$

or $0{\cdot}204 I_A = 12{\cdot}58$

$\therefore\ I_A = 12{\cdot}58/0{\cdot}204 = 61{\cdot}7$ A

The *actual distribution of currents in the various sections of the distributor is shown in Fig. 13.18. It is clear that currents are coming to load point E from both sides *i.e.* from point D and point F. Hence, E is the point of minimum potential.

$\therefore$ Minimum consumer voltage,

$$V_E = V_A - [I_{AC} R_{AC} + I_{CD} R_{CD} + I_{DE} R_{DE}]$$

* Knowing the value of I_A, current in any section can be determined. Thus,

Current in section CD, $I_{CD} = I_A - 20 = 61{\cdot}7 - 20 = 41{\cdot}7\ A$ from C to D

Current in section EF, $I_{EF} = I_A - 110 = 61{\cdot}7 - 110 = -48{\cdot}3$ A from E to F

$= 48{\cdot}3$ A from F to E

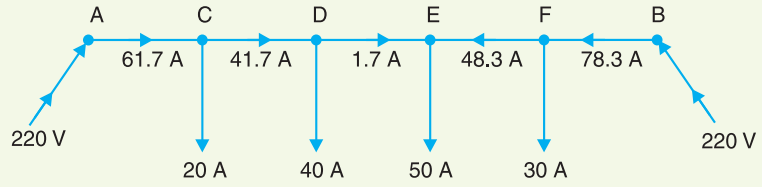


Fig. 13.18

$= 220 - [61{\cdot}7 \times 0{\cdot}034 + 41{\cdot}7 \times 0{\cdot}051 + 1{\cdot}7 \times 0{\cdot}051]$

$= 220 - 4{\cdot}31 =$ **215·69 V**

Example 13.11. *A 2-wire d.c. distributor AB is fed from both ends. At feeding point A, the voltage is maintained as at 230 V and at B 235 V. The total length of the distributor is 200 metres and loads are tapped off as under :*

25 A at 50 metres from A ; 50 A at 75 metres from A

30 A at 100 metres from A ; 40 A at 150 metres from A

The resistance per kilometre of one conductor is 0·3 Ω. Calculate :

(*i*) *currents in various sections of the distributor*

(*ii*) *minimum voltage and the point at which it occurs*

Solution. Fig. 13.19 shows the distributor with its tapped currents. Let I_A amperes be the current supplied from the feeding point A. Then currents in the various sections of the distributor are as shown in Fig 13.19.

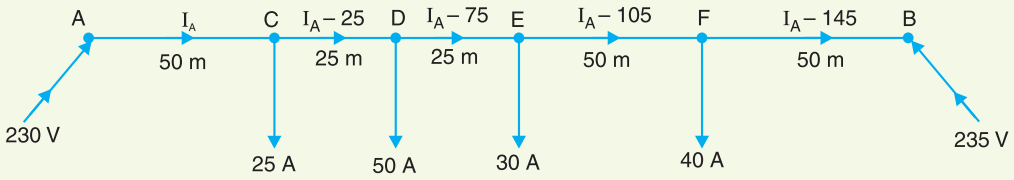


Fig. 13.19

Resistance of 1000 m length of distributor (both wires)

$= 2 \times 0{\cdot}3 = 0{\cdot}6\,\Omega$

Resistance of section AC, $R_{AC} = 0{\cdot}6 \times 50/1000 = 0{\cdot}03\,\Omega$

Resistance of section CD, $R_{CD} = 0{\cdot}6 \times 25/1000 = 0{\cdot}015\,\Omega$

Resistance of section DE, $R_{DE} = 0{\cdot}6 \times 25/1000 = 0{\cdot}015\,\Omega$

Resistance of section EF, $R_{EF} = 0{\cdot}6 \times 50/1000 = 0{\cdot}03\,\Omega$

Resistance of section FB, $R_{FB} = 0{\cdot}6 \times 50/1000 = 0{\cdot}03\,\Omega$

Voltage at B = Voltage at A – Drop over AB

or $V_B = V_A - [I_A R_{AC} + (I_A - 25) R_{CD} + (I_A - 75) R_{DE} + (I_A - 105) R_{EF} + (I_A - 145) R_{FB}]$

or $235 = 230 - [0{\cdot}03 I_A + 0{\cdot}015 (I_A - 25) + 0{\cdot}015 (I_A - 75) + 0{\cdot}03 (I_A - 105) + 0{\cdot}03 (I_A - 145)]$

or $235 = 230 - [0{\cdot}12 I_A - 9]$

$\therefore \quad I_A = \dfrac{239 - 235}{0{\cdot}12} = 33{\cdot}34$ A

(*i*) $\therefore$ Current in section AC, $I_{AC} = I_A =$ **33·34 A**

Current in section CD, $I_{CD} = I_A - 25 = 33{\cdot}34 - 25 =$ **8·34 A**

Current in section DE, I_{DE} = $I_A - 75 = 33{\cdot}34 - 75 = -41{\cdot}66$ A *from D to E*
= **41·66 A** *from E to D*

Current in section EF, I_{EF} = $I_A - 105 = 33{\cdot}34 - 105 = -71{\cdot}66$ A *from E to F*
= **71·66 A** *from F to E*

Current in section FB, I_{FB} = $I_A - 145 = 33{\cdot}34 - 145 = -111{\cdot}66$ A *from F to B*
= **111·66 A** *from B to F*

(ii) The actual distribution of currents in the various sections of the distributor is shown in Fig. 13.20. The currents are coming to load point D from both sides of the distributor. Therefore, load point D is the point of minimum potential.

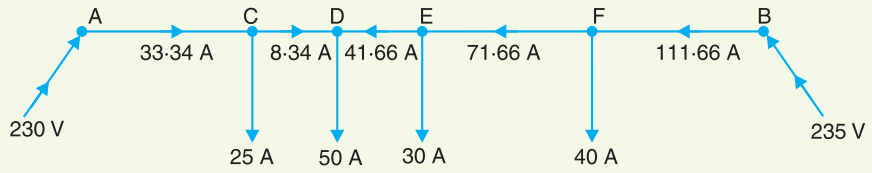


Fig. 13.20

Voltage at D, V_D = $V_A - [I_{AC} R_{AC} + I_{CD} R_{CD}]$
= $230 - [33{\cdot}34 \times 0{\cdot}03 + 8{\cdot}34 \times 0{\cdot}015]$
= $230 - 1{\cdot}125 =$ **228·875 V**

Example 13.12. *A two-wire d.c. distributor AB, 600 metres long is loaded as under :*

Distance from A (metres) :	*150*	*300*	*350*	*450*
Loads in Amperes :	*100*	*200*	*250*	*300*

The feeding point A is maintained at 440 V and that of B at 430 V. If each conductor has a resistance of 0·01 Ω per 100 metres, calculate :

(*i*) *the currents supplied from A to B,* (*ii*) *the power dissipated in the distributor.*

Solution. Fig. 13.21 shows the distributor with its tapped currents. Let I_A amperes be the current supplied from the feeding point A. Then currents in the various sections of the distributor are as shown in Fig.13.21.

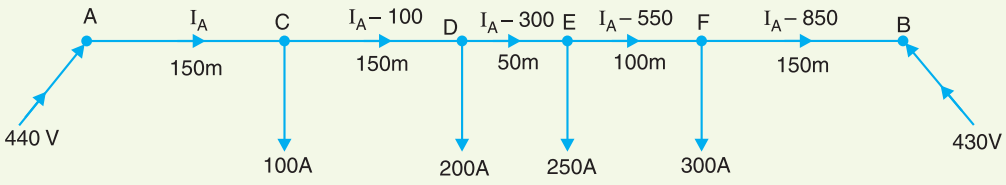


Fig. 13.21

Resistance of 100 m length of distributor (both wires)
= $2 \times 0{\cdot}01 = 0{\cdot}02\,\Omega$

Resistance of section AC, R_{AC} = $0{\cdot}02 \times 150/100 = 0{\cdot}03\,\Omega$

Resistance of section CD, R_{CD} = $0{\cdot}02 \times 150/100 = 0{\cdot}03\,\Omega$

Resistance of section DE, R_{DE} = $0{\cdot}02 \times 50/100 = 0{\cdot}01\,\Omega$

Resistance of section EF, R_{EF} = $0{\cdot}02 \times 100/100 = 0{\cdot}02\,\Omega$

Resistance of section FB, R_{FB} = $0{\cdot}02 \times 150/100 = 0{\cdot}03\,\Omega$

Voltage at B = Voltage at A — Drop over AB

or V_B = $V_A - [I_A R_{AC} + (I_A - 100) R_{CD} + (I_A - 300) R_{DE} + (I_A - 550) R_{EF} + (I_A - 850) R_{FB}]$

or $430 = 440 - [0{\cdot}03\,I_A + 0{\cdot}03\,(I_A - 100) + 0{\cdot}01\,(I_A - 300) + 0{\cdot}02\,(I_A - 550) + 0{\cdot}03\,(I_A - 850)]$

or $430 = 440 - [0{\cdot}12\,I_A - 42{\cdot}5]$

$\therefore \quad I_A = \dfrac{482{\cdot}5 - 430}{0{\cdot}12} = 437{\cdot}5$ A

The actual distribution of currents in the various sections of the distributor is shown in Fig.13.22. Incidentally, E is the point of minimum potential.

(i) Referring to Fig. 13.22, it is clear that

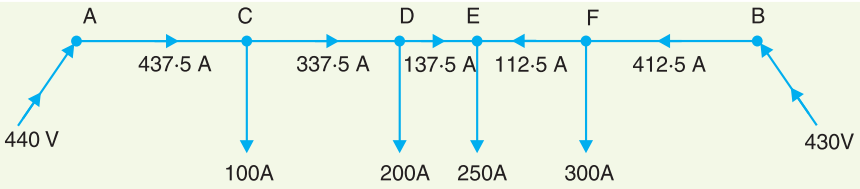


Fig. 13.22

Current supplied from end A, I_A = **437·5 A**

Current supplied from end B, I_B = **412·5 A**

(ii) Power loss in the distributor

$= I_{AC}^2 R_{AC} + I_{CD}^2 R_{CD} + I_{DE}^2 R_{DE} + I_{EF}^2 R_{EF} + I^2_{FB} R_{FB}$

$= (437{\cdot}5)^2 \times 0{\cdot}03 + (337{\cdot}5)^2 \times 0{\cdot}03 + (137{\cdot}5)^2 \times 0{\cdot}01 + (112{\cdot}5)^2 \times 0{\cdot}02 + (412{\cdot}5)^2 \times 0{\cdot}03$

$= 5742 + 3417 + 189 + 253 + 5104 = 14{,}705$ watts = **14·705 kW**

Example 13.13. *An electric train runs between two sub-stations 6 km apart maintained at voltages 600 V and 590 V respectively and draws a constant current of 300 A while in motion. The track resistance of go and return path is 0·04 Ω/km. Calculate :*

(*i*) *the point along the track where minimum potential occurs*

(*ii*) *the current supplied by the two sub-stations when the train is at the point of minimum potential*

Solution. The single line diagram is shown in Fig. 13.23 where substation A is at 600 V and sub-station B at 590 V. Suppose that minimum potential occurs at point M at a distance x km from the substation A. Let I_A amperes be the current supplied by the sub-station A. Then current supplied by sub-station B is 300 — I_A as shown in Fig 13.23.

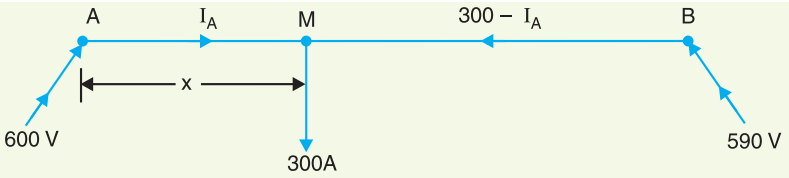


Fig. 13.23

Resistance of track (go and return path) per km $= 0{\cdot}04\,\Omega$

Track resistance for section AM, $R_{AM} = 0{\cdot}04\,x\,\Omega$

Track resistance for section MB, $R_{MB} = 0{\cdot}04\,(6 - x)\Omega$

Potential at M, $V_M = V_A - I_A R_{AM}$ ... (*i*)

Also, Potential at M, $V_M = V_B - (300 - I_A) R_{MB}$ ... (*ii*)

From equations (*i*) and (*ii*), we get,

$$V_A - I_A R_{AM} = V_B - (300 - I_A) R_{MB}$$

or $$600 - 0{\cdot}04\,x\,I_A = 590 - (300 - I_A) \times 0{\cdot}04\,(6 - x)$$

or $$600 - 0{\cdot}04\,x\,I_A = 590 - 0{\cdot}04\,(1800 - 300\,x - 6\,I_A + I_A \times x)$$

or $$600 - 0{\cdot}04\,x\,I_A = 590 - 72 + 12\,x + 0{\cdot}24\,I_A - 0{\cdot}04\,x I_A$$

or $$0{\cdot}24 I_A = 82 - 12\,x$$

or $$I_A = 341{\cdot}7 - 50\,x$$

Substituting the value of I_A in eq. (*i*), we get,

$$V_M = V_A - (341{\cdot}7 - 50\,x) \times 0{\cdot}04\,x$$

$\therefore$ $$V_M = 600 - 13{\cdot}7\,x + 2x^2 \quad ...(iii)$$

(*i*) For V_M to be minimum, its differential coefficient *w.r.t.* x must be zero *i.e.*

$$\frac{d}{dx}(600 - 13{\cdot}7\,x + 2x^2) = 0$$

or $$0 - 13{\cdot}7 + 4x = 0$$

$\therefore$ $$x = 13{\cdot}7/4 = \mathbf{3{\cdot}425\ km}$$

i.e. minimum potential occurs at a distance of 3·425 km from the sub-station A.

(*ii*) $\therefore$ Current supplied by sub-station A

$$= 341{\cdot}7 - 50 \times 3{\cdot}425 = 341{\cdot}7 - 171{\cdot}25 = \mathbf{170{\cdot}45\ A}$$

Current supplied by sub-station B $= 300 - I_A = 300 - 170{\cdot}45 = \mathbf{129{\cdot}55\ A}$

TUTORIAL PROBLEMS

1. A 2-wire d.c. distributor AB is fed at both ends at the same voltage of 230 V. The length of the distributor is 500 metres and the loads are tapped off from the end A as shown below :

Load :	100 A	60 A	40 A	100 A
Distance :	50 m	150 m	250 m	400 m

If the maximum voltage drop of 5·5 V is to be allowed, find the X-sectional area of each conductor and point of minimum potential. Specific resistance of conductor material may be taken as $1{\cdot}73 \times 10^{-8}\ \Omega$ m.
[1·06 cm^2 ; 250 m from A]

2. A d.c. distributor AB is fed at both ends. At feeding point A, the voltage is maintained at 235 V and at B at 236 V. The total length of the distributor is 200 metres and loads are tapped off as under :

20 A at 50 m from A
40 A at 75 m from A
25 A at 100 m from A
30 A at 150 m from A

The resistance per kilometre of one conductor is 0·4 Ω. Calculate the minimum voltage and the point at which it occurs. **[232·175 V ; 75 m from point A]**

3. A two conductor main AB, 500 m in length is fed from both ends at 250 volts. Loads of 50 A, 60 A, 40 A and 30 A are tapped at distance of 100 m, 250 m, 350 m and 400 m from end A respectively. If the X-section of conductor be 1 cm^2 and specific resistance of the material of the conductor is 1·7 μ Ω cm, determine the minimum consumer voltage. **[245·07 V]**

13.6 Uniformly Loaded Distributor Fed at Both Ends

We shall now determine the voltage drop in a uniformly loaded distributor fed at both ends. There can be two cases *viz.* the distributor fed at both ends with (*i*) equal voltages (*ii*) unequal voltages. The two cases shall be discussed separately.

(*i*) Distributor fed at both ends with equal voltages. Consider a distributor AB of length l metres, having resistance r ohms per metre run and with uniform loading of i amperes per

metre run as shown in Fig. 13.24. Let the distributor be fed at the feeding points A and B at equal voltages, say V volts. The total current supplied to the distributor is $i\,l$. As the two end voltages are equal, therefore, current supplied from each feeding point is $i\,l/2$ *i.e.*

Current supplied from each feeding point

$$= \frac{i\,l}{2}$$

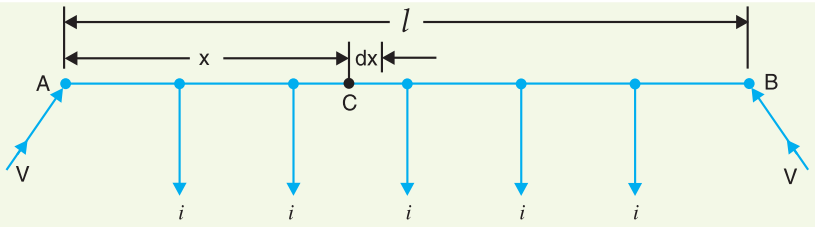


Fig. 13.24

Consider a point C at a distance x metres from the feeding point A. Then current at point C is

$$= \frac{i\,l}{2} - i\,x = i\left(\frac{l}{2} - x\right)$$

Now, consider a small length dx near point C. Its resistance is $r\,dx$ and the voltage drop over length dx is

$$dv = i\left(\frac{l}{2} - x\right) r\,dx = i\,r\left(\frac{l}{2} - x\right) dx$$

$$\therefore \quad \text{Voltage drop upto point } C = \int_0^x i\,r\left(\frac{l}{2} - x\right) dx = i\,r\left(\frac{l\,x}{2} - \frac{x^2}{2}\right)$$

$$= \frac{i\,r}{2}\left(l\,x - x^2\right)$$

Obviously, the point of minimum potential will be the mid-point. Therefore, maximum voltage drop will occur at mid-point *i.e.* where $x = l/2$.

$$\therefore \quad \text{Max. voltage drop} = \frac{i\,r}{2}\left(l\,x - x^2\right)$$

$$= \frac{i\,r}{2}\left(l \times \frac{l}{2} - \frac{l^2}{4}\right) \qquad [\text{Putting } x = l/2]$$

$$= \frac{1}{8}\,i\,r\,l^2 = \frac{1}{8}\,(i\,l)\,(r\,l) = \frac{1}{8}\,I\,R$$

where $i\,l$ = I, the total current fed to the distributor from both ends

$r\,l$ = R, the total resistance of the distributor

$$\text{Minimum voltage} = V - \frac{I\,R}{8} \text{ volts}$$

(ii) **Distributor fed at both ends with unequal voltages.** Consider a distributor AB of length l metres having resistance r ohms per metre run and with a uniform loading of i amperes per metre run as shown in Fig. 13.25. Let the distributor be fed from feeding points A and B at voltages V_A and V_B respectively.

Suppose that the point of minimum potential C is situated at a distance x metres from the feeding point A. Then current supplied by the feeding point A will be $*i\,x$.

* As C is at minimum potential, therefore, there is no current at this point. Consequently, current in section AC (*i.e.* $i\,x$) will be the current supplied by feeding point A.

$$\therefore \quad \text{Voltage drop in section } AC = \frac{i\,r\,x^2}{2} \text{ volts}$$

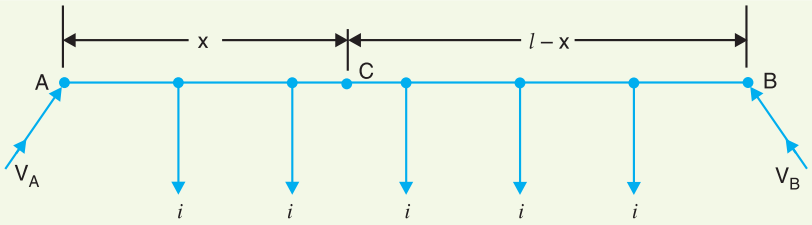


Fig. 13.25

As the distance of C from feeding point B is $(l - x)$, therefore, current fed from B is $i\,(l - x)$.

$$\therefore \quad \text{Voltage drop in section } BC = \frac{i\,r\,(l-x)^2}{2} \text{ volts}$$

$$\text{Voltage at point } C,\ V_C = V_A - \text{Drop over } AC$$

$$= V_A - \frac{i\,r\,x^2}{2} \qquad ...(i)$$

$$\text{Also, voltage at point } C,\ V_C = V_B - \text{Drop over } BC$$

$$= V_B - \frac{i\,r\,(l-x)^2}{2} \qquad ...(ii)$$

From equations (i) and (ii), we get,

$$V_A - \frac{i\,r\,x^2}{2} = V_B - \frac{i\,r\,(l-x)^2}{2}$$

Solving the equation for x, we get,

$$x = \frac{V_A - V_B}{i\,r\,l} + \frac{l}{2}$$

As all the quantities on the right hand side of the equation are known, therefore, the point on the distributor where minimum potential occurs can be calculated.

Example 13.14. *A two-wire d.c. distributor cable 1000 metres long is loaded with 0·5 A/metre. Resistance of each conductor is 0·05 Ω/km. Calculate the maximum voltage drop if the distributor is fed from both ends with equal voltages of 220 V. What is the minimum voltage and where it occurs ?*

Solution.

Current loading, $i = 0{\cdot}5$ A/m

Resistance of distributor/m, $r = 2 \times 0{\cdot}05/1000 = 0{\cdot}1 \times 10^{-3}\ \Omega$

Length of distributor, $l = 1000$ m

Total current supplied by distributor, $I = i\,l = 0{\cdot}5 \times 1000 = 500$ A

Total resistance of the distributor, $R = r\,l = 0{\cdot}1 \times 10^{-3} \times 1000 = 0{\cdot}1\ \Omega$

$$\therefore \quad \text{Max. voltage drop} = \frac{I\,R}{8} = \frac{500 \times 0{\cdot}1}{8} = \mathbf{6{\cdot}25\ V}$$

Minimum voltage will occur at the mid-point of the distributor and its value is

$$= 220 - 6{\cdot}25 = \mathbf{213{\cdot}75\ V}$$

Example 13.15. *A 2-wire d.c. distributor AB 500 metres long is fed from both ends and is loaded uniformly at the rate of 1·0 A/metre. At feeding point A, the voltage is maintained at 255 V and at B at 250 V. If the resistance of each conductor is 0·1 Ω per kilometre, determine :*

(i) the minimum voltage and the point where it occurs

(ii) the currents supplied from feeding points A and B

Solution. Fig. 13.26 shows the single line diagram of the distributor.

Voltage at feeding point A, $V_A = 255$ V

Voltage at feeding point B, $V_B = 250$ V

Length of distributor, $l = 500$ m

Current loading, $i = 1$ A/m

Resistance of distributor/m, $r = 2 \times 0{\cdot}1/1000 = 0{\cdot}0002\,\Omega$

(*i*) Let the minimum potential occur at a point C distant x metres from the feeding point A. As proved in Art. 13.6,

$$x = \frac{V_A - V_B}{i\,r\,l} + \frac{l}{2} = \frac{255 - 250}{1 \times 0{\cdot}0002 \times 500} + 500/2$$

$$= 50 + 250 = \mathbf{300\ m}$$

i.e. minimum potential occurs at 300 m from point A.

Minimum voltage, $V_C = V_A - \frac{ir\,x^2}{2} = 255 - \frac{1 \times 0{\cdot}0002 \times (300)^2}{2}$

$$= 255 - 9 = \mathbf{246\ V}$$

(*ii*) Current supplied from $A = i\,x = 1 \times 300 = \mathbf{300\ A}$

Current supplied from $B = i\,(l - x) = 1\,(500 - 300) = \mathbf{200\ A}$

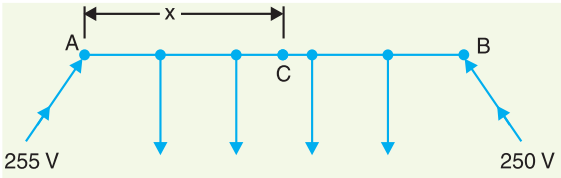


Fig. 13.26

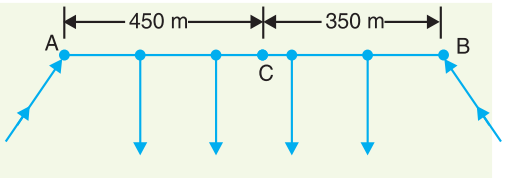


Fig. 13.27

Example 13.16. *A 800 metres 2-wire d.c. distributor AB fed from both ends is uniformly loaded at the rate of 1·25 A/metre run. Calculate the voltage at the feeding points A and B if the minimum potential of 220 V occurs at point C at a distance of 450 metres from the end A. Resistance of each conductor is 0·05 Ω/km.*

Solution. Fig. 13.27 shows the single line diagram of the distributor.

Current loading, $i = 1{\cdot}25$ A/m

Resistance of distributor/m, $r = 2 \times 0{\cdot}05/1000 = 0{\cdot}0001\,\Omega$

Voltage at C, $V_C = 220$ V

Length of distributor, $l = 800$ m

Distance of point C from A, $x = 450$ m

Voltage drop in section $AC = \frac{i\,r\,x^2}{2} = \frac{1{\cdot}25 \times 0{\cdot}0001 \times (450)^2}{2} = 12{\cdot}65$ V

$\therefore$ Voltage at feeding point A, $V_A = 220 + 12{\cdot}65 = \mathbf{232{\cdot}65\ V}$

Voltage drop in section $BC = \frac{i\,r\,(l - x)^2}{2} = \frac{1{\cdot}25 \times 0{\cdot}0001 \times (800 - 450)^2}{2}$

$= 7{\cdot}65$ V

$\therefore$ Voltage at feeding point B, $V_B = 220 + 7{\cdot}65 = \mathbf{227{\cdot}65\ V}$

Example 13.17.

(*i*) *A uniformly loaded distributor is fed at the centre. Show that maximum voltage drop = I R/8 where I is the total current fed to the distributor and R is the total resistance of the distributor.*

(*ii*) *A 2-wire d.c. distributor 1000 metres long is fed at the centre and is loaded uniformly at the rate of 1·25 A/metre. If the resistance of each conductor is 0·05 Ω/km, find the maximum voltage drop in the distributor.*

Solution. (*i*) Fig. 13.28 shows distributor *AB* fed at centre *C* and uniformly loaded with *i* amperes/metre. Let *l* metres be the length of the distributor and *r* ohms be the resistance per metre run. Obviously, maximum voltage drop will occur at either end.

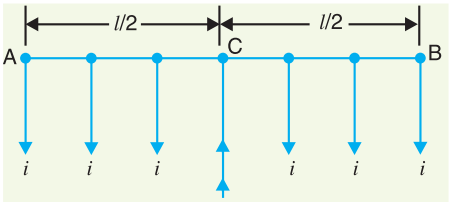


Fig. 13.28

$\therefore$ Max. voltage drop = Voltage drop in half distributor

$$= \frac{1}{2}\left(\frac{i\,l}{2}\right)\left(\frac{r\,l}{2}\right) = \frac{1}{8}(i\,l)(r\,l)$$

$$= \frac{1}{8} I R$$

where $i\,l$ = I, the total current fed to the distributor

$r\,l$ = R, the total resistance of the distributor

(*ii*) Total current fed to the distributor is

$$I = i\,l = 1{\cdot}25 \times 1000 = 1250 \text{ A}$$

Total resistance of the distributor is

$$R = r\,l = 2 \times 0{\cdot}05 \times 1 = 0{\cdot}1\ \Omega$$

$$\text{Max. voltage drop} = \frac{1}{8} I R = \frac{1}{8} \times 1250 \times 0{\cdot}1 = \mathbf{15.62\ V}$$

Example 13.18. *Derive an expression for the power loss in a uniformly loaded distributor fed at both ends with equal voltages.*

Solution. Consider a distributor *AB* of length *l* metres, having resistance *r* ohms per metre run with uniform loading of *i* amperes per metre run as shown in Fig.13.29. Let the distributor be fed at the feeding points *A* and *B* at equal voltages, say V volts. The total current supplied by the distributor is *i l*. As the two end voltages are equal, therefore, current supplied from each feeding point is *i l*/2.

Current supplied from each feeding point = $\frac{i\,l}{2}$

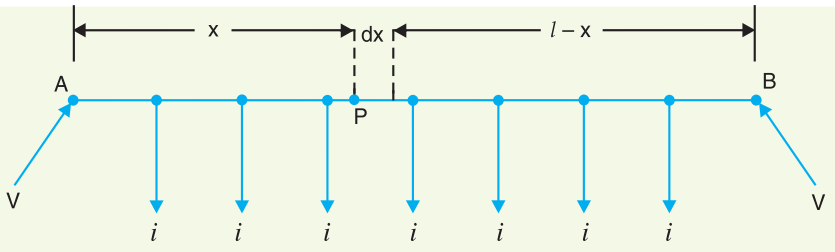


Fig. 13.29

Consider a small length *dx* of the distributor at point *P* which is at a distance *x* from the feeding end *A*.

Resistance of length *dx* = *r dx*

$$\text{Current in length } dx = \frac{i\,l}{2} - i\,x = i\left(\frac{l}{2} - x\right)$$

$$\text{Power loss in length } dx = (\text{current in } dx)^2 \times \text{Resistance of } dx$$

$$= \left[i\left(\frac{l}{2} - x\right)\right]^2 \times r\,dx$$

Total power loss in the distributor is

$$P = \int_0^l \left[i\left(\frac{l}{2} - x\right)\right]^2 r\,dx = i^2 r \int_0^l \left(\frac{l^2}{4} - l\,x + x^2\right) dx$$

$$= i^2 r\left[\frac{l^2 x}{4} - \frac{l\,x^2}{2} + \frac{x^3}{3}\right]_0^l = i^2 r\left[\frac{l^3}{4} - \frac{l^3}{2} + \frac{l^3}{3}\right]$$

$$\therefore \quad P = \frac{i^2\,r\,l^3}{12}$$

13.7 Distributor with Both Concentrated and Uniform Loading

There are several problems where a distributor has both concentrated and uniform loadings. In such situations, the total drop over any section of the distributor is equal to the sum of drops due to concentrated and uniform loading in that section. We shall solve a few problems by way of illustration.

Example 13.19. *A 2-wire d.c. distributor AB, 900 metres long is fed at A at 400 V and loads of 50 A, 100 A and 150 A are tapped off from C, D and E which are at a distance of 200 m, 500 m and 800 m from point A respectively. The distributor is also loaded uniformly at the rate of 0.5 A/m. If the resistance of distributor per metre (go and return) is 0.0001 Ω, calculate voltage (i) at point B and (ii) at point D.*

Solution. This problem can be solved in two stages. First, the drop at any point due to concentrated loading is found. To this is added the voltage drop due to uniform loading.

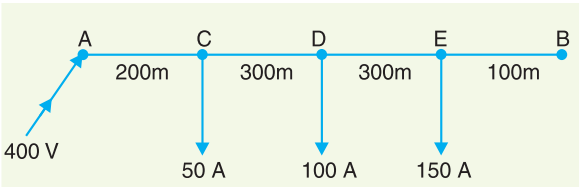


Fig. 13.30

Drops due to concentrated loads. Fig. 13.30 shows only the concentrated loads tapped off from the various points. The currents in the various sections are :

$$I_{AC} = 300 \text{ A} ; I_{CD} = 250 \text{ A} ; I_{DE} = 150 \text{ A}$$

$$\text{Drop in section } AC = I_{AC} R_{AC} = 300 \times (200 \times 0{\cdot}0001) = 6 \text{ V}$$

$$\text{Drop in section } CD = 250 \times (300 \times 0{\cdot}0001) = 7{\cdot}5 \text{ V}$$

$$\text{Drop in section } DE = 150 \times (300 \times 0{\cdot}0001) = 4{\cdot}5 \text{ V}$$

$$\text{Total drop over } AB = 6 + 7{\cdot}5 + 4{\cdot}5 = 18 \text{ V}$$

Drops due to uniform loading

$$\text{Drop over } AB = \frac{i\,r\,l^2}{2} = \frac{0{\cdot}5 \times 0{\cdot}0001 \times (900)^2}{2} = 20{\cdot}25 \text{ V}$$

$$\text{Drop over } AD = i\,r\left(l\,x - \frac{x^2}{2}\right)$$

Here, $l = 900$ m ; $x = 500$ m

$$\therefore \quad \text{Drop over } AD = 0{\cdot}5 \times 0{\cdot}0001\left(900 \times 500 - \frac{500^2}{2}\right) = 16{\cdot}25 \text{ V}$$

(*i*) Voltage at point B = V_A – Drop over AB due to conc. and uniform loadings
= 400 – (18 + 20·25) = **361·75 V**

(*ii*) Voltage at point D = V_A – Drop over AD due to conc. and uniform loadings
= 400 – (6 + 7·5 + 16·25) = **370·25 V**

Example 13.20. *Two conductors of a d.c. distributor cable AB 1000 m long have a total resistance of 0·1 Ω. The ends A and B are fed at 240 V. The cable is uniformly loaded at 0·5 A per metre length and has concentrated loads of 120 A, 60 A, 100 A and 40 A at points distant 200 m, 400 m, 700 m and 900 m respectively from the end A. Calculate (i) the point of minimum potential (ii) currents supplied from ends A and B (iii) the value of minimum potential.*

Solution.

Distributor resistance per metre length, $r = 0{\cdot}1/1000 = 10^{-4}\ \Omega$

Uniform current loading, $i = 0{\cdot}5$ A/m

(*i*) Point of minimum potential. The point of minimum potential is not affected by the uniform loading of the distributor. Therefore, let us consider the concentrated loads first as shown in Fig. 13.31. Suppose the current supplied by end A is I. Then currents in the various sections will be as shown in Fig. 13.31.

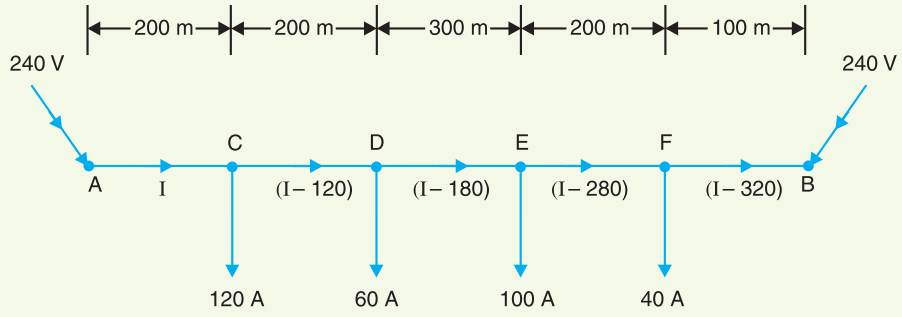


Fig. 13.31

$V_A - V_B$ = Drop over the distributor AB

$240 - 240 = I_{AC} R_{AC} + I_{CD} R_{CD} + I_{DE} R_{DE} + I_{EF} R_{EF} + I_{FB} R_{FB}$

or $0 = 10^{-4} [I \times 200 + (I - 120)\ 200 + (I - 180)\ 300 + (I - 280)\ 200 + (I - 320) \times 100]$

or $0 = 1000\ I - 166000 \quad \therefore \quad I = 166000/1000 = 166$ A

The actual distribution of currents in the various sections of the distributor due to concentrated loading is shown in Fig. 13.32. It is clear from this figure that *D is the point of minimum potential.*

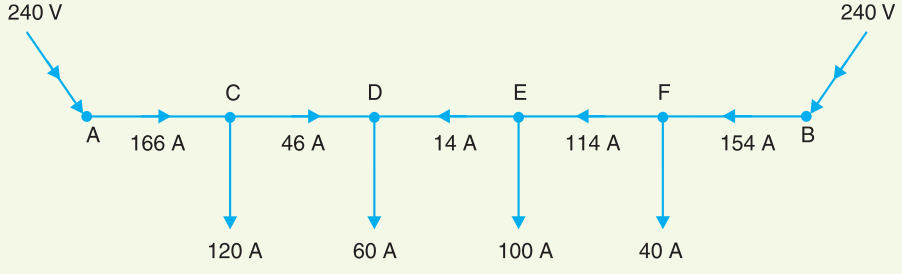


Fig. 13.32

(*ii*) The feeding point A will supply 166 A due to concentrated loading *plus* 0·5 × 400 = 200 A due to uniform loading.

$\therefore$ Current supplied by A, I_A = 166 + 200 = **366 A**

The feeding point B will supply a current of 154 A due to concentrated loading *plus* 0·5 × 600 = 300 A due to uniform loading.

$\therefore$ Current supplied by B, I_B = 154 + 300 = **454 A**

(iii) As stated above, D is the point of minimum potential.

$\therefore$ Minimum potential, V_D = V_A – Drop in AD due to conc. loading – Drop in AD due to uniform loading

Now, Drop in AD due to conc. loading $= I_{AC} R_{AC} + I_{CD} R_{CD}$

$= 166 \times 10^{-4} \times 200 + 46 \times 10^{-4} \times 200$

$= 3{\cdot}32 + 0{\cdot}92 = 4{\cdot}24$ V

Drop in AD due to uniform loading $= \dfrac{i\, r^{*}\, l^2}{2} = \dfrac{0{\cdot}5 \times 10^{-4} \times (400)^2}{2} = 4\,V$

$\therefore$ $V_D = 240 - 4{\cdot}24 - 4$ = **231·76 V**

Example 13.21. *A d.c. 2-wire distributor AB is 500m long and is fed at both ends at 240 V. The distributor is loaded as shown in Fig 13.33. The resistance of the distributor (go and return) is 0·001Ω per metre. Calculate (i) the point of minimum voltage and (ii) the value of this voltage.*

Solution. Let D be the point of **minimum potential and let x be the current flowing in section CD as shown in Fig 13.33. Then current supplied by end B will be $(60 - x)$.

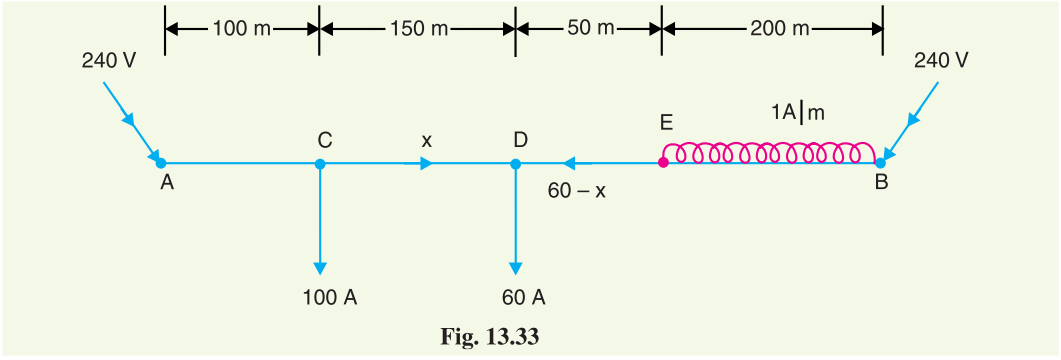


Fig. 13.33

(i) If r is the resistance of the distributor (go and return) per metre length, then,

Voltage drop in length AD $= I_{AC} R_{AC} + I_{CD} R_{CD}$

$= (100 + x) \times 100\, r + x \times 150\, r$

Voltage drop in length BD $= \dfrac{i\, r\, l^2}{2} + (60 - x) \times 250\, r$

$= \dfrac{1 \times r \times (200)^2}{2} + (60 - x) \times 250\, r$

Since the feeding points A and B are at the same potential,

$\therefore$ $(100 + x) \times 100\, r + x \times 150\, r = \dfrac{1 \times r \times (200)^2}{2} + (60 - x)\, 250\, r$

or $100x + 10000 + 150x = 20000 + 15000 - 250x$

or $500x = 25000$ $\quad \therefore x = 50$ A

* Drop due to uniform loading can be determined by imagining that the distributor is cut into two at point D so that AD can be thought as a distributor fed at one end and loaded uniformly.

** You may carry out the calculation by assuming C to be point of minimum potential. The answer will be unaffected.

The actual directions of currents in the various sections of the distributor are shown in Fig. 13.34. Note that currents supplied by *A* and *B* meet at *D*. *Hence point D is the point of minimum potential.*

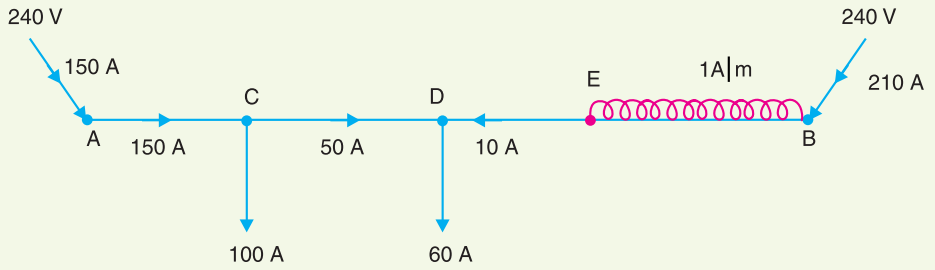


Fig. 13.34

(ii) Total current = $160 + 1 \times 200 = 360$ A

Current supplied by *A*, I_A = $100 + x = 100 + 50 = 150$ A

Current supplied by *B*, I_B = $360 - 150 = 210$ A

Minimum potential, V_D = $V_A - I_{AC} R_{AC} - I_{CD} R_{CD}$

= $240 - 150 \times (100 \times 0{\cdot}001) - 50 \times (150 \times 0{\cdot}001)$

= $240 - 15 - 7{\cdot}5 =$ **217·5 V**

TUTORIAL PROBLEMS

1. A 2-wire d.c. distributor *AB*, 1000 m long has a total resistance of 0·1 Ω. The ends *A* and *B* are fed at 240 V. The distributor is uniformly loaded at 0.5 A/metre length and has concentrated loads of 120 A, 60 A, 100 A and 40 A at points distant 200, 400, 700 and 900 m respectively from end A. Calculate:

(*i*) the point of minimum potential

(*ii*) value of minimum potential

(*iii*) current fed at both ends **[(*i*) 400 m from A (*ii*) 231·76 V (*iii*) I_A = 366 A ; I_B = 454 A]**

2. A 2-wire d.c. distributor *AB* is 300 metres long. The end *A* is fed at 205 V and end B at 200 V. The distributor is uniformly loaded at 0·15 A/metre length and has concentrated loads of 50 A, 60 A and 40 A at points distant 75, 175, 225 m respectively from the end *A*. The resistance of each conductor is 0·15 Ω per kilometre. Calculate :

(*i*) the point of minimum potential

(*ii*) currents fed at ends *A* and *B* **[(*i*) 175 m from A (*ii*) I_A = 150 A ; I_B = 45 A]**

3. A d.c. 2-wire distributor *AB* is 450 m long and is fed at both ends at 250 V. The distributor is loaded as shown in Fig. 13.35. The resistance of each conductor is 0·05 Ω per km. Find the point of minimum potential and its potential. **[261·74 m from A ; 247·35 V]**

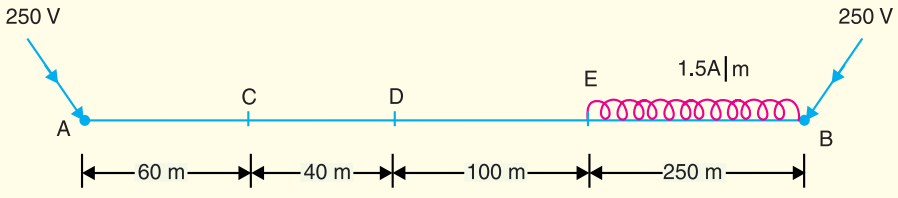


Fig. 13.35

13.8 Ring Distributor

A distributor arranged to form a closed loop and fed at one or more points is called a *ring distributor*. Such a distributor starts from one point, makes a loop through the area to be served, and returns to the

original point. For the purpose of calculating voltage distribution, the distributor can be considered as consisting of a series of open distributors fed at both ends. The principal advantage of ring distributor is that by proper choice in the number of feeding points, great economy in copper can be affected.

The most simple case of a ring distributor is the one having only one feeding point as shown in Fig. 13.36(*ii*). Here A is the feeding point and tappings are taken from points B and C. For the purpose of calculations, it is equivalent to a straight distributor fed at both ends with equal voltages.

Example 13.22. *A 2-wire d.c. ring distributor is 300 m long and is fed at 240 V at point A. At point B, 150 m from A, a load of 120 A is taken and at C, 100 m in the opposite direction, a load of 80 A is taken. If the resistance per 100 m of single conductor is 0·03 Ω, find :*

(*i*) *current in each section of distributor*

(*ii*) *voltage at points B and C*

Solution.

Resistance per 100 m of distributor

$= 2 \times 0{\cdot}03 = 0{\cdot}06\,\Omega$

Resistance of section AB, $R_{AB} = 0{\cdot}06 \times 150/100 = 0{\cdot}09\,\Omega$

Resistance of section BC, $R_{BC} = 0{\cdot}06 \times 50/100 = 0{\cdot}03\,\Omega$

Resistance of section CA, $R_{CA} = 0{\cdot}06 \times 100/100 = 0{\cdot}06\,\Omega$

(*i*) Let us suppose that a current I_A flows in section AB of the distributor. Then currents in sections BC and CA will be $(I_A - 120)$ and $(I_A - 200)$ respectively as shown in Fig. 13.36 (*i*).

According to Kirchhoff's voltage law, the voltage drop in the closed loop $ABCA$ is zero *i.e.*

$$I_{AB} R_{AB} + I_{BC} R_{BC} + I_{CA} R_{CA} = 0$$

or $$0{\cdot}09\, I_A + 0{\cdot}03\,(I_A - 120) + 0{\cdot}06\,(I_A - 200) = 0$$

or $$0{\cdot}18\, I_A = 15{\cdot}6$$

$\therefore$ $$I_A = 15{\cdot}6/0{\cdot}18 = 86{\cdot}67 \text{ A}$$

The actual distribution of currents is as shown in Fig. 13.36 (*ii*) from where it is seen that B is the point of minimum potential.

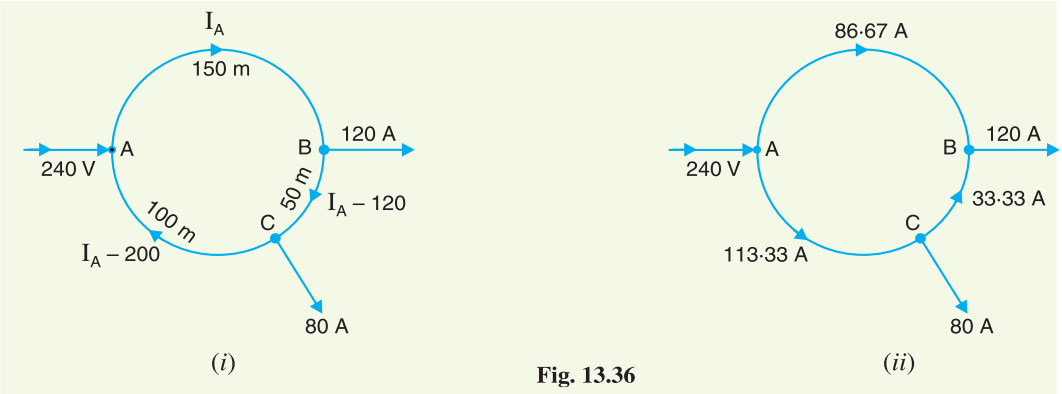


Fig. 13.36

Current in section AB, $I_{AB} = I_A = \mathbf{86{\cdot}67\ A}$ *from A to B*

Current in section BC, $I_{BC} = I_A - 120 = 86{\cdot}67 - 120 = -33{\cdot}33$ A

$= \mathbf{33.33\ A}$ *from C to B*

Current in section CA, $I_{CA} = I_A - 200 = 86{\cdot}67 - 200 = -113{\cdot}33$ A

$= \mathbf{113 \cdot 33\ A}$ *from A to C*

(*ii*) Voltage at point B, $V_B = V_A - I_{AB} R_{AB} = 240 - 86{\cdot}67 \times 0{\cdot}09 = \mathbf{232{\cdot}2\ V}$

Voltage at point C, $V_C = V_B + I_{BC} R_{BC}$
$= 232{\cdot}2 + 33{\cdot}33 \times 0{\cdot}03 =$ **233·2 V**

Example 13.23. *A 2-wire d.c. distributor ABCDEA in the form of a ring main is fed at point A at 220 V and is loaded as under :*

10A at B ; 20A at C ; 30A at D and 10 A at E.

The resistances of various sections (go and return) are : AB = 0·1 Ω ; BC = 0·05 Ω ; CD = 0·01 Ω ; DE = 0·025 Ω and EA = 0·075 Ω. Determine :

(*i*) *the point of minimum potential*

(*ii*) *current in each section of distributor*

Solution. Fig. 13.37 (*i*) shows the ring main distributor. Let us suppose that current I flows in section AB of the distributor. Then currents in the various sections of the distributor are as shown in Fig. 13.37 (*i*).

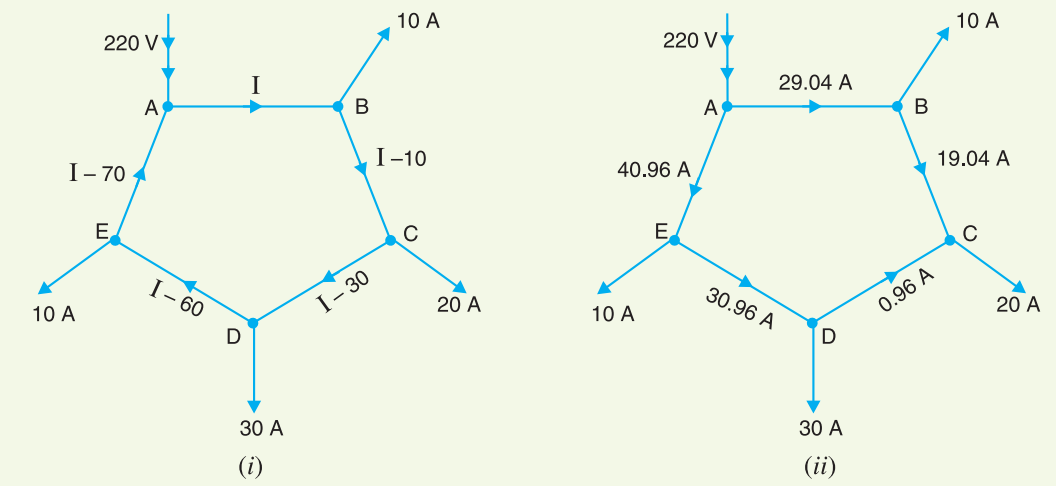


Fig. 13.37

(*i*) According to Kirchhoff's voltage law, the voltage drop in the closed loop $ABCDEA$ is zero *i.e.*

$I_{AB} R_{AB} + I_{BC} R_{BC} + I_{CD} R_{CD} + I_{DE} R_{DE} + I_{EA} R_{EA} = 0$

or $0{\cdot}1I + 0{\cdot}05\,(I - 10) + 0{\cdot}01\,(I - 30) + 0{\cdot}025\,(I - 60) + 0{\cdot}075\,(I - 70) = 0$

or $0{\cdot}26\,I = 7{\cdot}55$

$\therefore \quad I = 7{\cdot}55/0{\cdot}26 = 29{\cdot}04$ A

The actual distribution of currents is as shown in Fig. 13.37 (*ii*) from where it is clear that C is the point of minimum potential.

$\therefore$ C is the point of minimum potential.

(*ii*) Current in section AB $= I =$ **29·04 A** *from A to B*

Current in section BC $= I - 10 = 29{\cdot}04 - 10 =$ **19·04 A** *from B to C*

Current in section CD $= I - 30 = 29{\cdot}04 - 30 = -0{\cdot}96$ A $=$ **0·96 A** *from D to C*

Current in section DE $= I - 60 = 29{\cdot}04 - 60 = -30{\cdot}96$ A $=$ **30·96 A** *from E to D*

Current in section EA $= I - 70 = 29{\cdot}04 - 70 = -40{\cdot}96$ A $=$ **40·96 A** *from A to E*

13.9 Ring Main Distributor with Interconnector

Sometimes a ring distributor has to serve a large area. In such a case, voltage drops in the various sections of the distributor may become excessive. In order to reduce voltage drops in various sections, distant points of the distributor are joined through a conductor called *interconnector*. Fig.

13.38 shows the ring distributor *ABCDEA*. The points *B* and *D* of the ring distributor are joined through an interconnector *BD*. There are several methods for solving such a network. However, the solution of such a network can be readily obtained by applying Thevenin's theorem. The steps of procedure are :

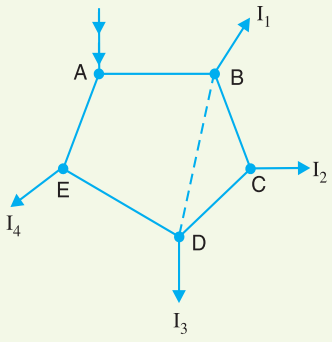
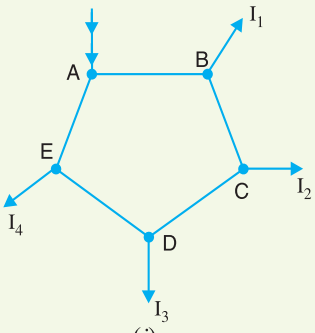
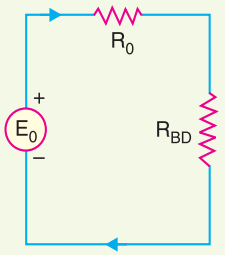


Fig. 13.38



(*i*)



(*ii*)

Fig. 13.39

(*i*) Consider the interconnector *BD* to be disconnected [See Fig. 13.39 (*i*)] and find the potential difference between *B* and *D*. This gives Thevenin's equivalent circuit voltage E_0.

(*ii*) Next, calculate the resistance viewed from points *B* and *D* of the network composed of distribution lines only. This gives Thevenin's equivalent circuit series resistance R_0.

(*iii*) If R_{BD} is the resistance of the interconnector *BD*, then Thevenin's equivalent circuit will be as shown in Fig. 13.39 (*ii*).

$$\therefore \quad \text{Current in interconnector } BD = \frac{E_0}{R_0 + R_{BD}}$$

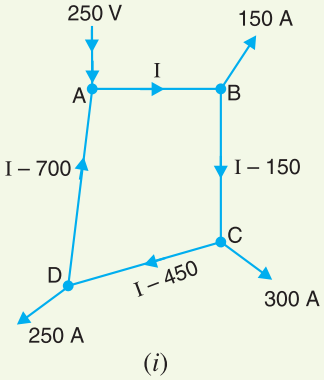
Therefore, current distribution in each section and the voltage of load points can be calculated.

Example 13.24. *A d.c. ring main ABCDA is fed from point A from a 250 V supply and the resistances (including both lead and return) of various sections are as follows : AB = 0·02 Ω ; BC = 0·018 Ω ; CD = 0·025 Ω and DA = 0·02 Ω. The main supplies loads of 150 A at B ; 300 A at C and 250 A at D. Determine the voltage at each load point.*

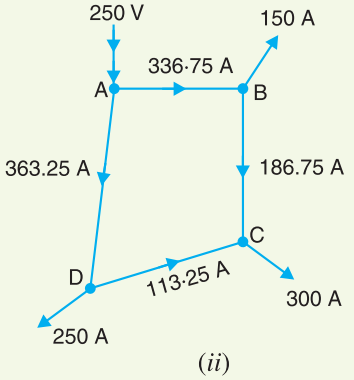
If the points A and C are linked through an interconnector of resistance 0·02 Ω, determine the new voltage at each load point.

Solution.

Without Interconnector. Fig. 13.40 (*i*) shows the ring distributor without interconnector. Let us suppose that a current *I* flows in section *AB* of the distributor. Then currents in various sections of the distributor will be as shown in Fig. 13.40 (*i*).



(*i*)



(*ii*)

Fig. 13.40

According to Kirchhoff's voltage law, the voltage drop in the closed loop *ABCDA* is zero *i.e.*

$$I_{AB}\,R_{AB} + I_{BC}\,R_{BC} + I_{CD}\,R_{CD} + I_{DA}\,R_{DA} = 0$$

or $0{\cdot}02I + 0{\cdot}018\,(I - 150) + 0{\cdot}025\,(I - 450) + 0{\cdot}02\,(I - 700) = 0$

or $0{\cdot}083\,I = 27{\cdot}95$

$\therefore$ $I = 27{\cdot}95/0{\cdot}083 = 336{\cdot}75$ A

The actual distribution of currents is as shown in Fig. 13.40 (*ii*).

Voltage drop in $AB = 336{\cdot}75 \times 0{\cdot}02 = 6{\cdot}735$ V

Voltage drop in $BC = 186{\cdot}75 \times 0{\cdot}018 = 3{\cdot}361$ V

Voltage drop in $CD = 113{\cdot}25 \times 0{\cdot}025 = 2{\cdot}831$ V

Voltage drop in $DA = 363{\cdot}25 \times 0{\cdot}02 = 7{\cdot}265$ V

$\therefore$ Voltage at point $B = 250 - 6{\cdot}735 =$ **243·265 V**

Voltage at point $C = 243{\cdot}265 - 3{\cdot}361 =$ **239·904 V**

Voltage at point $D = 239{\cdot}904 + 2{\cdot}831 =$ **242·735 V**

With Interconnector. Fig. 13.41 (*i*) shows the ring distributor with interconnector *AC*. The current in the interconnector can be found by applying Thevenin's theorem.

E_0 = Voltage between points *A* and *C*

$= 250 - 239{\cdot}904 = 10{\cdot}096$ V

R_0 = Resistance viewed from points *A* and *C*

$$= \frac{(0{\cdot}02 + 0{\cdot}018)\,(0{\cdot}02 + 0{\cdot}025)}{(0{\cdot}02 + 0{\cdot}018) + (0{\cdot}02 + 0{\cdot}025)} = 0{\cdot}02\ \Omega$$

R_{AC} = Resistance of interconnector $= 0{\cdot}02\,\Omega$

Thevenin's equivalent circuit is shown in Fig. 13.41 (*ii*). Current in interconnector *AC*

$$= \frac{E_0}{R_0 + R_{AC}} = \frac{10{\cdot}096}{0{\cdot}02 + 0{\cdot}02} = 252{\cdot}4\ A \text{ from } A \text{ to } C$$

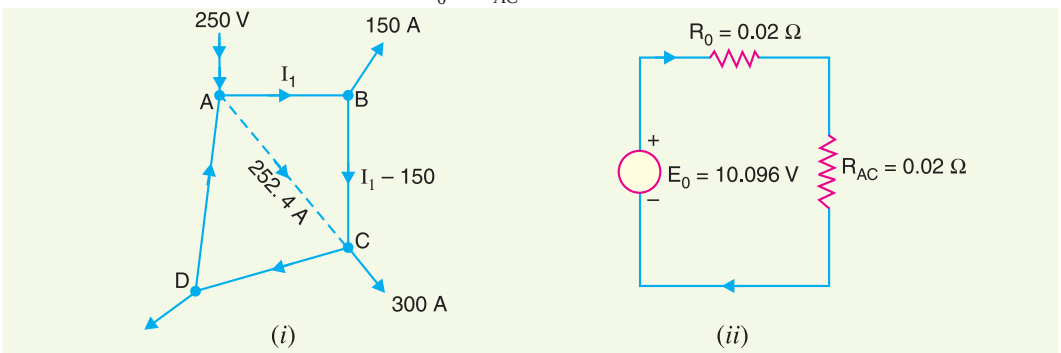


Fig. 13.41

Let us suppose that current in section *AB* is I_1. Then current in section *BC* will be $I_1 - 150$. As the voltage drop round the closed mesh *ABCA* is zero,

$\therefore$ $0{\cdot}02\,I_1 + 0{\cdot}018\,(I_1 - 150) - 0{\cdot}02 \times 252{\cdot}4 = 0$

or $0{\cdot}038\,I_1 = 7{\cdot}748$

$\therefore$ $I_1 = 7{\cdot}748/0{\cdot}038 = 203{\cdot}15$ A

The actual distribution of currents in the ring distributor with interconnector will be as shown in Fig. 13.42.

Drop in $AB = 203{\cdot}15 \times 0{\cdot}02 = 4{\cdot}063$ V

Drop in BC = $53{\cdot}15 \times 0{\cdot}018$

= $0{\cdot}960$ V

Drop in AD = $244{\cdot}45 \times 0{\cdot}02 = 4{\cdot}9$ V

$\therefore$ Potential of B = $250 - 4{\cdot}063$

= **245·93 V**

Potential of C = $245{\cdot}93 - 0{\cdot}96$

= **244·97 V**

Potential of D = $250 - 4{\cdot}9 =$ **245·1 V**

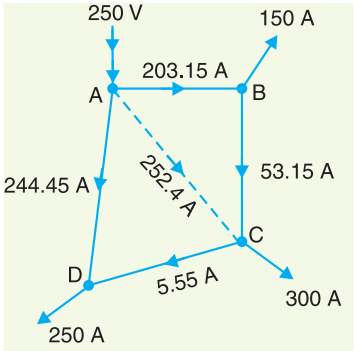


Fig. 13.42

It may be seen that with the use of interconnector, the voltage drops in the various sections of the distributor are reduced.

Example 13.25. *Fig. 13.43 shows a ring distributor with interconnector BD. The supply is given at point A. The resistances of go and return conductors of various sections are indicated in the figure. Calculate :*

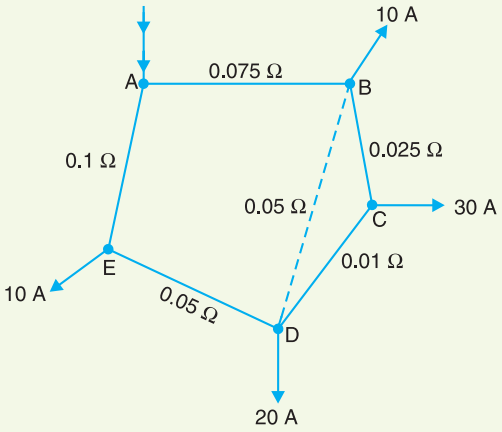


Fig. 13.43

(*i*) *current in the interconnector*

(*ii*) *voltage drop in the interconnector*

Solution. When interconnector BD is removed, let the current in branch AB be I. Then current distribution will be as shown in Fig. 13.44 (*i*). As the total drop round the ring $ABCDEA$ is zero,

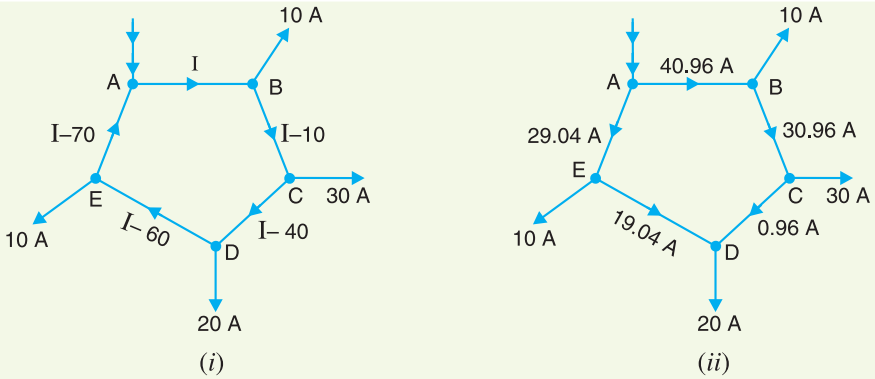


Fig. 13.44

$\therefore$ $0{\cdot}075\,I + 0{\cdot}025\,(I - 10) + 0{\cdot}01\,(I - 40) + 0{\cdot}05\,(I - 60) + 0{\cdot}1\,(I - 70) = 0$

or $0{\cdot}26\,I = 10{\cdot}65$

$\therefore \quad I = \dfrac{10{\cdot}65}{0{\cdot}26} = 40{\cdot}96$ A

The actual distribution of currents will be as shown in Fig. 13.44 (*ii*).

Voltage drop along $BCD = 30{\cdot}96 \times 0{\cdot}025 + 0{\cdot}96 \times 0{\cdot}01$

$= 0{\cdot}774 + 0{\cdot}0096 = 0{\cdot}7836$ V

This is equal to Thevenin's open circuited voltage E_0 *i.e.*

$E_0 = 0{\cdot}7836$ V

R_0 = Resistance viewed from *B* and *D*

$$= \frac{(0{\cdot}075 + 0{\cdot}1 + 0{\cdot}05)(0{\cdot}025 + 0{\cdot}01)}{(0{\cdot}075 + 0{\cdot}1 + 0{\cdot}05) + (0{\cdot}025 + 0{\cdot}01)}$$

$$= \frac{(0{\cdot}225)(0{\cdot}035)}{0{\cdot}225 + 0.035} = 0{\cdot}03\,\Omega$$

(*i*) Current in interconnector *BD* is

$$I_{BD} = \frac{E_0}{R_0 + R_{BD}} = \frac{0{\cdot}7836}{0{\cdot}03 + 0.05} = \mathbf{9.8\ A}$$

(*ii*) Voltage drop along interconnector *BD*

$= I_{BD}\,R_{BD} = 9{\cdot}8 \times 0{\cdot}05 = \mathbf{0{\cdot}49\ V}$

TUTORIAL PROBLEMS

1. A 300 m ring distributor has loads as shown in Fig. 13.45 where distances are in metres. The resistance of each conductor is 0·2 W per 1000 metres and the loads are tapped off at points *B*, *C* and *D* as shown. If the distributor is fed at A at 240 V, find voltages at *B*, *C* and *D*.

[V_B = 236·9 V ; V_C = 235·97 V ; V_D = 237·45 V]

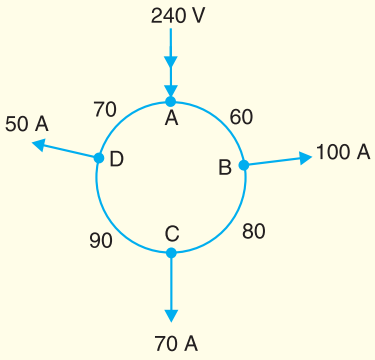


Fig. 13.45

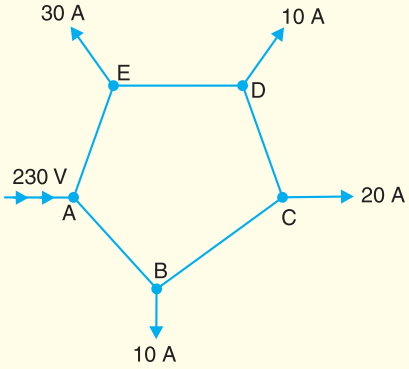


Fig. 13.46

2. A d.c. 2-wire ring main *ABCDEA* is fed from 230 V supply as shown in Fig. 13.46. The resistance of each section (go and return) *AB*, *BC*, *CD*, *DE* and *EA* is 0·1 W. The loads are tapped off as shown. Find the voltage at each load point.

[V_B = 227 V ; V_C = 225 V ; V_D = 225 V ; V_E = 226 V]

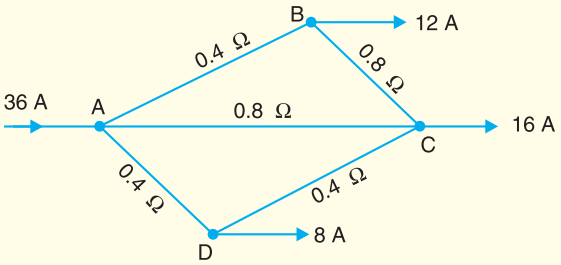


Fig. 13.47

3. In the d.c. network shown in Fig.13.47, A is the feeding point and is maintained at 250 V. The resistances

of the various branches (go and return) are indicated in the figure. Determine the current in each branch. **[*AB* = 144A ; *BC* = 2A ; *DC* = 5A ; *AD* = 13A]**

13.10 3-Wire D.C. System

The great disadvantage of direct current for general power purposes lies in the fact that its voltage cannot readily be changed, except by the use of rotating machinery, which in most cases is too expensive. The problem can be solved to a limited extent by the use of 3-wire d.c. system which makes available two voltages *viz*. V volts between any outer and neutral and 2V volts between the outers. Motor loads requiring high voltage are connected between the outers whereas lighting and heating loads requiring less voltage are connected between any one outer and the neutral. Due to the availability of two voltages, 3-wire system is preferred over 2-wire system for d.c. distribution.

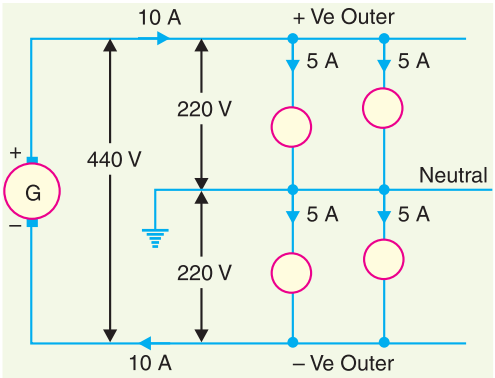


Fig. 13.48

Fig. 13.48 shows the general principles of a 3-wire d.c. system. It consists of two outers and a middle or neutral wire which is earthed at the generator end. The potential of the neutral wire is *half-way between the potentials of the outers. Thus, if p.d. between the outers is 440 V, then positive outer is at 220 V *above* the neutral and negative outer is 220 V *below* the neutral. The current in the neutral wire will depend upon the loads applied to the two sides.

(*i*) If the loads applied on both sides of the neutral are equal (*i.e*. balanced) as shown in Fig 13.48, the current in the neutral wire will be zero. Under these conditions, the potential of the neutral will be exactly half-way between the potential difference of the outers.

(*ii*) If the load on the positive outer (I_1) is greater than on the negative outer (I_2), then out of balance current $I_1 - I_2$ will flow in the neutral wire from load end to supply end as shown in Fig. 13.49 (*i*). Under this condition, the potential of neutral wire will no longer be midway between the potentials of the outers.

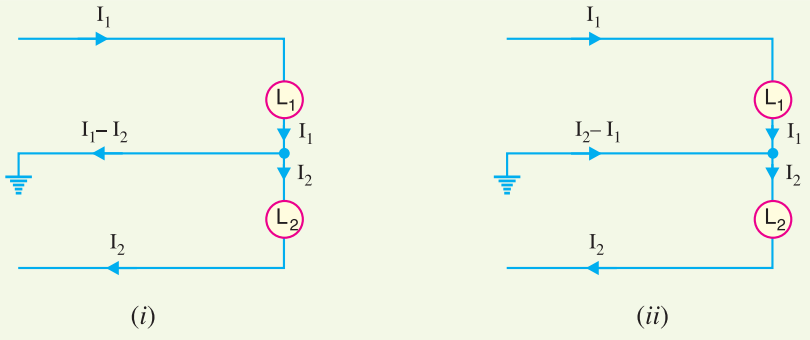


Fig. 13.49

(*iii*) If the load on the negative outer (I_2) is greater than on the positive outer (I_1), then out of balance current $I_2 - I_1$ will flow in the neutral from supply end to load end as shown in Fig. 13.49 (*ii*). Again, the neutral potential will not remain half-way between that of the outers.

(*iv*) As the neutral carries only the out of balance current which is generally small, therefore, area of X-section of neutral is taken half as compared to either of the outers.

It may be noted that it is desirable that voltage between any outer and the neutral should have the same value. This is achieved by distributing the loads equally on both sides of the neutral.

* For balanced loads *i.e*. equal loads on both sides of the neutral wire.

13.11 Current Distribution in 3-Wire D.C. System

Fig. 13.50 shows a 3-wire 500/250 V d.c. distributor. Typical values of loads have been assumed to make the treatment more illustrative. The motor requiring 500 V is connected across the outers and takes a current of 75 A. Other loads requiring lower voltage of 250 V are connected on both sides of the neutral.

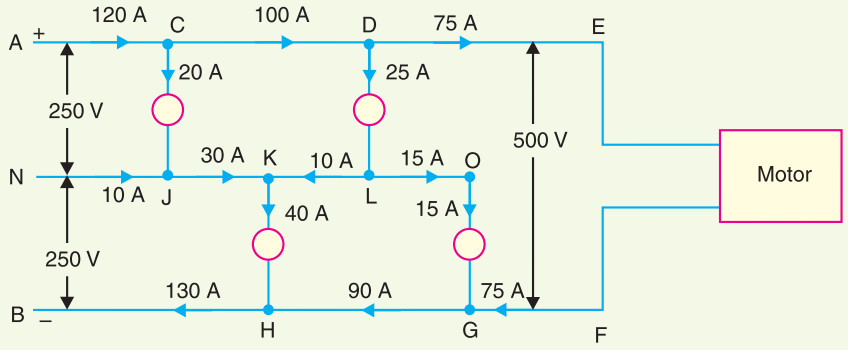


Fig. 13.50

Applying Kirchhoff's current law, it is clear that a current of 120 A enters the positive outer while 130 A comes out of the negative outer. Therefore, 130 – 120 = 10 A must flow *in* the neutral at point *N*. Once the magnitude and direction of current in the section *NJ* is known, the directions and magnitudes of currents in the other sections of the neutral can be easily determined. For instance, the currents meeting at point *K* must add up to 40 A to supply the load *KH*. As seen in Fig. 13.50, 20A of *CJ* and 10A of *NJ* flow towards *K*, the remaining 10A coming from point *L*. The current of 25A of load *DL* is divided into two parts ; 10A flowing along section *LK* and the remaining 15 A along the section *LO* to supply the load *OG*.

Load-point voltages. Knowing the currents in the various sections of the outers and neutral, the voltage at any load point can be determined provided resistances are known. As an illustration, let us calculate the voltage across load *CJ* of Fig.13.50. Applying Kirchhoff's voltage law to the loop *ACJNA*, we have,

[Algebraic sum of voltage drops] + [Algebraic sum of e.m.f.s.] = 0

or *[– drop in *AC* – voltage across *CJ* + drop in *NJ*] + [250] = 0

or Voltage across *CJ* = 250 – drop in *AC* + drop in *NJ*

Example 13.26. *A load supplied on 3-wire d.c. system takes a current of 50 A on the +ve side and 40 A on the negative side. The resistance of each outer wire is 0·1 Ω and the cross-section of middle wire is one-half of that of outer. If the system is supplied at 500/250 V, find the voltage at the load end between each outer and middle wire.*

Solution. Fig. 13.51 shows the current loading. Obviously, current in the neutral wire is 50 – 40 = 10A. As the X-sectional area of neutral is half that of outer, therefore, its resistance = 2 × 0·1 = 0·2Ω.

Voltage at the load end on the +ve side,

$$V_{EL} = 250 - I_1 R_{AE} - (I_1 - I_2) R_{NL}$$
$$= 250 - 50 \times 0{\cdot}1 - (10) \times 0{\cdot}2 = \mathbf{243\ V}$$

Voltage at the load end on the –ve side,

---

* Remember, rise in potential should be considered positive while fall in potential should be considered negative. In section *AC*, current flows from *A* to *C* and hence there is fall in potential. In section *JN*, obviously, there is rise in potential.

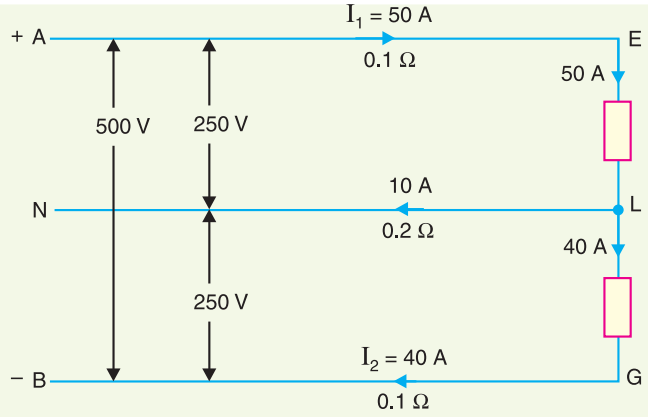


Fig. 13.51

$$V_{LG} = 250 + (I_1 - I_2)\, R_{NL} - I_2\, R_{BG}$$
$$= 250 + 10 \times 0{\cdot}2 - 40 \times 0{\cdot}1 = \mathbf{248\ V}$$

Example 13.27. *A 3-wire d.c. distribution system supplies a load of 5 Ω resistance across the +ve outer and neutral and a load of 6 Ω resistance across –ve outer and neutral at the far end of the distributor. The resistance of each conductor is 0·1 Ω. If the voltage between any outer and neutral at the load end is to be kept at 240 V, find the feeding end voltages.*

Solution. Fig. 13.52 shows the 3-wire distribution system.

Current on +ve outer, $I_1 = 240/5 = 48\text{A}$

Current on –ve outer, $I_2 = 240/6 = 40\text{A}$

Current in neutral $= I_1 - I_2 = 48 - 40 = 8\text{A}$

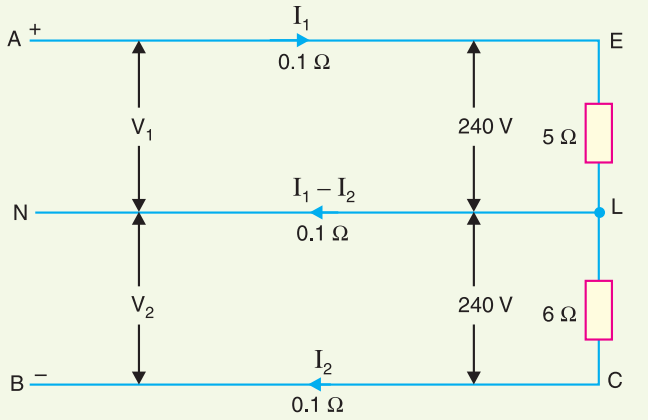


Fig. 13.52

Voltage between +ve outer and neutral at feeding end is

$$V_1 = V_{EL} + I_1\, R_{AE} + (I_1 - I_2)\, R_{NL}$$
$$= 240 + 48 \times 0{\cdot}1 + 8 \times 0{\cdot}1 = \mathbf{245{\cdot}6\ V}$$

Voltage between –ve outer and neutral at feeding end is

$$V_2 = V_{LC} - (I_1 - I_2)\, R_{NL} + I_2\, R_{BC}$$
$$= 240 - 8 \times 0{\cdot}1 + 40 \times 0{\cdot}1 = \mathbf{243{\cdot}2\ V}$$

Example 13.28. *A 3-wire 500/250 V d.c. system has a load of 35 kW between the positive lead and the middle wire and a load of 20 kW between the negative lead and the middle wire. If there is a break in the middle wire, calculate the voltage between the outers and the middle wire.*

Solution. Fig. 13.53 (*i*) shows the arrangement before the disconnection of the middle wire. The voltages on the two sides of the middle wire are equal *i.e.* 250 V. Let R_1 be the resistance of load on the +ve side and R_2 be the resistance of load on the negative side.

$$R_1 = \frac{(250)^2}{35 \times 10^3} = 1{\cdot}785\ \Omega \quad ; \quad R_2 = \frac{(250)^2}{20 \times 10^3} = 3{\cdot}125\ \Omega \qquad [\because R = V^2/P]$$

When there is a break in the middle wire, two resistances R_1 and R_2 are put in series across 500 V as shown in Fig. 13.53 (*ii*).

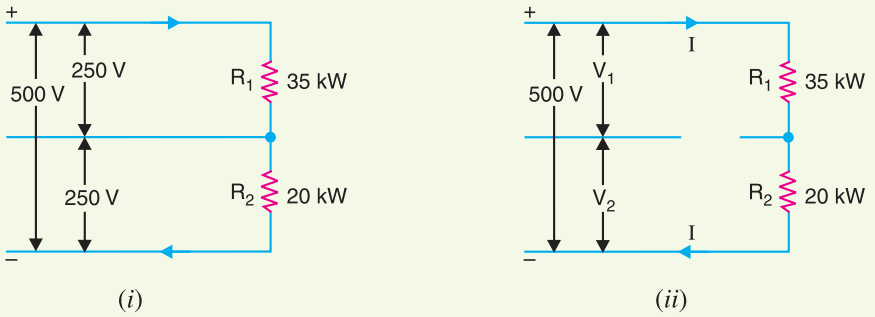


Fig. 13.53

$\therefore$ Circuit current, $I = \dfrac{500}{R_1 + R_2} = \dfrac{500}{1{\cdot}785 + 3{\cdot}125} = 101{\cdot}83$ A

$\therefore$ Voltage across +ve outer and middle wire,

$$V_1 = I R_1 = 101{\cdot}83 \times 1{\cdot}785 = \mathbf{181{\cdot}8\ V}$$

Voltage across –ve outer and middle wire,

$$V_2 = I R_2 = 101{\cdot}83 \times 3{\cdot}125 = \mathbf{318{\cdot}2\ V}$$

Example 13.29. *A 3-wire, 500/250 V distributor is loaded as shown in Fig. 13.54. The resistance of each section is given in ohm. Find the voltage across each load point.*

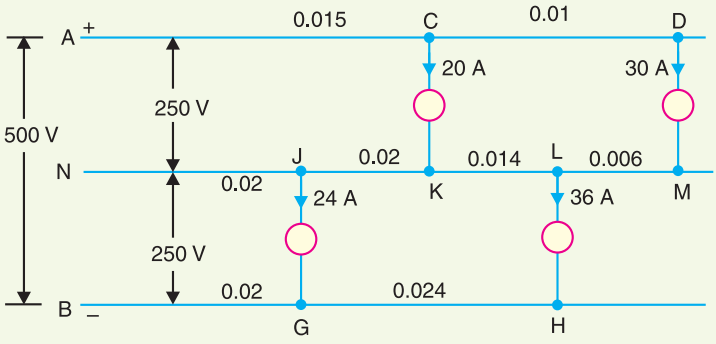


Fig. 13.54

Solution. From the current loading given in Fig. 13.54, we can find the magnitudes and directions of currents in the various sections by applying Kirchhoff's current law. Fig. 13.55 shows the magnitudes and directions of currents in the various sections.

The voltage drops in the various sections are worked out below.

Section	*Resistance* (Ω)	*Current* (*A*)	*Drop* (*V*)
AC	0·015	50	0·75
CD	0·01	30	0·3
ML	0·006	30	0·18
KL	0·014	6	0·084
KJ	0·02	14	0·28
NJ	0·02	10	0·2
HG	0·024	36	0·864
GB	0·02	60	1·2

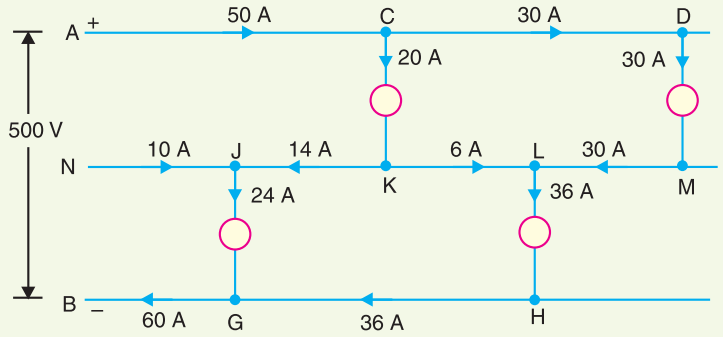


Fig. 13.55

Voltage across load *CK* = 250 – Drop in *AC* – Drop in *KJ* + Drop in *NJ*
= 250 – 0·75 – 0·28 + 0·2 = **249·17 V**

Voltage across load *DM* = 249·17 – Drop in *CD* – Drop in *ML* + Drop in *KL*
= 249·17 – 0·3 – 0·18 + 0·084 = **248·774 V**

Voltage across load *JG* = 250 – Drop in *NJ* – Drop in *GB*
= 250 – 0·2 – 1·2 = **248·6 V**

Voltage across load *LH* = 248·6 + Drop in *KJ* – Drop in *KL* – Drop in *HG*
= 248·6 + 0·28 – 0·084 – 0·864 = **247·932 V**

Example 13.30. *A 3-wire d.c. distributor AE 600 m long is supplied at end A at 500/250 V and is loaded as under :*

Positive side : 60A, 200 m from A ; 40 A, 360 m from A

Negative side : 20A, 100 m from B ; 60A, 260 m from B and 15A, 600 m from B

The resistance of each outer is 0·02 Ω per 100 metres and the cross-section of the neutral wire is the same as that of the outer. Find the voltage across each load point.

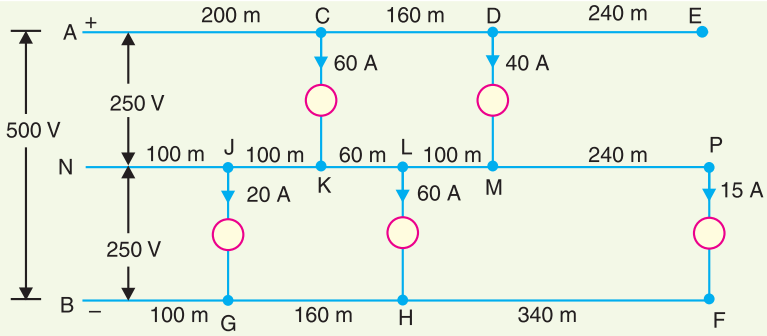


Fig. 13.56

Solution. From the current loading given in Fig. 13.56, we can find the magnitudes and directions of currents in the various sections by Kirchhoff's current law as shown in Fig. 13.57. As the neutral is of same cross-section, its resistance is 0·02 Ω per 100 metres. The voltage drops in the various sections are worked out below :

Section	*Resistance* (Ω)	*Current* (*A*)	*Drop* (*V*)
AC	0·02 × 200/100 = 0·04	100	4
CD	0·02 × 160/100 = 0·032	40	1·28
MP	0·02 × 240/100 = 0·048	15	0·72
ML	0·02 × 100/100 = 0·02	25	0·5
KL	0·02 × 60/100 = 0·012	35	0·42
KJ	0·02 × 100/100 = 0·02	25	0·5
JN	0·02 × 100/100 = 0·02	5	0·1
FH	0·02 × 340/100 = 0·068	15	1·02
HG	0·02 × 160/100 = 0·032	75	2·4
GB	0·02 × 100/100 = 0·02	95	1·9

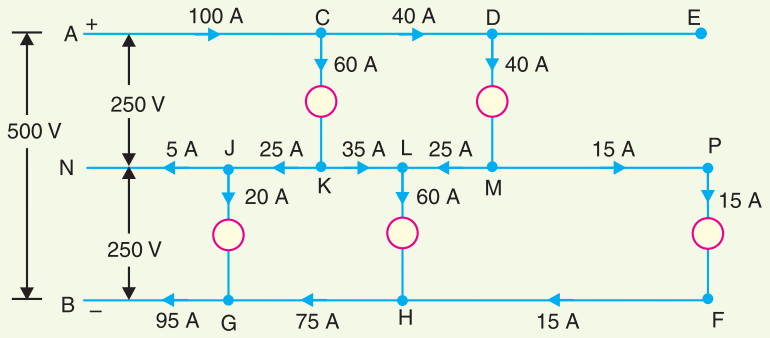


Fig. 13.57

Voltage across *CK* = 250 – Drop in *AC* – Drop in *KJ* – Drop in *JN*
= 250 – 4 – 0·5 – 0·1 = **245·4 V**

Voltage across *DM* = 245·4 – Drop in *CD* – Drop in *ML* + Drop in *KL*
= 245·4 – 1·28 – 0·5 + 0·42 = **244·04 V**

Voltage across *JG* = 250 + Drop in *JN* – Drop in *GB*
= 250 + 0·1 – 1·9 = **248·2 V**

Voltage across *LH* = 248·2 + Drop in *KJ* – Drop in *KL* – Drop in *HG*
= 248·2 + 0·5 – 0·42 – 2·4 = **245·88 V**

Voltage across *PF* = 245·88 + Drop in *ML* – Drop in *MP* – Drop in *FH*
= 245·88 + 0·5 – 0·72 – 1·02 = **244·64 V**

Example 13.31. *The 3-wire d.c. system supplies a load of 4 Ω resistance across +ve wire and the neutral wire and a load of 6 Ω resistance across –ve outer and the neutral at the far end of the distributor. The resistance of each conductor is 0.15 Ω and voltage across each outer and neutral is 240 V at the load end. Determine the load current and load voltages when there is a break in the* (*i*) *neutral wire* (*ii*) *positive outer* (*iii*) *negative outer. Assume that the load resistances and the feeding end voltages remain the same.*

Solution. Fig. 13.58 shows the conditions of the problem when the system is healthy. Let us find the feeding end voltages.

Current in the positive outer, $I_1 = 240/4 = 60$ A

Current in the negative outer, $I_2 = 240/6 = 40$ A

Current in the neutral wire $= I_1 - I_2 = 60 - 40 = 20$ A

Voltage between +ve outer and neutral at feeding point is

$$V_1 = V_{EL} + I_1 R_{AE} + (I_1 - I_2) R_{NL}$$
$$= 240 + 60 \times 0{\cdot}15 + 20 \times 0{\cdot}15 = 252 \text{ V}$$

Voltage between –ve outer and neutral at feeding point is

$$V_2 = V_{LC} - (I_1 - I_2) R_{NL} + I_2 R_{BC}$$
$$= 240 - 20 \times 0{\cdot}15 + 40 \times 0{\cdot}15 = 243 \text{ V}$$

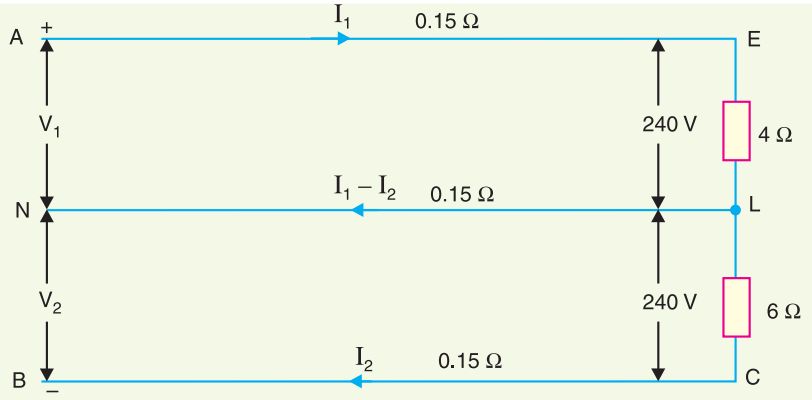


Fig. 13.58

(*i*) When neutral breaks. When there is a break in the neutral, the system is equivalent to 2-wire d.c. system having load resistance = 4 + 6 = 10 Ω and p.d. = 252 + 243 = 495 V at the feeding end. If I is the load current, then,

Total circuit resistance $= 10 + 0{\cdot}15 + 0{\cdot}15 = 10{\cdot}3\Omega$

$\therefore$ Load current, $I = 495/10{\cdot}3 =$ **48·06 A**

Voltage across 4 Ω resistance $= I \times 4 = 48{\cdot}06 \times 4 =$ **192·24 V**

Voltage across 6 Ω resistance $= I \times 6 = 48{\cdot}06 \times 6 =$ **288·36 V**

(*ii*) When +ve outer breaks. When there is a break in the +ve outer, there will be no current in 4 Ω load. The circuit is again 2-wire d.c. system but now load is 6 Ω and p.d. at the feeding point is 243 V.

Total circuit resistance $= 6 + 0{\cdot}15 + 0{\cdot}15 = 6{\cdot}3\Omega$

If I' is the load current, then,

$I' = 243/6{\cdot}3 =$ **38·57 A**

Voltage across 6 Ω $= I' \times 6 = 38{\cdot}57 \times 6 =$ **231·42 V**

(*iii*) When –ve outer breaks. When there is a break in the negative outer, there will be no current in 6 Ω load. The circuit is again 2-wire d.c. system but now load is 4 Ω and p.d. at the feeding point is 252 V.

Total circuit resistance $= 4 + 0{\cdot}15 + 0{\cdot}15 = 4{\cdot}3\Omega$

If I'' is the load current, then,

$I'' = 252/4{\cdot}3 =$ **58·6 A**

Voltage across 4 Ω $= I'' \times 4 = 58{\cdot}6 \times 4 =$ **234·42 V**

TUTORIAL PROBLEMS

1. A load supplied by a 3-wire d.c. distribution system takes a current of 600 A on the +ve side and 480 A on the –ve side. The resistance of the outer conductor is 0·015 Ω and that of the middle wire is 0·03 Ω. Determine the voltage at the load end between each outer and middle wire. Supply voltage is 500/250 V.
[+ve side : 237·4 V ; –ve side : 246·4 V]

2. A 3-wire d.c. distributor, 250 m long, is supplied at end *P* at 500/250 V and is loaded as under :
Positive side : 20 A, 150 m from *P* ; 30 A, 250 m from *P*
Negative side : 24 A, 100 m from *P* ; 36 A, 220 m from *P*
The resistance of each outer wire is 0·02 Ω per 100 m and the cross-section of the middle wire is one half that of the outer. Find the voltage across each load point.
[Positive side : 248·62 V ; 247·83 V ; Negative side : 248·4 V ; 247·65 V]

3. The 3-wire d.c. system supplies a load of 4 Ω resistance across the +ve wire and the neutral and a load of 6 Ω resistance across –ve outer and the neutral at the far end of the distributor. The resistance of each conductor is 0·15 Ω and voltage across each outer and neutral is 240 V at the load end. Calculate the feeding end voltage. **[+ve side : 252 V ; –ve side : 243 V]**

13.12 Balancers in 3-Wire D.C. System

Although in a 3-wire d.c. system every effort is made to distribute the various loads equally on both sides of the neutral, yet it is difficult to achieve the exact balance. The result is that some current does flow in the neutral wire and consequently the voltages on the two sides of the neutral do not remain equal to each other. In order to maintain voltages on the two sides of the neutral equal to each other, a *balancer set* is used.

Circuit details. Fig. 13.59 shows the use of a balancer set in a 3-wire d.c. system. The balancer consists of two identical shunt wound machines *A* and *B* coupled mechanically and having their armature and field circuits connected in series across the outers. The neutral wire is connected to the junction of the armatures as shown. The circuit arrangement has two obvious advantages. Firstly, only one generator (*G*) is required which results in a great saving in cost. Secondly, the balancer set tends to equalise the voltages on the two sides of the neutral.

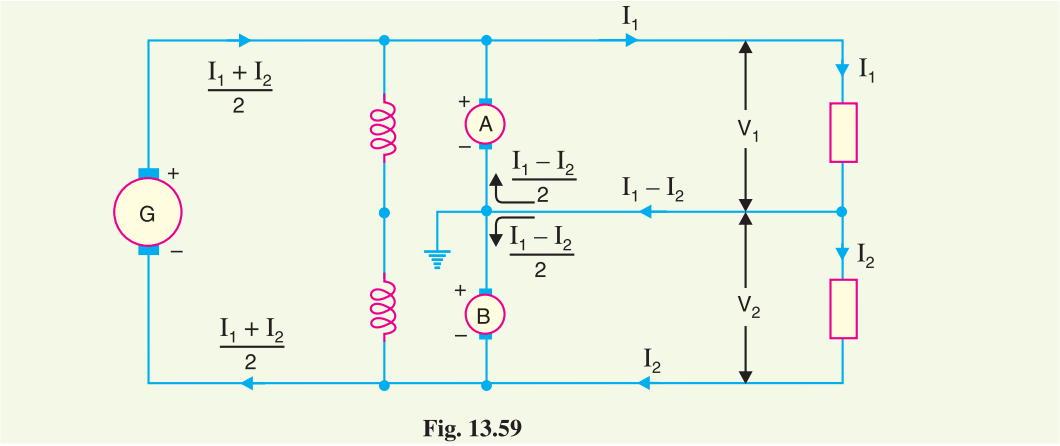


Fig. 13.59

Theory. Since the speeds and field currents of the two machines are equal, their back e.m.f.s have the same value. When the system is unloaded or when the loads on the two sides are the same (*i.e.* balanced), no current flows in the neutral wire. Hence, the two machines run as unloaded motors

When the load is unbalanced, the current supplied by the +ve outer will be different from that supplied by the negative outer. Suppose that load I_1 on the +ve outer is greater that the load I_2 on the –ve outer. Since the +ve side is more heavily loaded, p.d. on this side tends to fall below the e.m.f. of the balancer set. Therefore, machine A runs as a generator. On the other hand, p.d. on the lightly

loaded –ve side rises above the e.m.f. of the balancer so that machine *B* runs as a motor. The result is that energy is transferred from lightly loaded side to the heavily loaded side, preventing the voltage across heavily loaded side form dropping very much below the normal value.

For the condition shown in Fig. 13.59, the machine *B* acts as a motor and machine *A* as a generator. The out of balance current $I_1 - I_2$ flows through the middle wire towards the balancers. Assuming the internal losses of the two machines to be negligible, then armature currents will be equal. Hence, one-half of the current in neutral *i.e.* $(I_1 - I_2)/2$ will flow through each machine as shown.

Let R_a = armature resistance of each machine

$*E$ = induced e.m.f. of each machine

V_1 = terminal p.d. of machine *A* running as a generator

V_2 = terminal p.d. of machine *B* running as a motor

$$\therefore \quad V_1 = E - \left(\frac{I_1 - I_2}{2}\right) R_a$$

and $$V_2 = E + \left(\frac{I_1 - I_2}{2}\right) R_a$$

Difference of voltages between two sides

$$= V_2 - V_1$$

$$= \left[E + \left(\frac{I_1 - I_2}{2}\right) R_a\right] - \left[E - \left(\frac{I_1 - I_2}{2}\right) R_a\right]$$

$$= (I_1 - I_2)\, R_a$$

It is clear that difference of voltages between the two sides of the system is proportional to

(*i*) the out-of-balance current $I_1 - I_2$

(*ii*) the armature resistance of balancer

Therefore, in order to keep the voltages on the two sides equal, R_a is kept small and loads are arranged on the two sides in such a way that out of balance current is as small as possible.

The difference of voltages $(V_2 - V_1)$ on the two sides can be further reduced by cross-connecting the shunt fields of the balancer set as shown in Fig. 13.60. As the generating machine *A* draws its excitation from lightly loaded side which is at a higher voltage, therefore, induced e.m.f. of the machine is increased. On the other hand, induced e.m.f. of machine *B* is decreased since it draws its excitation from the heavily loaded side. The result is that the difference $V_2 - V_1$ is decreased considerably. It may be noted that a perfect balance cannot be obtained because the operation of the balancer set depends upon a slight unbalancing of the voltages on the two sides.

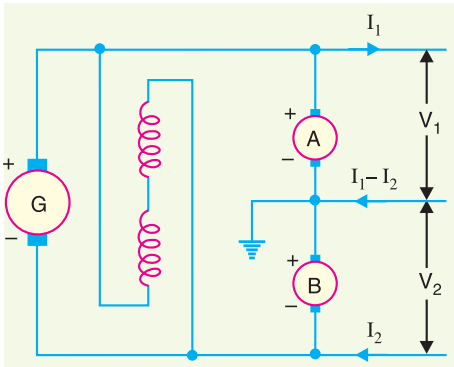


Fig. 13.60

Example 13.32. *A d.c. 3-wire system with 500 V between the outers supplies 1500 kW on the +ve outer and 2000 kW on the negative outer. If the losses in the machines are negligible, calculate:*

(*i*) *current in the neutral wire*

(*ii*) *total current supplied by main generator*

(*iii*) *current in each balancer armature*

Solution. The connections are shown in Fig. 13.61. As the negative side is more heavily loaded, therefore, machine *B* acts as a generator and machine *A* as a motor.

* Since both machines have the same excitation and run at the same speed, their induced e.m.f.s will be same.

Load current on +ve outer, $I_1 = 1500 \times 10^3/250 = 6000$ A

Load current on –ve outer, $I_2 = 2000 \times 10^3/250 = 8000$ A

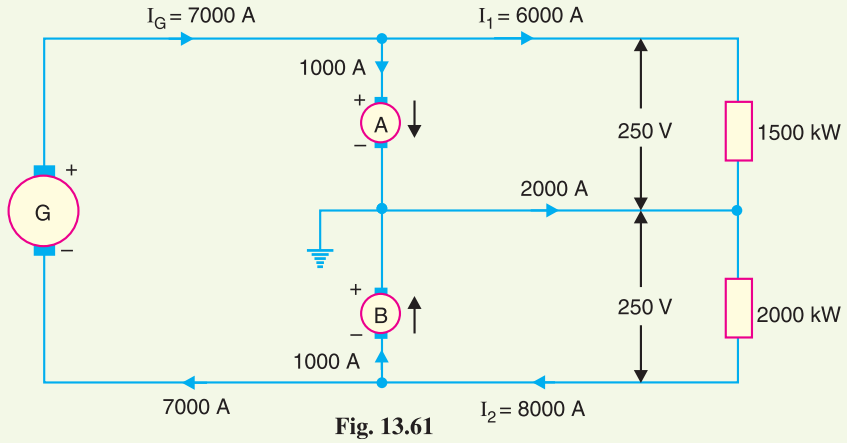


Fig. 13.61

(i) Current in the neutral $= I_2 - I_1 = 8000 - 6000 =$ **2000 A**

(ii) Total load on main generator $= 1500 + 2000 = 3500$ kW

$\therefore$ Current supplied by main generator, $I_G = 3500 \times 10^3/500 =$ **7000 A**

(iii) Current in machine *A* $= I_G - I_1 = 7000 - 6000 =$ **1000 A**

Current in machine *B* $= I_2 - I_G = 8000 - 7000 =$ **1000 A**

Example 13.33. *A d.c. 3-wire system with 500 V between outers has lighting loads of 150 kW on the positive side and 100 kW on the negative side. The loss in each balancer machine is 3 kW. Calculate* :

(*i*) *total load on the main generator*

(*ii*) *kW loading of each balancer machine*

Solution. The connections are shown in Fig. 13.62. As the positive side is more heavily loaded, therefore, machine *A* acts as a generator and machine *B* as a motor.

(i) Total load on the main generator

= load on +ve side + load on –ve side + losses

= $150 + 100 + 2 \times 3 =$ **256 kW**

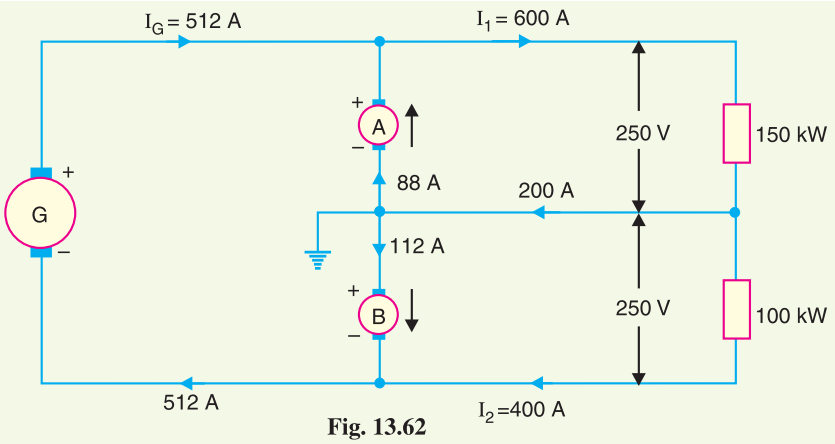


Fig. 13.62

(ii) Current supplied by the main generator,

$I_G = 256 \times 10^3/500 = 512$ A

Load current on +ve side, $I_1 = 150 \times 10^3/250 = 600$ A

Load current on –ve side, $I_2 = 100 \times 10^3/250 = 400$ A

Current in neutral wire $= I_1 - I_2 = 600 - 400 = 200$ A

Current through machine *A* $= I_1 - I_G = 600 - 512 = 88$ A

Current through machine *B* $= I_G - I_2 = 512 - 400 = 112$ A

∴ Load on machine *A* $= 88 \times 250/1000 =$ **22 kW**

Load on machine *B* $= 112 \times 250/1000 =$ **28 kW**

Example 13.34. *In a 500/250 V d.c. 3-wire system, there is a current of 1200 A on the +ve side and 1000 A on the –ve side and a motor load of 200 kW across the outers. The loss in each balancer machine is 5 kW. Calculate :*

(*i*) *current of the main generator*

(*ii*) *load on each balancer machine*

Solution. The connections are shown in Fig. 13.63. As the positive side is more heavily loaded, therefore, machine *A* acts as a generator and machine *B* as a motor.

Load on +ve side, $P_1 = 250 \times 1200/1000 = 300$ kW

Load on –ve side, $P_2 = 250 \times 1000/1000 = 250$ kW

Load on outers, $P_3 = 200$ kW

(i) Total load on the main generator $= P_1 + P_2 + P_3$ + loss in balancer machines

$= 300 + 250 + 200 + 10 = 760$ kW

Current of main generator, $I_G = 760 \times 10^3/500 =$ **1520 A**

(ii) Current in neutral $= 1200 - 1000 = 200$ A

Current through machine *A* $= 1600 - 1520 = 80$ A

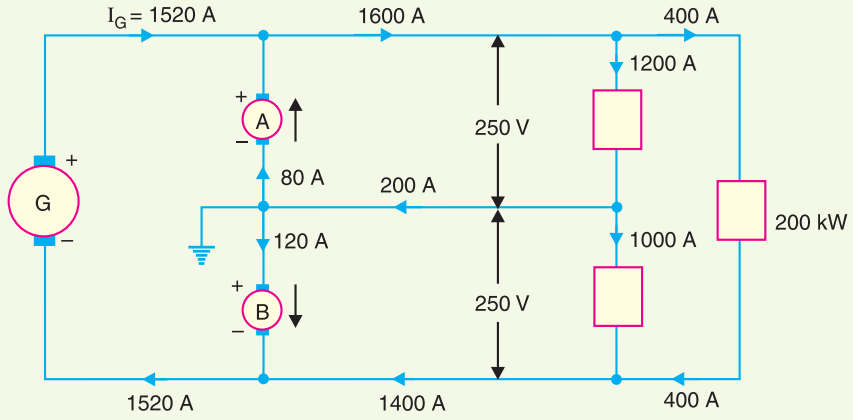


Fig. 13.63

Current thro' machine *B* $= 1520 - 1400 = 120$ A

Load on machine *A* $= 80 \times 250/1000 =$ **20 kW**

Load on machine *B* $= 120 \times 250/1000 =$ **30 kW**

Example 13.35. *A d.c. 3-wire system with 500 volts across outers supplies 800 A on the positive side and 550 A on the negative side and 1500 A across the outers. The rotary balancer has each an*

armature resistance of 0·2 Ω and takes 5 A on no load. Find :

(*i*) *current loading of each balancer machine*

(*ii*) *the voltage across each balancer machine*

(*iii*) *total load on the main generator*

Solution. The connections are shown in Fig. 13.64. As the positive side is more heavily loaded, therefore, machine *A* acts as a generator and machine *B* as a motor.

Total current on +ve side = 800 + 1500 = 2300 A

Total current on –ve side = 550 + 1500 = 2050 A

Current in neutral wire = 800 – 550 = 250 A

Let the current through machines *A* and *B* be I_A and I_B respectively. Then $I_A + I_B$ must be equal to current in the neutral wire *i.e.*

$$I_A + I_B = 250$$

or $$I_A = 250 - I_B$$

(*i*) Let V_A and V_B be the p.d.s. of machines *A* and *B* respectively. Since machine *B* is driving the machine *A*, output of *B* supplies the losses in the set plus the output of machine *A* *i.e.*

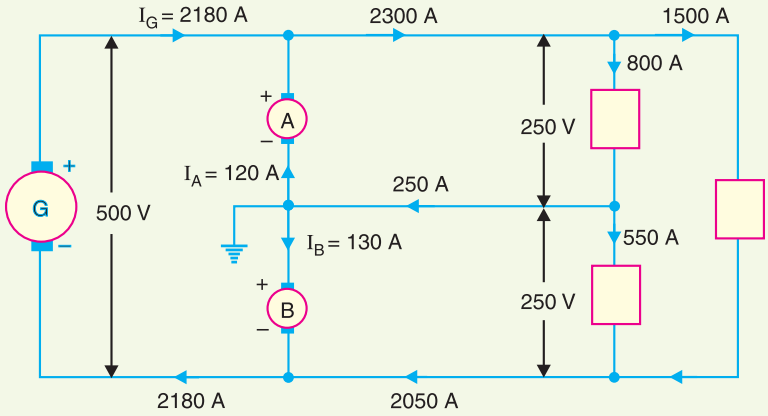


Fig. 13.64

Output of machine *B* = Output of machine A + *Losses in the set

or $$V_B I_B = V_A I_A + 500 \times 5 + I^2_A R_A + I^2_B R_B$$

$\therefore$ $$V_B I_B = V_A (250 - I_B) + 2500 + (250 - I_B)^2 \times 0{\cdot}2 + 0{\cdot}2 I^2_B \quad ...(i)$$

Each machine has same value of back e.m.f. *E* since their field currents and speeds are the same.

Back e.m.f., $$E = 250 - 0{\cdot}2 \times 5 = 249 \text{ V}$$

Terminal p.d. across *A*, $$V_A = E - I_A R_A = 249 - 0{\cdot}2\,(250 - I_B)$$

$\therefore$ $$V_A = 199 + 0{\cdot}2\, I_B \quad ...(ii)$$

Terminal p.d. across *B*, $$V_B = E + I_B R_B$$

$$= 249 + 0{\cdot}2\, I_B \quad ...(iii)$$

Substituting the values of V_A and V_B in exp. (*i*), we get,

$$(249 + 0{\cdot}2\, I_B)\, I_B = (199 + 0{\cdot}2\, I_B)(250 - I_B) + 2500 + (250 - I_B)^2 \times 0{\cdot}2 + 0{\cdot}2 I^2_B$$

or $$249\, I_B + 0{\cdot}2\, I_B^2 = 49{,}750 - 199\, I_B + 50\, I_B - 0{\cdot}2\, I^2_B + 2500 + 12{,}500 + 0{\cdot}2\, I^2_B - 100\, I_B + 0{\cdot}2\, I^2_B$$

* Losses in the set = No load losses + Copper losses

$= 500 \times 5 + I^2_A R_A + I^2_B R_B$

or $498\ I_B = 64{,}750$

$\therefore\ I_B = 64750/498 = \mathbf{130\ A}$

and $I_A = 250 - 130 = \mathbf{120\ A}$

(ii) Voltage across machine *A*, $V_A = 199 + 0{\cdot}2\ I_B = 199 + 0{\cdot}2 \times 130 = \mathbf{225\ V}$

Voltage across machine *B*, $V_B = 249 + 0{\cdot}2\ I_B = 249 + 0{\cdot}2 \times 130 = \mathbf{275\ V}$

(iii) Load on main generator $= 2300 - I_A = 2300 - 120 = \mathbf{2180\ A}$

TUTORIAL PROBLEMS

1. A load on 3-wire d.c. system with 500 V between the outers consists of 1500 A on the positive side and 1300A on the negative side while motors connected across outers absorb 500 kW. Assuming each balancer machine has a loss of 3·75 kW, calculate the load on the main generator and each balancer machine. **[1207·5 kW ; 21·25 kW ; 28·75 kW]**
2. In a 500/250 V d.c. 3-wire system, there is a current of 2000 A on the +ve side, 1600A on the negative side and a load of 300 kW across the outers. The loss in each balancer set is 8 kW. Calculate the current in each armature of the balancer set and total load on the main generator. **[168A ; 232A ; 1216 kW]**
3. In a 500/250 volt 3-wire d.c. system, there is an out of balance load of 200 kW on the positive side. The loss in each balancer set is 10 kW and the current in the negative main is 2800A. Calculate the current in each armature of the balancer set and the total load on the generator. **[440A ; 360A ; 1620 kW]**

13.13 Boosters

A **booster** *is a d.c. generator whose function is to inject or add certain voltage into a circuit so as to compensate the IR drop in the feeders etc.*

A booster is essentially a series d.c. generator of large current capacity and is connected in series with the feeder whose voltage drop is to be compensated as shown in Fig. 13.65. It is driven at constant speed by a shunt motor working from the bus-bars. As the booster is a series generator, therefore, voltage generated by it is directly proportional to the field current which is here the feeder current. When the feeder current increases, the voltage drop in the feeder also increases. But increased feeder current results in greater field excitation of booster which injects higher voltage into the feeder to compensate the voltage drop. For exact compensation of voltage drop, the booster must be marked on the straight or linear portion of its voltage-current characteristics.

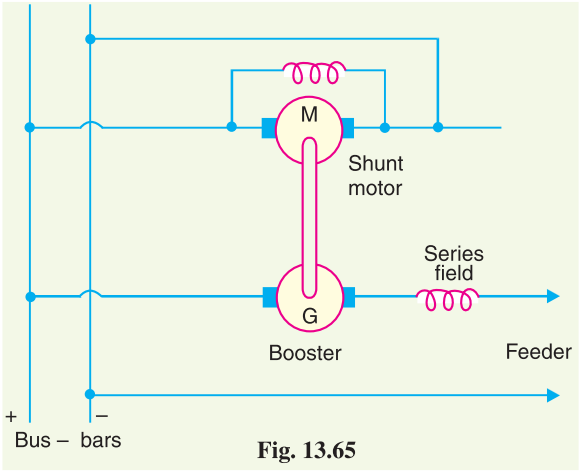


Fig. 13.65

It might be suggested to compensate the voltage drop in the feeder by overcompounding the generators instead of using a booster. Such a method is not practicable for feeders of different lengths because it will disturb the voltage of other feeders. The advantage of using a booster is that each feeder can be regulated independently — a great advantage if the feeders are of different lengths.

Example 13.36. *A 2-wire system has the voltage at the supply end maintained at 500 V. The line is 3 km long. If the full-load current is 120 A, what must be the booster voltage and output in order that the far end voltage may also be 500 V ? Take the resistance of the cable at the working temperature as 0·5 Ω/km.*

Solution.

Total resistance of line = $0{\cdot}5 \times 3 = 1{\cdot}5\ \Omega$

F.L. voltage drop in the line = $1{\cdot}5 \times 120 = 180$ V

$\therefore$ Terminal voltage of booster = **180 V**

$$\text{Output of booster} = \frac{120 \times 180}{1000}\ \text{kW} = \mathbf{21{\cdot}6\ kW}$$

13.14 Comparison of 3-Wire and 2-Wire D.C. Distribution

It is worthwhile to make a comparison between 3-wire and 2-wire systems for d.c. distribution. It will be shown that there is a great saving of conductor material if we use 3-wire system instead of 2-wire system for d.c. distribution. For comparison, it will be assumed that :

(*i*) the amount of power P transmitted is the same

(*ii*) the *voltage V at the consumer's terminals is the same

(*iii*) the distance of transmission is the same

(*iv*) the efficiency of transmission (and hence losses) is the same

(*v*) the 3-wire system is balanced *i.e.* no current in the neutral wire

(*vi*) the area of X-section of neutral wire is half the cross-section of outers in 3-wire system

Let R_2 = resistance of each conductor in 2-wire system

R_3 = resistance of each outer in 3-wire system

Current through outers in case of 3-wire system is

$I_3 = P/2V$

Total loss in two outers $= 2\,I_3^2\,R_3 = 2\,(P/2V)^2\,R_3$

Current in 2-wire system, $I_2 = P/V$

Total loss $= 2\,I_2^2\,R_2 = 2\,(P/V)^2\,R_2$

Since efficiency of transmission is the same, it means losses are the same *i.e.*

$$2\,(P/2V)^2\,R_3 = 2\,(P/V)^2\,R_2$$

$\therefore$ $R_3 = 4\,R_2$

Therefore, the area of X-section of outers in 3-wire case will be one-fourth of each conductor in 2-wire case.

Let a = area of X-section of each conductor is 2-wire case

Then $a/4$ = area of X-section of each outer in 3-wire case

and $a/8$ = area of X-section of neutral wire [assumption (*vi*) above]

If l is the length of the line, then,

$$\text{Volume of Cu for 3-wire system} = l\left(\frac{a}{4}+\frac{a}{4}+\frac{a}{8}\right) = \frac{5}{8}\,a\,l$$

$$\text{Volume of Cu for 2-wire system} = l\,(a+a) = 2\,a\,l$$

$$\therefore \quad \frac{\text{Volume of Cu for 3-wire system}}{\text{Volume of Cu for 2-wire system}} = \frac{5}{8}\,a\,l \times \frac{1}{2\,a\,l} = \frac{5}{16}$$

Hence a 3-wire system requires only 5/16 th (or 31·25%) as much copper as a 2-wire system.

Note. If the neutral has the same X-section as the outer, then,

$$\text{Volume of Cu for 3-wire system} = l\left(\frac{a}{4}+\frac{a}{4}+\frac{a}{4}\right) = \frac{3}{4}\,a\,l$$

$$\text{Volume of Cu for 2-wire system} = l\,(a+a) = 2\,a\,l$$

* Note that in case of 3-wire system, the voltage between the outers will be 2V.

$$\therefore \quad \frac{\text{Volume of Cu for 3-wire system}}{\text{Volume of Cu for 2-wire system}} = \frac{3}{4}\,a\,l \times \frac{1}{2\,a\,l} = \frac{3}{8} \text{ or } 37{\cdot}5\%$$

13.15 Ground Detectors

Ground detectors are the devices that are used to detect/indicate the ground fault for ungrounded d.c. systems. When a ground fault occurs on such a system, immediate steps should be taken to clear it. If this is not done and a second ground fault happens, a short circuit occurs. Lamps are generally used for the detection of ground faults. They are connected for ungrounded 2-wire system as shown in Fig. 13.66. Each lamp should have a voltage rating equal to the line voltage. The two lamps in series, being subjected to half their rated voltage, will glow dimly. If a ground fault occurs on either wires, the lamp connected to the grounded wire will not glow while the other lamp will glow brightly.

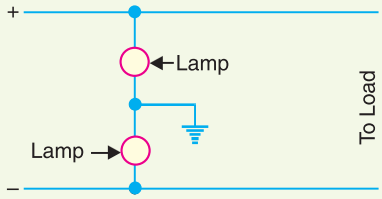


Fig. 13.66

SELF - TEST

1. Fill in the blanks by inserting appropriate words/figures.

(*i*) In a singly fed distributor, if fault occurs on any section, the supply to all consumers has to be .....

(*ii*) A ring main distributor fed at one end is equivalent to ..... fed at both ends with equal voltages.

(*iii*) A distributor is designed from ..... considerations.

(*iv*) The point of minimum potential of a uniformly loaded distributor fed at both ends with equal voltages will occur at .....

(*v*) The d.c. interconnector is used ..... the voltage drops in the various sections of the distributor.

(*vi*) In a 3 wire d.c. system, the load on +ve side is 400A and on negative side it is 300A. Then current in neutral wire is .....

(*vii*) In a balanced 3-wire d.c. system, the potential of neutral is ..... between that of outers.

(*viii*) A booster is used to ..... voltage drop in feeders *etc.*

(*ix*) Balancer set is used to maintain voltage on the two sides of the neutral .....

(*x*) In a balanced 3-wire d.c. system, if voltage across the outers is 500 V, then voltage between any outer and neutral is .....

2. Pick up the correct words/figures from brackets and fill in the blanks.

(*i*) The voltage drop in a doubly fed distributor is ..... than the equivalent singly fed distributor. (*less, more*)

(*ii*) In a 3-wire system, the area of X-section of neutral is generally ..... of either outer. (*half, double*)

(*iii*) If in a 3-wire d.c. system, the current in the neutral wire is zero, then voltage between any outer and neutral is ..... (*the same, different*)

(*iv*) A booster is connected in ..... with the feeder. (*series, parallel*)

(*v*) For exact compensation of voltage drop in the feeder, the booster must work on ..... portion of its V—I characteristic. (*linear, non-linear*)

(*vi*) The balancer machine connected to the heavily loaded side works as a ..... (*generator, motor*)

ANSWERS TO SELF-TEST

1. (*i*) shut off (*ii*) straight distributor (*iii*) voltage drop (*iv*) mid-point (*v*) to reduce (*vi*) 100 A (*vii*) midway (*viii*) compensate (*ix*) equal to each other (*x*) 250 V

2. (*i*) less (*ii*) half (*iii*) the same (*iv*) series (*v*) linear (*vi*) generator

situated near consumers' localities. Each distribution transformer steps down the voltage to 400 V and power is distributed to ultimate consumers' by 400/230 V, 3-phase, 4-wire system. In this chapter, we shall focus our attention on the various aspects of a.c. distribution.

14.1 A.C. Distribution Calculations

A.C. distribution calculations differ from those of d.c. distribution in the following respects :

(*i*) In case of d.c. system, the voltage drop is due to resistance alone. However, in a.c. system, the voltage drops are due to the combined effects of resistance, inductance and capacitance.

(*ii*) In a d.c. system, additions and subtractions of currents or voltages are done arithmetically but in case of a.c. system, these operations are done vectorially.

(*iii*) In an a.c. system, power factor (p.f.) has to be taken into account. Loads tapped off form the distributor are generally at different power factors. There are two ways of referring power factor *viz*

(*a*) It may be referred to supply or receiving end voltage which is regarded as the reference vector.

(*b*) It may be referred to the voltage at the load point itself.

There are several ways of solving a.c. distribution problems. However, symbolic notation method has been found to be most convenient for this purpose. In this method, voltages, currents and impedances are expressed in complex notation and the calculations are made exactly as in d.c. distribution.

14.2 Methods of Solving A.C. Distribution Problems

In a.c. distribution calculations, power factors of various load currents have to be considered since currents in different sections of the distributor will be the vector sum of load currents and not the arithmetic sum. The power factors of load currents may be given (*i*) *w.r.t.* receiving or sending end voltage or (*ii*) *w.r.t.* to load voltage itself. Each case shall be discussed separately.

(*i*) Power factors referred to receiving end voltage. Consider an a.c. distributor *AB* with concentrated loads of I_1 and I_2 tapped off at points *C* and *B* as shown in Fig. 14.1. Taking the receiving end voltage V_B as the reference vector, let lagging power factors at *C* and *B* be cos ϕ_1 and cos ϕ_2 *w.r.t.* V_B. Let R_1, X_1 and R_2, X_2 be the resistance and reactance of sections *AC* and *CB* of the distributor.

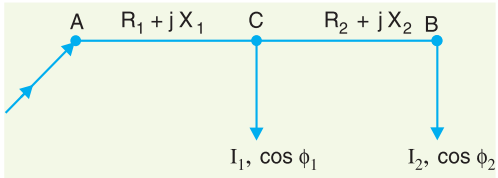


Fig. 14.1

Impedance of section *AC*, $\overrightarrow{Z_{AC}} = R_1 + j\,X_1$

Impedance of section *CB*, $\overrightarrow{Z_{CB}} = R_2 + j\,X_2$

Load current at point *C*, $\vec{I_1} = I_1 (\cos \phi_1 - j \sin \phi_1)$

Load current at point *B*, $\vec{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2)$

Current in section *CB*, $\overrightarrow{I_{CB}} = \vec{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2)$

Current in section *AC*, $\overrightarrow{I_{AC}} = \vec{I_1} + \vec{I_2}$

$= I_1 (\cos \phi_1 - j \sin \phi_1) + I_2 (\cos \phi_2 - j \sin \phi_2)$

Voltage drop in section *CB*, $\overrightarrow{V_{CB}} = \overrightarrow{I_{CB}}\,\overrightarrow{Z_{CB}} = I_2 (\cos \phi_2 - j \sin \phi_2)(R_2 + j\,X_2)$

Voltage drop in section *AC*, $\overrightarrow{V_{AC}} = \overrightarrow{I_{AC}}\,\overrightarrow{Z_{AC}} = \left(\vec{I_1} + \vec{I_2}\right) Z_{AC}$

$= [I_1(\cos \phi_1 - j \sin \phi_1) + I_2 (\cos \phi_2 - j \sin \phi_2)] [R_1 + jX_1]$

Sending end voltage, $\overrightarrow{V_A} = \overrightarrow{V_B} + \overrightarrow{V_{CB}} + \overrightarrow{V_{AC}}$

Sending end current, $\overrightarrow{I_A} = \overrightarrow{I_1} + \overrightarrow{I_2}$

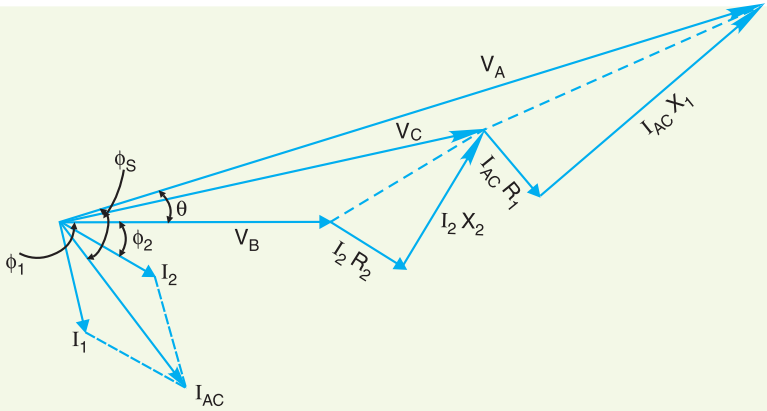


Fig. 14.2

The vector diagram of the a.c. distributor under these conditions is shown in Fig. 14.2. Here, the receiving end voltage V_B is taken as the reference vector. As power factors of loads are given *w.r.t.* V_B, therefore, I_1 and I_2 lag behind V_B by ϕ_1 and ϕ_2 respectively.

***(ii)* Power factors referred to respective load voltages.** Suppose the power factors of loads in the previous Fig. 14.1 are referred to their respective load voltages. Then ϕ_1 is the phase angle between V_C and I_1 and ϕ_2 is the phase angle between V_B and I_2. The vector diagram under these conditions is shown in Fig. 14.3.

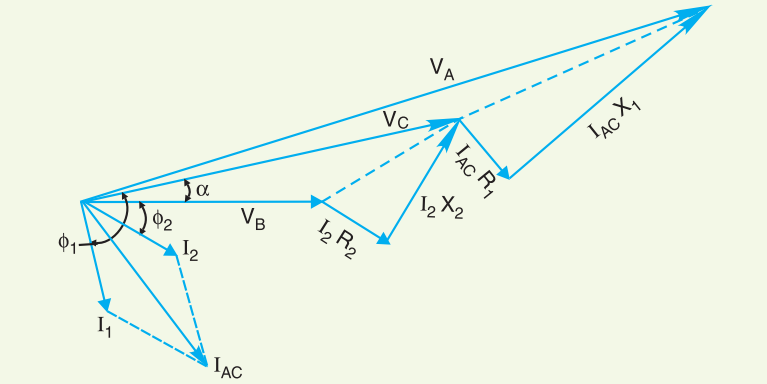


Fig. 14.3

Voltage drop in section $CB = \overrightarrow{I_2}\,\overrightarrow{Z_{CB}} = I_2 (\cos \phi_2 - j \sin \phi_2) (R_2 + j\, X_2)$

Voltage at point $C = \overrightarrow{V_B}$ + Drop in section $CB = V_C \angle \alpha$ (say)

Now $\overrightarrow{I_1} = I_1 \angle -\phi_1$ *w.r.t.* voltage V_C

$\therefore$ $\overrightarrow{I_1} = I_1 \angle -(\phi_1 - \alpha)$ *w.r.t.* voltage V_B

i.e. $\overrightarrow{I_1} = I_1 [\cos (\phi_1 - \alpha) - j \sin (\phi_1 - \alpha)]$

Now $\overrightarrow{I_{AC}} = \overrightarrow{I_1} + \overrightarrow{I_2}$

$= I_1 [\cos(\phi_1 - \alpha) - j \sin(\phi_1 - \alpha)] + I_2 (\cos \phi_2 - j \sin \phi_2)$

Voltage drop in section AC $= \overrightarrow{I_{AC}}\,\overrightarrow{Z_{AC}}$

$\therefore$ Voltage at point A $= V_B$ + Drop in CB + Drop in AC

Example 14.1. *A single phase a.c. distributor AB 300 metres long is fed from end A and is loaded as under :*

(*i*) *100 A at 0·707 p.f. lagging 200 m from point A*

(*ii*) *200 A at 0·8 p.f. lagging 300 m from point A*

The load resistance and reactance of the distributor is 0·2 Ω and 0·1 Ω per kilometre. Calculate the total voltage drop in the distributor. The load power factors refer to the voltage at the far end.

Solution. Fig. 14.4 shows the single line diagram of the distributor.

Impedance of distributor/km $= (0{\cdot}2 + j\,0{\cdot}1)\,\Omega$

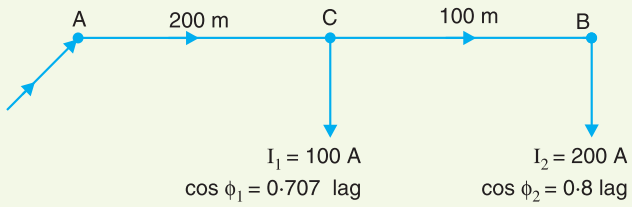


Fig. 14.4

Impedance of section AC, $\overrightarrow{Z_{AC}} = (0{\cdot}2 + j\,0{\cdot}1) \times 200/1000 = (0{\cdot}04 + j\,0{\cdot}02)\,\Omega$

Impedance of section CB, $\overrightarrow{Z_{CB}} = (0{\cdot}2 + j\,0{\cdot}1) \times 100/1000 = (0{\cdot}02 + j\,0{\cdot}01)\,\Omega$

Taking voltage at the far end B as the reference vector, we have,

Load current at point B, $\overrightarrow{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2) = 200\,(0{\cdot}8 - j\,0{\cdot}6)$

$= (160 - j\,120)$ A

Load current at point C, $\overrightarrow{I_1} = I_1 (\cos \phi_1 - j \sin \phi_1) = 100\,(0{\cdot}707 - j\,0{\cdot}707)$

$= (70{\cdot}7 - j\,70{\cdot}7)$ A

Current in section CB, $\overrightarrow{I_{CB}} = \overrightarrow{I_2} = (160 - j\,120)$ A

Current in section AC, $\overrightarrow{I_{AC}} = \overrightarrow{I_1} + \overrightarrow{I_2} = (70{\cdot}7 - j\,70{\cdot}7) + (160 - j\,120)$

$= (230{\cdot}7 - j\,190{\cdot}7)$ A

Voltage drop in section CB, $\overrightarrow{V_{CB}} = \overrightarrow{I_{CB}}\,\overrightarrow{Z_{CB}} = (160 - j\,120)\,(0{\cdot}02 + j\,0{\cdot}01)$

$= (4{\cdot}4 - j\,0{\cdot}8)$ volts

Voltage drop in section AC, $\overrightarrow{V_{AC}} = \overrightarrow{I_{AC}}\,\overrightarrow{Z_{AC}} = (230{\cdot}7 - j\,190{\cdot}7)\,(0{\cdot}04 + j\,0{\cdot}02)$

$= (13{\cdot}04 - j\,3{\cdot}01)$ volts

Voltage drop in the distributor $= \overrightarrow{V_{AC}} + \overrightarrow{V_{CB}} = (13{\cdot}04 - j\,3{\cdot}01) + (4{\cdot}4 - j\,0{\cdot}8)$

$= (17{\cdot}44 - j\,3{\cdot}81)$ volts

Magnitude of drop $= \sqrt{(17{\cdot}44)^2 + (3{\cdot}81)^2} =$ **17·85 V**

Example 14.2. *A single phase distributor 2 kilometres long supplies a load of 120 A at 0·8 p.f. lagging at its far end and a load of 80 A at 0·9 p.f. lagging at its mid-point. Both power factors are*

referred to the voltage at the far end. The resistance and reactance per km (go and return) are 0·05 Ω and 0·1 Ω respectively. If the voltage at the far end is maintained at 230 V, calculate :

(i) voltage at the sending end

(ii) phase angle between voltages at the two ends.

Solution. Fig. 14.5 shows the distributor *AB* with *C* as the mid-point

Impedance of distributor/km = $(0{\cdot}05 + j\,0{\cdot}1)\,\Omega$

Impedance of section *AC*, $\vec{Z}_{AC} = (0{\cdot}05 + j\,0{\cdot}1) \times 1000/1000 = (0{\cdot}05 + j\,0{\cdot}1)\,\Omega$

Impedance of section *CB*, $\vec{Z}_{CB} = (0{\cdot}05 + j\,0{\cdot}1) \times 1000/1000 = (0{\cdot}05 + j\,0{\cdot}1)\,\Omega$

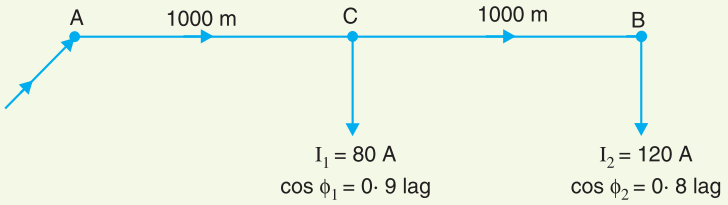


Fig. 14.5

Let the voltage V_B at point *B* be taken as the reference vector.

Then, $\overrightarrow{V_B} = 230 + j\,0$

(i) Load current at point *B*, $\overrightarrow{I_2} = 120\,(0{\cdot}8 - j\,0{\cdot}6) = 96 - j\,72$

Load current at point *C*, $\overrightarrow{I_1} = 80\,(0{\cdot}9 - j\,0{\cdot}436) = 72 - j\,34{\cdot}88$

Current in section *CB*, $\overrightarrow{I_{CB}} = \overrightarrow{I_2} = 96 - j\,72$

Current in section *AC*, $\overrightarrow{I_{AC}} = \overrightarrow{I_1} + \overrightarrow{I_2} = (72 - j\,34{\cdot}88) + (96 - j\,72)$

$= 168 - j\,106{\cdot}88$

Drop in section *CB*, $\overrightarrow{V_{CB}} = \overrightarrow{I_{CB}}\,\overrightarrow{Z_{CB}} = (96 - j\,72)\,(0{\cdot}05 + j\,0{\cdot}1)$

$= 12 + j\,6$

Drop in section *AC*, $\overrightarrow{V_{AC}} = \overrightarrow{I_{AC}}\,\overrightarrow{Z_{AC}} = (168 - j\,106{\cdot}88)\,(0{\cdot}05 + j\,0{\cdot}1)$

$= 19{\cdot}08 + j\,11{\cdot}45$

$\therefore$ Sending end voltage, $\overrightarrow{V_A} = \overrightarrow{V_B} + \overrightarrow{V_{CB}} + \overrightarrow{V_{AC}}$

$= (230 + j\,0) + (12 + j\,6) + (19.08 + j\,11.45)$

$= 261.08 + j\,17.45$

Its magnitude is $= \sqrt{(261{\cdot}08)^2 + (17{\cdot}45)^2} = \mathbf{261{\cdot}67\ V}$

(ii) The phase difference θ between V_A and V_B is given by :

$$\tan\theta = \frac{17{\cdot}45}{261{\cdot}08} = 0{\cdot}0668$$

$\therefore$ $\theta = \tan^{-1} 0{\cdot}0668 = \mathbf{3{\cdot}82^o}$

Example 14.3. *A single phase distributor one km long has resistance and reactance per conductor of 0·1 Ω and 0·15 Ω respectively. At the far end, the voltage V_B = 200 V and the current is 100 A at a p.f. of 0·8 lagging. At the mid-point M of the distributor, a current of 100 A is tapped at a p.f.*

of 0·6 lagging with reference to the voltage V_M at the mid-point. Calculate :

(*i*) *voltage at mid-point*

(*ii*) *sending end voltage* V_A

(*iii*) *phase angle between* V_A *and* V_B

Solution. Fig. 14.6 shows the single line diagram of the distributor *AB* with *M* as the mid-point.

Total impedance of distributor $= 2(0{\cdot}1 + j\,0{\cdot}15) = (0{\cdot}2 + j\,0{\cdot}3)\,\Omega$

Impedance of section *AM*, $\overrightarrow{Z_{AM}} = (0{\cdot}1 + j\,0{\cdot}15)\,\Omega$

Impedance of section *MB*, $\overrightarrow{Z_{MB}} = (0{\cdot}1 + j\,0{\cdot}15)\,\Omega$

Let the voltage V_B at point *B* be taken as the reference vector.

Then, $\overrightarrow{V_B} = 200 + j\,0$

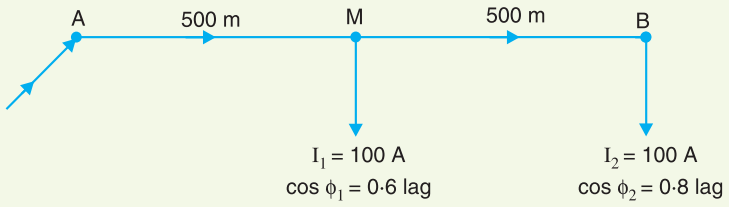


Fig. 14.6

(*i*) Load current at point *B*, $\overrightarrow{I_2} = 100\,(0{\cdot}8 - j\,0{\cdot}6) = 80 - j\,60$

Current in section *MB*, $\overrightarrow{I_{MB}} = \overrightarrow{I_2} = 80 - j\,60$

Drop in section *MB*, $\overrightarrow{V_{MB}} = \overrightarrow{I_{MB}}\,\overrightarrow{Z_{MB}}$

$= (80 - j\,60)\,(0{\cdot}1 + j\,0{\cdot}15) = 17 + j\,6$

$\therefore$ Voltage at point *M*, $\overrightarrow{V_M} = \overrightarrow{V_B} + \overrightarrow{V_{MB}} = (200 + j\,0) + (17 + j\,6)$

$= 217 + j\,6$

Its magnitude is $= \sqrt{(217)^2 + (6)^2} = \mathbf{217{\cdot}1\ V}$

Phase angle between V_M and V_B, $\alpha = \tan^{-1} 6/217 = \tan^{-1} 0{\cdot}0276 = 1{\cdot}58^\circ$

(*ii*) The load current I_1 has a lagging p.f. of 0·6 *w.r.t.* V_M. It lags behind V_M by an angle $\phi_1 = \cos^{-1} 0{\cdot}6 = 53{\cdot}13^\circ$

$\therefore$ Phase angle between I_1 and V_B, $\phi'_1 = \phi_1 - \alpha = 53{\cdot}13^\circ - 1{\cdot}58 = 51{\cdot}55^\circ$

Load current at *M*, $\overrightarrow{I_1} = I_1\,(\cos\phi_1' - j\sin\phi_1') = 100\,(\cos 51{\cdot}55^\circ - j\sin 51{\cdot}55^\circ)$

$= 62{\cdot}2 - j\,78{\cdot}3$

Current in section *AM*, $\overrightarrow{I_{AM}} = \overrightarrow{I_1} + \overrightarrow{I_2} = (62{\cdot}2 - j\,78{\cdot}3) + (80 - j\,60)$

$= 142{\cdot}2 - j\,138{\cdot}3$

Drop in section *AM*, $\overrightarrow{V_{AM}} = \overrightarrow{I_{AM}}\,\overrightarrow{Z_{AM}} = (142{\cdot}2 - j\,138{\cdot}3)\,(0{\cdot}1 + j\,0{\cdot}15)$

$= 34{\cdot}96 + j\,7{\cdot}5$

Sending end voltage, $\overrightarrow{V_A} = \overrightarrow{V_M} + \overrightarrow{V_{AM}} = (217 + j\,6) + (34{\cdot}96 + j\,7{\cdot}5)$

$= 251{\cdot}96 + j\ 13{\cdot}5$

Its magnitude is $= \sqrt{(251{\cdot}96)^2 + (13{\cdot}5)^2} = \mathbf{252{\cdot}32\ V}$

(iii) The phase difference θ between V_A and V_B is given by :

$\tan\theta = 13{\cdot}5/251{\cdot}96 = 0{\cdot}05358$

$\therefore \quad \theta = \tan^{-1} 0{\cdot}05358 = \mathbf{3{\cdot}07^o}$

Hence supply voltage is 252·32 V and leads V_B by 3·07°.

Example 14.4. *A single phase ring distributor ABC is fed at A. The loads at B and C are 20 A at 0.8 p.f. lagging and 15 A at 0.6 p.f. lagging respectively ; both expressed with reference to the voltage at A. The total impedance of the three sections AB, BC and CA are (1 + j 1), (1+ j2) and (1 + j3) ohms respectively. Find the total current fed at A and the current in each section. Use Thevenin's theorem to obtain the results.*

Solution. Fig. 14.7 (*i*) shows the ring distributor *ABC*. Thevenin's theorem will be used to solve this problem. First, let us find the current in *BC*. For this purpose, imagine that section *BC* is removed as shown in Fig. 14.7 (*ii*).

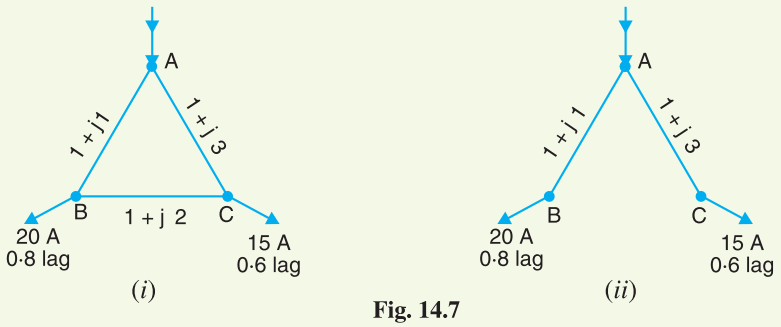


Fig. 14.7

Referring to Fig.14.7 (*ii*), we have,

Current in section *AB* $= 20\ (0{\cdot}8 - j\ 0.6) = 16 - j\ 12$

Current in section *AC* $= 15\ (0{\cdot}6 - j\ 0{\cdot}8) = 9 - j\ 12$

Voltage drop in section *AB* $= (16 - j\ 12)\ (1 + j1) = 28 + j\ 4$

Voltage drop in section *AC* $= (9 - j\ 12)\ (1 + j\ 3) = 45 + j\ 15$

Obviously, point *B* is at higher potential than point *C*. The p.d. between *B* and *C* is Thevenin's equivalent circuit e.m.f. E_0 *i.e.*

Thevenin's equivalent circuit e.m.f., E_0 = p.d. between *B* and *C*

$= (45 + j\ 15) - (28 + j\ 4) = 17 + j\ 11$

Thevenin's equivalent impedance Z_0 can be found by looking into the network from points *B* and *C*.

Obviously, $Z_0 = (1 + j1) + (1 + j\ 3) = 2 + j4$

$\therefore$ Current in $BC = \dfrac{E_0}{Z_0 + \text{Impedance of } BC}$

$= \dfrac{17 + j11}{(2 + j4) + (1 + j2)} = \dfrac{17 + j11}{3 + j6}$

$= 2{\cdot}6 - j\ 1{\cdot}53 = \mathbf{3\angle -30{\cdot}48^o\ A}$

Current in $AB = (16 - j\ 12) + (2{\cdot}6 - j\ 1{\cdot}53)$

$= 18{\cdot}6 - j\,13{\cdot}53 = \mathbf{23\angle -36{\cdot}03^{o}\ A}$

Current in AC $= (9 - j\,12) - (2{\cdot}6 - j\,1{\cdot}53)$

$= 6{\cdot}4 - j\,10{\cdot}47 = \mathbf{12{\cdot}27\angle -58{\cdot}56^{o}\ A}$

Current fed at A $= (16 - j\,12) + (9 - j\,12)$

$= 25 - j\,24 = \mathbf{34{\cdot}65\angle -43{\cdot}83^{o}\ A}$

Example 14.5. *A 3-phase, 400V distributor AB is loaded as shown in Fig.14.8. The 3-phase load at point C takes 5A per phase at a p.f. of 0·8 lagging. At point B, a 3-phase, 400 V induction motor is connected which has an output of 10 H.P. with an efficiency of 90% and p.f. 0·85 lagging.*

If voltage at point B is to be maintained at 400 V, what should be the voltage at point A ? The resistance and reactance of the line are 1Ω and 0·5Ω per phase per kilometre respectively.

Solution. It is convenient to consider one phase only. Fig.14.8 shows the single line diagram of the distributor. Impedance of the distributor per phase per kilometre = $(1 + j\,0{\cdot}5)\,\Omega$.

Impedance of section AC, $\overrightarrow{Z_{AC}} = (1 + j\,0{\cdot}5) \times 600/1000 = (0{\cdot}6 + j\,0{\cdot}3)\ \Omega$

Impedance of section CB, $\overrightarrow{Z_{CB}} = (1 + j\,0{\cdot}5) \times 400/1000 = (0{\cdot}4 + j\,0{\cdot}2)\ \Omega$

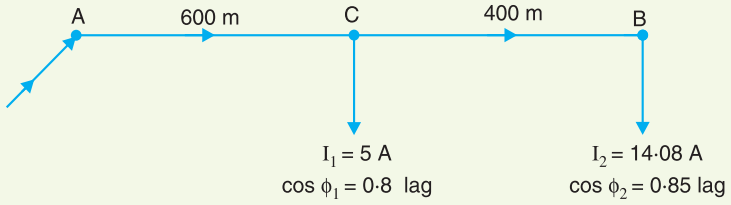


Fig. 14.8

Phase voltage at point B, $V_B = 400/\sqrt{3} = 231$ V

Let the voltage V_B at point B be taken as the reference vector.

Then, $\overrightarrow{V_B} = 231 + j\,0$

Line current at B $= \dfrac{\text{H.P.} \times 746}{\sqrt{3} \times \text{line voltage} \times \text{p.f.} \times \text{efficiency}}$

$= \dfrac{10 \times 746}{\sqrt{3} \times 400 \times 0{\cdot}85 \times 0{\cdot}9} = 14{\cdot}08$ A

$\therefore$ *Current/phase at B, $I_2 = 14{\cdot}08$ A

Load current at B, $\overrightarrow{I_2} = 14{\cdot}08\,(0{\cdot}85 - j\,0{\cdot}527) = 12 - j\,7{\cdot}4$

Load current at C, $\overrightarrow{I_1} = 5\,(0{\cdot}8 - j\,0{\cdot}6) = 4 - j\,3$

Current in section AC, $\overrightarrow{I_{AC}} = \overrightarrow{I_1} + \overrightarrow{I_2} = (4 - j\,3) + (12 - j\,7{\cdot}4)$

$= 16 - j\,10{\cdot}4$

Current in section CB, $\overrightarrow{I_{CB}} = \overrightarrow{I_2} = 12 - j\,7{\cdot}4$

Voltage drop in CB, $\vec{V}_{CB} = \vec{I}_{CB}\vec{Z}_{CB} = (12 - j\,7{\cdot}4)\,(0{\cdot}4 + j\,0{\cdot}2)$

$= 6{\cdot}28 - j\,0{\cdot}56$

Voltage drop in AC, $\vec{V}_{AC} = \vec{I}_{AC}\vec{Z}_{AC} = (16 - j\,10{\cdot}4)\,(0{\cdot}6 + j\,0{\cdot}3)$

$= 12{\cdot}72 - j\,1{\cdot}44$

* In a 3-phase system, if the type of connection is not mentioned, then star connection is understood.

Voltage at A per phase, $\vec{V}_A = \vec{V}_B + \vec{V}_{CB} + \vec{V}_{AC}$

$= (231 + j\,0) + (6{\cdot}28 - j\,0{\cdot}56) + (12{\cdot}72 - j\,1{\cdot}44)$

$= 250 - j\,2$

Magnitude of V_A/phase $= \sqrt{(250)^2 + (2)^2} = 250$ V

$\therefore$ Line voltage at A $= \sqrt{3} \times 250 =$ **433 V**

Example 14.6. *A 3-phase ring main ABCD fed at A at 11 kV supplies balanced loads of 50 A at 0.8 p.f. lagging at B, 120 A at unity p.f. at C and 70 A at 0·866 lagging at D, the load currents being referred to the supply voltage at A. The impedances of the various sections are :*

Section AB $= (1 + j\,0{\cdot}6)\ \Omega$; *Section BC* $= (1{\cdot}2 + j\,0{\cdot}9)\ \Omega$

Section CD $= (0{\cdot}8 + j\,0{\cdot}5)\ \Omega$; *Section DA* $= (3 + j\,2)\ \Omega$

Calculate the currents in various sections and station bus-bar voltages at B, C and D.

Solution. Fig.14.9 shows one phase of the ring main. The problem will be solved by Kirchhoff's laws. Let current in section AB be $(x + j\,y)$.

$\therefore$ Current in section BC, $\overrightarrow{I_{BC}} = (x + j\,y) - 50\,(0{\cdot}8 - j\,0{\cdot}6) = (x - 40) + j\,(y + 30)$

Current in section CD, $\overrightarrow{I_{CD}} = [(x - 40) + j\,(y + 30)] - [120 + j\,0]$

$= (x - 160) + j\,(y + 30)$

Current in section DA, $\overrightarrow{I_{DA}} = [(x - 160) + j\,(y + 30)] - [70\,(0{\cdot}866 - j\,0{\cdot}5)]$

$= (x - 220{\cdot}6) + j\,(y + 65)$

Drop in section AB $= \overrightarrow{I_{AB}}\,\overrightarrow{Z_{AB}} = (x + jy)\,(1 + j0{\cdot}6)$

$= (x - 0{\cdot}6y) + j\,(0{\cdot}6x + y)$

Drop in section BC $= \overrightarrow{I_{BC}}\,\overrightarrow{Z_{BC}}$

$= [(x - 40) + j\,(y + 30)]\,[(1{\cdot}2 + j\,0{\cdot}9)]$

$= (1{\cdot}2x - 0{\cdot}9\,y - 75) + j\,(0{\cdot}9x + 1{\cdot}2\,y)$

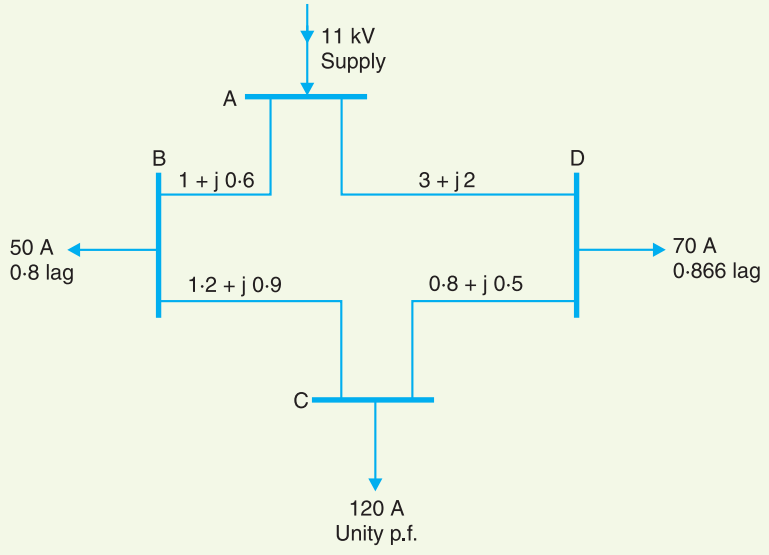


Fig. 14.9

Drop in section $CD = \overrightarrow{I_{CD}}\,\overrightarrow{Z_{CD}}$

$= [(x - 160) + j\,(y + 30)]\,[(0{\cdot}8 + j\,0{\cdot}5)]$

$= (0{\cdot}8x - 0{\cdot}5y - 143) + j\,(0{\cdot}5x + 0{\cdot}8y - 56)$

Drop in section $DA = \overrightarrow{I_{DA}}\,\overrightarrow{Z_{DA}}$

$= [(x - 220{\cdot}6) + j\,(y + 65)]\,[(3 + j\,2)]$

$= (3x - 2y - 791{\cdot}8) + j\,(2x + 3y - 246{\cdot}2)$

Applying Kirchhoff's voltage law to mesh $ABCDA$, we have,

Drop in AB + Drop in BC + Drop in CD + Drop in $DA = 0$

or $[(x - 0{\cdot}6y) + j\,(0{\cdot}6x + y)] + [(1{\cdot}2x - 0{\cdot}9y - 75) + j\,(0{\cdot}9x + 1{\cdot}2y)]$

$+ [(0{\cdot}8x - 0{\cdot}5y - 143) + j\,(0{\cdot}5x + 0{\cdot}8y - 56)]$

$+ [(3x - 2y - 791{\cdot}8) + j\,(2x + 3y - 246{\cdot}2)] = 0$

or $(6x - 4y - 1009{\cdot}8) + j\,(4x + 6y - 302{\cdot}2) = 0$

As the real (or active) and imaginary (or reactive) parts have to be separately zero,

$\therefore \quad 6x - 4y - 1009{\cdot}8 = 0$

and $\quad 4x + 6y - 302{\cdot}2 = 0$

Solving for x and y, we have,

$x = 139{\cdot}7$ A ; $y = -42{\cdot}8$ A

Current in section AB $= \mathbf{(139{\cdot}7 - j\,42{\cdot}8)\ A}$

Current in section BC $= (x - 40) + j\,(y + 30)$

$= (139{\cdot}7 - 40) + j\,(-42{\cdot}8 + 30) = \mathbf{(99{\cdot}7 - j\,12{\cdot}8)\ A}$

Current in section CD $= (x - 160) + j\,(y + 30)$

$= (139{\cdot}7 - 160) + j\,(-42{\cdot}8 + 30)$

$= \mathbf{(-20{\cdot}3 - j\,12{\cdot}8)\ A}$

Current in section DA $= (x - 220{\cdot}6) + j\,(y + 65)$

$= (139{\cdot}7 - 220{\cdot}6) + j\,(-42{\cdot}8 + 65)$

$= \mathbf{(-80{\cdot}9 + j\,22{\cdot}2)\ A}$

Voltage at supply end A, $V_A = 11000/\sqrt{3} = 6351$ V/phase

$\therefore$ Voltage at station B, $\overrightarrow{V_B} = \overrightarrow{V_A} - \overrightarrow{I_{AB}}\,\overrightarrow{Z_{AB}}$

$= (6351 + j\,0) - (139{\cdot}7 - j\,42{\cdot}8)\,(1 + j\,0{\cdot}6)$

$= \mathbf{(6185{\cdot}62 - j\,41{\cdot}02)\ volts/phase}$

Voltage at station C, $\overrightarrow{V_C} = \overrightarrow{V_B} - \overrightarrow{I_{BC}}\,\overrightarrow{Z_{BC}}$

$= (6185{\cdot}62 - j\,41{\cdot}02) - (99{\cdot}7 - j\,12{\cdot}8)\,(1{\cdot}2 + j\,0{\cdot}9)$

$= \mathbf{(6054{\cdot}46 - j\,115{\cdot}39)\ volts/phase}$

Voltage at station D, $\overrightarrow{V_D} = \overrightarrow{V_C} - \overrightarrow{I_{CD}}\,\overrightarrow{Z_{CD}}$

$= (6054{\cdot}46 - j\,115{\cdot}39) - (-20{\cdot}3 - j\,12{\cdot}8) \times (0{\cdot}8 + j\,0{\cdot}5)$

$= \mathbf{(6064{\cdot}3 - j\,95)\ volts/phase}$

TUTORIAL PROBLEMS

1. A single phase distributor *AB* has a total impedance of (0·1 + *j* 0·2) ohm. At the far end *B*, a current of 80 A at 0·8 p.f. lagging and at mid-point *C* a current of 100 A at 0·6 p.f. lagging are tapped. If the voltage of the far end is maintained at 200 V, determine :

(*i*) Supply end voltage V_A

(*ii*) Phase angle between V_A and V_B

The load power factors are w.r.t. the voltage at the far end. **[(*i*) 227·22 V (*ii*) 2°31′]**

2. A single-phase a.c. distributor *AB* is fed from end *A* and has a total impedance of (0·2 + *j* 03) ohm. At the far end, the voltage V_B = 240 V and the current is 100 A at a p.f. of 0·8 lagging. At the mid-point *M*, a current of 100 A is tapped at a p.f. of 0·6 lagging with reference to the voltage V_M at the mid-point. Calculate the supply voltage V_A and phase angle between V_A and V_B. **[292 V, 2·6°]**

3. A single phase ring distributor *ABC* is fed at A. The loads at *B* and *C* are 40 A at 0·8 p.f. lagging and 60 A at 0·6 p.f. lagging respectively. Both power factors expressed are referred to the voltage at point A. The total impedance of sections *AB*, *BC* and *CA* are 2 + *j*1, 2 + *j*3 and 1 + *j*2 ohms respectively. Determine the current in each section.

[Current in *AB* = (39·54 – *j* 25·05) amp ; *BC* = (7·54 – *j* 1·05) amp ; *CA* = (28·46 – *j* 46·95) amp.]

4. A 3-phase ring distributor *ABCD* fed at A at 11 kV supplies balanced loads of 40 A at 0·8 p.f. lagging at *B*, 50 A at 0·707 p.f. lagging at C and 30 A at 0·8 p.f. lagging at D, the load currents being referred to the supply voltage at A.

The impedances per phase of the various sections are :

Section *AB* = (1 + *j* 2) Ω ; Section *BC* = (2 + *j* 3) Ω

Section *CD* = (1 + *j* 1) Ω ; Section *DA* = (3 + *j* 4) Ω

Calculate the currents in various sections and station bus-bar voltages at *B*, *C* and *D*.

[Current in *AB* = (53·8 – *j* 46) amp ; *BC* = (21·8 – *j* 22) amp.
CD = (–13·55 + *j* 13·35) amp ; *DA* = (–40·55 – *j* 26·45) amp.
V_B = (6212·5 – *j* 61·6) volts/phase ; V_C = (6103 – *j* 83) volts/phase
V_D = (6129·8 – *j* 82·8) volts/phase]



Phase Sequence Indicator

14.3 3-Phase Unbalanced Loads

The 3-phase loads that have the same impedance and power factor in each phase are called balanced loads. The problems on balanced loads can be solved by considering one phase only ; the conditions in the other two phases being similar. However, we may come across a situation when loads are unbalanced *i.e.* each load phase has different impedance and/or power factor. In that case, current and power in each phase will be different. In practice, we may come across the following unbalanced loads :

(*i*) Four-wire star-connected unbalanced load

(*ii*) Unbalanced Δ-connected load

(*iii*) Unbalanced 3-wire, *Y*-connected load

The 3-phase, 4-wire system is widely used for distribution of electric power in commercial and industrial buildings. The single phase load is connected between any line and neutral wire while a 3-phase load is connected across the three lines. The 3-phase, 4-wire system invariably carries *un-balanced loads. In this chapter, we shall only discuss this type of unbalanced load.

14.4 Four-Wire Star-Connected Unbalanced Loads

We can obtain this type of load in two ways. First, we may connect a 3-phase, 4-wire unbalanced load to a 3-phase, 4-wire supply as shown in Fig. 14.10. Note that star point *N* of the supply is connected to the load star point *N*′. Secondly, we may connect single phase loads between any line and the neutral wire as shown in Fig.14.11. This will also result in a 3-phase, 4-wire **unbalanced load because it is rarely possible that single phase loads on all the three phases have the same magnitude and power factor. Since the load is unbalanced, the line currents will be different in magnitude and displaced from one another by unequal angles. The current in the neutral wire will be the phasor sum of the three line currents *i.e.*

Current in neutral wire, $I_N = I_R + I_Y + I_B$ ...*phasor sum*

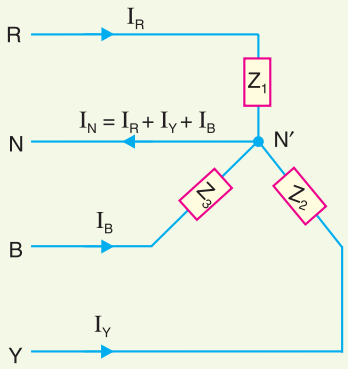


Fig. 14.10

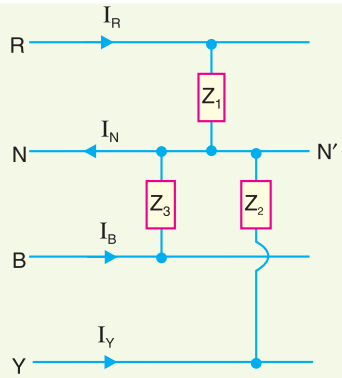


Fig. 14.11

The following points may be noted carefully :

(*i*) Since the neutral wire has negligible resistance, supply neutral *N* and load neutral *N*′ will be at the same potential. It means that voltage across each impedance is equal to the phase voltage of the supply. However, current in each phase (or line) will be different due to unequal impedances.

(*ii*) The amount of current flowing in the neutral wire will depend upon the magnitudes of line currents and their phasor relations. In most circuits encountered in practice, the neutral current is equal to or smaller than one of the line currents. The exceptions are those circuits having severe unbalance.

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* No doubt 3-phase loads (*e.g.* 3-phase motors) connected to this supply are balanced but when we add single phase loads (*e.g.* lights, fans etc.), the balance is lost. It is because it is rarely possible that single phase loads on all the three phases have the same magnitude and power factor.

** In actual practice, we never have an unbalanced 3-phase, 4-wire load. Most of the 3-phase loads (*e.g.* 3-phase motors) are 3-phase, 3-wire and are balanced loads. In fact, these are the single phase loads on the 3-phase, 4-wire supply which constitute unbalanced, 4-wire *Y*-connected load.

Example 14.7. *Non-reactive loads of 10 kW, 8 kW and 5 kW are connected between the neutral and the red, yellow and blue phases respectively of a 3-phase, 4-wire system. The line voltage is 400V. Calculate (i) the current in each line and (ii) the current in the neutral wire.*

Solution. This is a case of unbalanced load so that the line currents (and hence the phase currents) in the three lines will be different. The current in the *neutral wire will be equal to the phasor sum of three line currents as shown in Fig. 14.12.

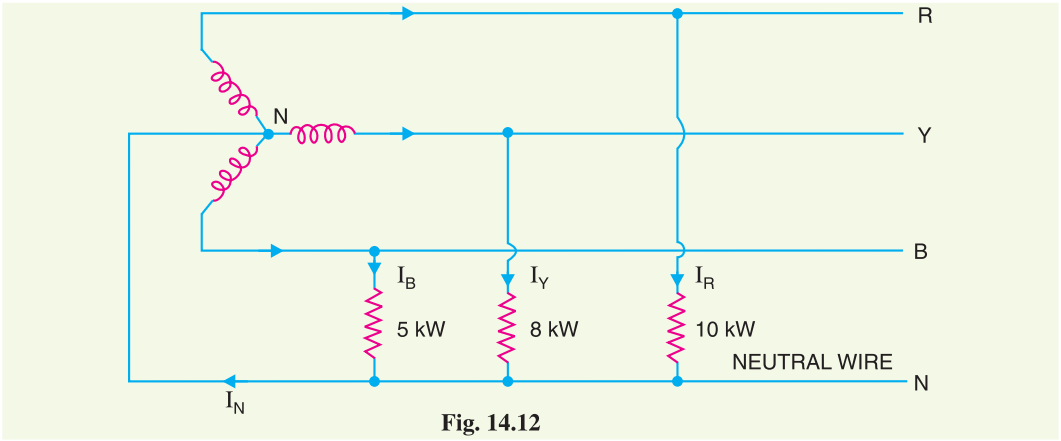


Fig. 14.12

(i) Phase voltage = $400/\sqrt{3} = 231$ V

$I_R = 10 \times 10^3/231 =$ **43·3 A**

$I_Y = 8 \times 10^3/231 =$ **34·6 A**

$I_B = 5 \times 10^3/231 =$ **21·65 A**

(ii) The three lines currents are represented by the respective phasors in Fig. 14.13. Note that the three line currents are of different magnitude but displaced 120° from one another. The current in the neutral wire will be the phasor sum of the three line currents.

Resolving the three currents along *x*-axis and *y*-axis, we have,

Resultant horizontal component $= I_Y \cos 30^o - I_B \cos 30^o$

$= 34{\cdot}6 \times 0{\cdot}866 - 21{\cdot}65 \times 0{\cdot}866 = 11{\cdot}22$ A

Resultant vertical component $= I_R - I_Y \cos 60^o - I_B \cos 60^o$

$= 43{\cdot}3 - 34{\cdot}6 \times 0{\cdot}5 - 21{\cdot}65 \times 0{\cdot}5 = 15{\cdot}2$ A

As shown in Fig. 14.14, current in neutral wire is

$$I_N = \sqrt{(11{\cdot}22)^2 + (15{\cdot}2)^2} = \textbf{18·9 A}$$

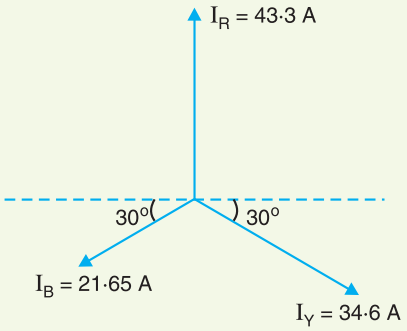


Fig. 14.13

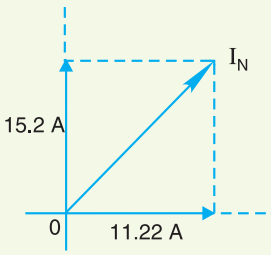


Fig. 14.14

* Had the load been balanced (*i.e.* each phase having identical load), the current in the neutral wire would have been zero.

Example 14.8. *A 3-phase, 4-wire system supplies power at 400 V and lighting at 230 V. If the lamps is use require 70, 84 and 33 amperes in each of the three lines, what should be the current in the neutral wire ? If a 3-phase motor is now started, taking 200 A from the lines at a p.f. of 0·2 lagging, what should be the total current in each line and the neutral wire ? Find also the total power supplied to the lamps and the motor.*

Solution. Fig. 14.15 shows the lamp load and motor load on 400 V/230 V, 3-phase, 4-wire sypply.

Lamp load alone. If there is lamp load alone, the line currents in phases *R*,*Y* and *B* are 70 *A*, 84 *A* and 33 *A* respectively. These currents will be 120° apart (assuming phase sequence *RYB*) as shown in Fig.14.16.

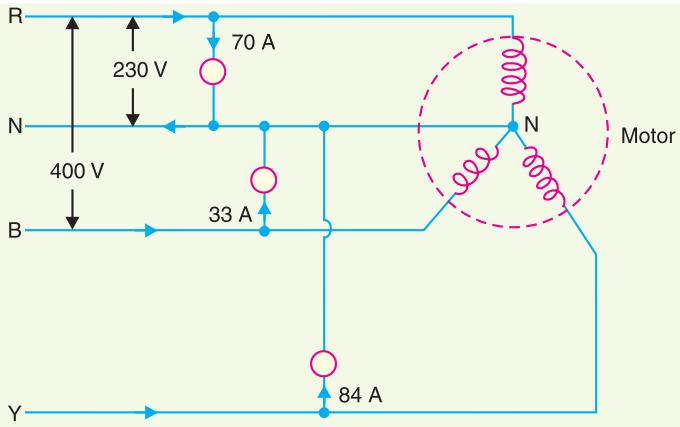


Fig. 14.15

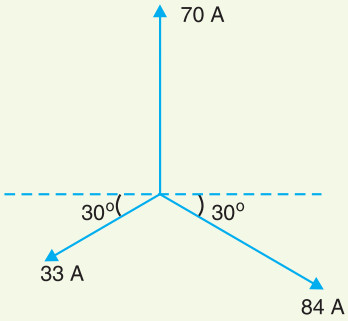


Fig. 14.16

Resultant *H*-component = $84 \cos 30^\circ - 33 \cos 30^\circ = 44{\cdot}17$ A

Resultant V-component = $70 - 33 \cos 60^\circ - 84 \cos 60^\circ = 11{\cdot}5$ A

∴ Neutral current, $I_N = \sqrt{(44{\cdot}17)^2 + (11{\cdot}5)^2}$ = **45·64 A**

Both lamp load and motor load

When motor load is also connected along with lighting load, there will be no change in current in the neutral wire. It is because the motor load is balanced and hence no current will flow in the neutral wire due to this load.

∴ Neutral current, I_N = **45·64 A** *...same as before*

The current in each line is the phasor sum of the line currents due to lamp load and motor load.

Active component of motor current = $200 \times \cos \phi_m = 200 \times 0{\cdot}2 = 40$ A

Reactive component of motor current = $200 \times \sin \phi_m = 200 \times 0{\cdot}98 = 196$ A

∴ $I_R = \sqrt{(\text{sum of active comp.})^2 + (\text{reactive comp.})^2}$

$= \sqrt{(40 + 70)^2 + (196)^2}$ = **224·8 A**

$I_Y = \sqrt{(40 + 84)^2 + (196)^2}$ = **232 A**

$I_B = \sqrt{(40 + 33)^2 + (196)^2}$ = **209·15 A**

Power supplied

Power supplied to lamps = $230\ (70 + 84 + 33) \times 1$ = **43010 W** ($\because \cos \phi_L = 1$)

Power supplied to motor $= \sqrt{3}\ V_L I_L \cos\phi_m$

$= \sqrt{3} \times 400 \times 200 \times 0{\cdot}2 =$ **27712 W**

Example 14.9. *The three line leads of a 400/230 V, 3-phase, 4-wire supply are designated as R, Y and B respectively. The fourth wire or neutral wire is designated as N. The phase sequence is RYB. Compute the currents in the four wires when the following loads are connected to this supply :*

From R to N : 20 kW, unity power factor

From Y to N : 28·75 kVA, 0·866 lag

From B to N : 28·75 kVA, 0·866 lead

If the load from B to N is removed, what will be the value of currents in the four wires ?

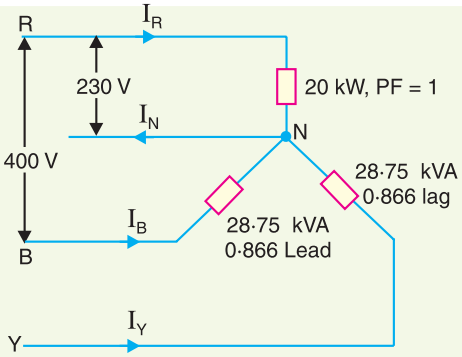


Fig. 14.17

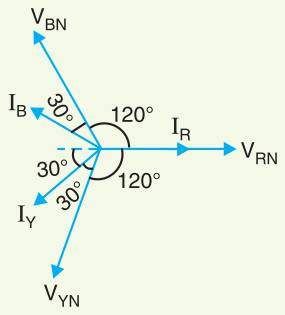


Fig. 14.18

Solution. Fig. 14.17 shows the circuit diagram whereas Fig.14.18 shows its phasor diagram. The current I_R is in phase with V_{RN}, current I_Y lags behind its phase voltage V_{YN} by $\cos^{-1} 0{\cdot}866 = 30^o$ and the current I_B leads its phase voltage V_{BN} by $\cos^{-1} 0{\cdot}866 = 30^o$.

$I_R = 20 \times 10^3/230 =$ **89·96 A**

$I_Y = 28{\cdot}75 \times 10^3/230 =$ **125 A**

$I_B = 28{\cdot}75 \times 10^3/230 =$ **125 A**

The current in the neutral wire will be equal to the phasor sum of the three line currents I_R, I_Y and I_B. Referring to the phasor diagram in Fig.14.18 and resolving these currents along *x*-axis and *y*-axis, we have,

Resultant *X*-component $= 86{\cdot}96 - 125 \cos 30^o - 125 \cos 30^o$

$= 86{\cdot}96 - 108{\cdot}25 - 108{\cdot}25 = -129{\cdot}54$ A

Resultant *Y*-component $= 0 + 125 \sin 30^o - 125 \sin 30^o = 0$

$\therefore$ Neutral current, $I_N = \sqrt{(-129{\cdot}54)^2 + (0)^2} =$ **129·54A**

When load from *B* to *N* removed. When the load from *B* to *N* is removed, the various line currents are :

$I_R =$ **86·96A** in phase with V_{RN} ; $I_Y =$ **125A** lagging V_{YN} by 30° ; $I_B =$ **0 A**

The current in the neutral wire is equal to the phasor sum of these three line currents. Resolving the currents along *x*-axis and *y*-axis, we have,

Resultant *X*-component $= 86{\cdot}96 - 125 \cos 30^o = 86{\cdot}96 - 108{\cdot}25 = -21{\cdot}29$ A

Resultant *Y*-component $= 0 - 125 \sin 30^o = 0 - 125 \times 0{\cdot}5 = -62{\cdot}5$ A

$\therefore$ Neutral current, $I_N = \sqrt{(-21{\cdot}29)^2 + (-62{\cdot}5)^2} =$ **66·03 A**

Example 14.10. *A 3-phase, 4-wire distributor supplies a balanced voltage of 400/230 V to a load consisting of 30 A at p.f. 0·866 lagging for R-phase, 30 A at p.f. 0·866 leading for Y phase and 30 A at unity p.f. for B phase. The resistance of each line conductor is 0·2 Ω. The area of X-section of neutral is half of any line conductor. Calculate the supply end voltage for R phase. The phase sequence is RYB.*

Solution. The circuit diagram is shown in Fig. 14.19. Since neutral is half the cross-section, its resistance is 0·4Ω. Considering the load end and taking V_R as the reference vector, the phase voltages can be written as :

$\overrightarrow{V_R} = 230 \angle 0^\circ$ volts ; $\overrightarrow{V_Y} = 230 \angle -120^\circ$ volts ; $\overrightarrow{V_B} = 230 \angle 120^\circ$ volts

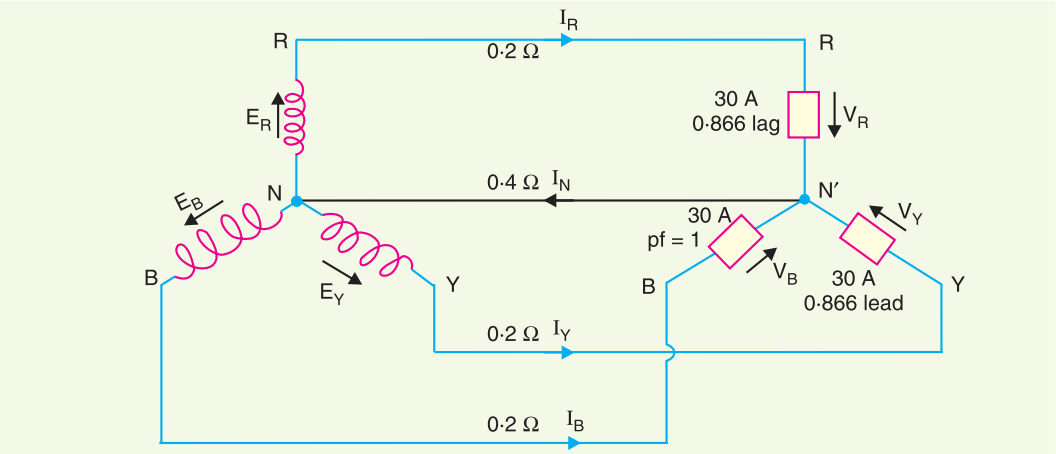


Fig. 14.19

The vector diagram of the circuit is shown in Fig. 14.20. The line current I_R lags behind V_R by an angle $\cos^{-1} 0{\cdot}866 = 30^\circ$. The current I_Y leads V_Y by 30° and the current I_B is in phase with V_B. Referring to the vector diagram of Fig.14.20, the line currents can be expressed as :

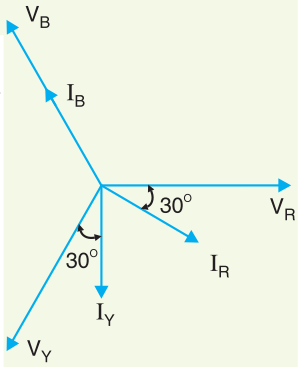


Fig. 14.20

$$\overrightarrow{I_R} = 30 \angle -30^\circ \text{ amperes}$$

$$\overrightarrow{I_Y} = 30 \angle -90^\circ \text{ amperes}$$

$$\overrightarrow{I_B} = 30 \angle 120^\circ \text{ amperes}$$

Current in neutral wire, $\overrightarrow{I_N} = \overrightarrow{I_R} + \overrightarrow{I_Y} + \overrightarrow{I_B}$

$$= 30 \angle -30^\circ + 30 \angle -90^\circ + 30 \angle 120^\circ$$

$$= 30\,(0{\cdot}866 - j\,0{\cdot}5) - 30\,(j) + 30\,(-0{\cdot}5 + j\,0{\cdot}866)$$

$$= 10{\cdot}98 - j\,19{\cdot}02$$

Let the supply voltage of phase R to neutral be $\overrightarrow{E_R}$. Then,

$\overrightarrow{E_R}$ = $\overrightarrow{V_R}$ + Drop in R phase + Drop in neutral

$$= (230 + j\,0) + 0{\cdot}2 \times 30 \angle -30^\circ + (10{\cdot}98 - j\,19{\cdot}02) \times 0{\cdot}4$$

$$= 230 + 6\,(0{\cdot}866 - j\,0{\cdot}5) + 0{\cdot}4\,(10{\cdot}98 - j\,19{\cdot}02)$$

$$= 239{\cdot}588 - j\,10{\cdot}608$$

$$= \mathbf{239{\cdot}8 \angle -2{\cdot}54^\circ \text{ volts}}$$

Example 14.11. *In a 3-phase, 4-wire, 400/230 V system, a lamp of 100 watts is connected to one phase and neutral and a lamp of 150 watts is connected to the second phase and neutral. If the neutral wire is disconnected accidentally, what will be the voltage across each lamp ?*

Solution. Fig. 14.21 (*i*) shows the lamp connections. The lamp L_1 of 100 watts is connected between phase *R* and neutral whereas lamp L_2 of 150 watts is connected between phase *Y* and the neutral.

$$\text{Resistance of lamp } L_1, \quad R_1 = \frac{(230)^2}{100} = 529\ \Omega$$

$$\text{Resistance of lamp } L_2, \quad R_2 = \frac{(230)^2}{150} = 352{\cdot}67\ \Omega$$

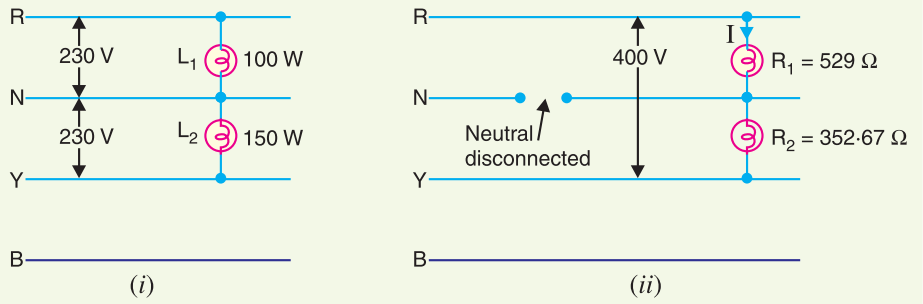


Fig. 14.21

When the neutral wire is disconnected as shown in Fig. 14.21 (*ii*), the two lamps are connected in series and the p.d. across the combination becomes equal to the line voltage E_L (= 400 V).

$$\text{Current through lamps, } I = \frac{E_L}{R_1 + R_2} = \frac{400}{529 + 352{\cdot}67} = 0{\cdot}454\ \text{A}$$

$$\text{Voltage across lamp } L_1 = I R_1 = 0{\cdot}454 \times 529 = \mathbf{240\ V}$$

$$\text{Voltage across lamp } L_2 = I R_2 = 0{\cdot}454 \times 352{\cdot}67 = \mathbf{160\ V}$$

Comments. The voltage across 100-watt lamp is increased to 240 V whereas that across 150-watt is decreased to 160 V. Therefore, 100-watt lamp becomes brighter and 150-watt lamp becomes dim. It may be noted here that if 100-watt lamp happens to be rated at 230 V, it may burn out due to 240 V coming across it.

TUTORIAL PROBLEMS

1. Non-reactive loads of 10 kW, 6kW and 4 kW are connected between the neutral and red, yellow and blue phases respectively of a 3-phase, 4-wire 400/230V supply. Find the current in each line and in the neutral wire. **[I_R = 43·3 A; I_Y = 26 A; I_B = 17·3 A; I_N = 22·9 A]**
2. A factory has the following loads with a power factor of 0·9 lagging in each case. Red phase 40 A, yellow phase 50 A and blue phase 60 A. If the supply is 400V, 3-phase, 4-wire, calculate the current in the neutral wire and the total power. **[17·3A, 31·2 kW]**
3. In a 3-phase, 4-wire system, two phases have currents of 10A and 6A at lagging power factors of 0·8 and 0·6 respectively, while the third phase is open-circuited. Calculate the current in the neutral wire. **[7A]**
4. A 3-phase, 4-wire system supplies a lighting load of 40A, 30A and 20A respectively in the three phases. If the line voltage is 400 V, determine the current in the neutral wire. **[17·32A]**

14.5. Ground Detectors

Ground detectors are the devices that are used to detect the ground fault for ungrounded a.c. systems.

traction, electroplating, *d.c. motors etc. This job is again performed by suitable apparatus (*e.g.* ignitron) called sub-station. It is clear that type of equipment needed in a sub-station will depend upon the service requirement. Although there can be several types of sub-stations, we shall mainly confine our attention to only those sub-stations where the incoming and outgoing supplies are a.c. *i.e.* sub-stations which change the voltage level of the electric supply.

25.1 Sub-Station

The assembly of apparatus used to change some characteristic (*e.g. voltage, a.c. to d.c., frequency, p.f. etc.*) *of electric supply is called a* **sub-station.**

Sub-stations are important part of power system. The continuity of supply depends to a considerable extent upon the successful operation of sub-stations. It is, therefore, essential to exercise utmost care while designing and building a sub-station. The following are the important points which must be kept in view while laying out a sub-station :

(*i*) It should be located at a proper site. As far as possible, it should be located at the centre of gravity of load.

(*ii*) It should provide safe and reliable arrangement. For safety, consideration must be given to the maintenance of regulation clearances, facilities for carrying out repairs and maintenance, abnormal occurrences such as possibility of explosion or fire etc. For reliability, consideration must be given for good design and construction, the provision of suitable protective gear *etc.*

(*iii*) It should be easily operated and maintained.

(*iv*) It should involve minimum capital cost.

25.2 Classification of Sub-Stations

There are several ways of classifying sub-stations. However, the two most important ways of classifying them are according to (1) service requirement and (2) constructional features.

1. According to service requirement. A sub-station may be called upon to change voltage level or improve power factor or convert a.c. power into d.c. power etc. According to the service requirement, sub-stations may be classified into :

(*i*) Transformer sub-stations. Those sub-stations which change the voltage level of electric supply are called transformer sub-stations. These sub-stations receive power at some voltage and deliver it at some other voltage. Obviously, transformer will be the main component in such sub-stations. Most of the sub-stations in the power system are of this type.

(*ii*) Switching sub-stations. These sub-stations do not change the voltage level *i.e.* incoming and outgoing lines have the same voltage. However, they simply perform the switching operations of power lines.

(*iii*) Power factor correction sub-stations. Those sub-stations which improve the power factor of the system are called power factor correction sub-stations. Such sub-stations are generally located at the receiving end of transmission lines. These sub-stations generally use synchronous condensers as the power factor improvement equipment.

(*iv*) Frequency changer sub-stations. Those sub-stations which change the supply frequency are known as frequency changer sub-stations. Such a frequency change may be required for industrial utilisation.

(*v*) Converting sub-stations. Those sub-stations which change a.c. power into d.c. power are called converting sub-stations. These sub-stations receive a.c. power and convert it into d.c. power

* Although most of the motors in operation in the world today are a.c. motors, yet one can find d.c. motors where fine speed control is required or where these were installed before the development of a.c. machinery.

with suitable apparatus (*e.g.* ignitron) to supply for such purposes as traction, electroplating, electric welding etc.

(*vi*) Industrial sub-stations. Those sub-stations which supply power to individual industrial concerns are known as industrial sub-stations.

2. According to constructional features. A sub-station has many components (*e.g.* circuit breakers, switches, fuses, instruments etc.) which must be housed properly to ensure continuous and reliable service. According to constructional features, the sub-stations are classified as :

(*i*) Indoor sub-station **(*ii*)** Outdoor sub-station
(*iii*) Underground sub-station **(*iv*)** Pole-mounted sub-station

(*i*) Indoor sub-stations. For voltages upto 11 kV, the equipment of the sub-station is installed indoor because of economic considerations. However, when the atmosphere is contaminated with impurities, these sub-stations can be erected for voltages upto 66 kV.

(*ii*) Outdoor sub-stations. For voltages beyond 66 kV, equipment is invariably installed outdoor. It is because for such voltages, the clearances between conductors and the space required for switches, circuit breakers and other equipment becomes so great that it is not economical to install the equipment indoor.

(*iii*) Underground sub-stations. In thickly populated areas, the space available for equipment and building is limited and the cost of land is high. Under such situations, the sub-station is created underground. The reader may find further discussion on underground sub-stations in Art. 25.6.

(*iv*) Pole-mounted sub-stations. This is an outdoor sub-station with equipment installed overhead on *H*-pole or 4-pole structure. It is the cheapest form of sub-station for voltages not exceeding 11kV (or 33 kV in some cases). Electric power is almost distributed in localities through such sub-stations. For complete discussion on pole-mounted sub-station, the reader may refer to Art. 25.5.

25.3 Comparison between Outdoor and Indoor Sub-Stations

The comparison between outdoor and indoor sub-stations is given below in the tabular form :

S.No.	Particular	Outdoor Sub-station	Indoor Sub-station
1	Space required	More	Less
2	Time required for erection	Less	More
3	Future extension	Easy	Difficult
4	Fault location	Easier beacuse the equipment is in full view	Difficult because the equipment is enclosed
5	Capital cost	Low	High
6	Operation	Difficult	Easier
7	Possibility of fault escalation	Less because greater clearances can be provided	More

From the above comparison, it is clear that each type has its own advantages and disadvantages. However, comparative economics (*i.e.* annual cost of operation) is the most powerful factor influencing the choice between indoor and outdoor sub-stations. The greater cost of indoor sub-station prohibits its use. But sometimes non-economic factors (*e.g.* public safety) exert considerable influence in choosing indoor sub-station. In general, most of the sub-stations are of outdoor type and the indoor sub-stations are erected only where outdoor construction is impracticable or prohibited by the local laws.

25.4 Transformer Sub-Stations

The majority of the sub-stations in the power system are concerned with the changing of voltage level of electric supply. These are known as transformer sub-stations because transformer is the main